

CHAPTER 17

ELECTRICITY

4.2 Electrical quantities

4.2.1 Electrical charge

- 1 State that there are positive and negative charges and that charge is measured in coulombs
- 2 State that unlike charges attract and like charges repel
- 3 Describe experiments to show electrostatic charging by friction
- 4 Explain that charging of solids by friction involves only a transfer of negative charge (electrons)
- 5 Describe an electric field as a region in which an electric charge experiences a force
- 6 State that the direction of an electric field line at a point is the direction of the force on a positive charge at that point
- 7 Describe simple electric field patterns, including the direction of the field:
 - (a) around a point charge
 - (b) around a charged conducting sphere
 - (c) between two oppositely charged parallel conducting plates (end effects will **not** be examined)
- 8 State examples of electrical conductors and insulators
- 9 Describe an experiment to distinguish between electrical conductors and insulators
- 10 Recall and use a simple electron model to explain the difference between electrical conductors and insulators

4.2.2 Electrical current

- 1 Define electric current as the charge passing a point per unit time; recall and use the equation
$$\text{electric current} = \frac{\text{charge}}{\text{time}}$$
$$I = \frac{Q}{t}$$
- 2 Describe electrical conduction in metals in terms of the movement of free electrons
- 3 Know that current is measured in amps (amperes) and that the amp is given by coulomb per second (C/s)
- 4 Know the difference between direct current (d.c.) and alternating current (a.c.)
- 5 State that conventional current is from positive to negative and that the flow of free electrons is from negative to positive
- 6 Describe the use of ammeters (analogue and digital) with different ranges

4.2.3 Electromotive force and potential difference

- 1 Define e.m.f. (electromotive force) as the electrical work done by a source in moving a unit charge around a complete circuit; recall and use the equation
$$\text{e.m.f.} = \frac{\text{work done (by a source)}}{\text{charge}}$$

$$E = \frac{W}{Q}$$

continued

4.2.3 Electromotive force and potential difference continued

- 2 Define p.d. (potential difference) as the work done by a unit charge passing through a component; recall and use the equation

$$\text{p.d.} = \frac{\text{work done (on a component)}}{\text{charge}}$$

$$V = \frac{W}{Q}$$

- 3 Know that e.m.f. and p.d. are measured in volts and that the volt is given by joule per coulomb (J/C)
- 4 Describe the use of voltmeters (analogue and digital) with different ranges
- 5 Calculate the total e.m.f. where several sources are arranged in series
- 6 State that the e.m.f. of identical sources connected in parallel is equal to the e.m.f. of one of the sources

4.2.4 Resistance

- 1 Recall and use the equation

$$\text{resistance} = \frac{\text{p.d.}}{\text{current}}$$

$$R = \frac{V}{I}$$

- 2 Describe an experiment to determine resistance using a voltmeter and an ammeter and do the appropriate calculations
- 3 Recall and use, for a wire, the direct proportionality between resistance and length, and the inverse proportionality between resistance and cross-sectional area
- 4 State Ohm's law, including reference to constant temperature
- 5 Sketch and explain the current-voltage graphs for a resistor of constant resistance, a filament lamp and a diode
- 6 Describe the effect of temperature increase on the resistance of a resistor, such as the filament in a filament lamp

4.3 Electric circuits

4.3.1 Circuit diagrams and circuit components

- 1 Draw and interpret circuit diagrams with cells, batteries, power supplies, generators, oscilloscopes, potential dividers, switches, resistors (fixed and variable), heaters, thermistors (NTC only), light-dependent resistors (LDRs), lamps, motors, ammeters, voltmeters, magnetising coils, transformers, fuses, relays, diodes and light-emitting diodes (LEDs), and know how these components behave in the circuit

4.3.2 Series and parallel circuits

- 1 Recall and use in calculations, the fact that:
 - (a) the current at every point in a series circuit is the same
 - (b) the sum of the currents entering a junction in a parallel circuit is equal to the sum of the currents that leave the junction
 - (c) the total p.d. across the components in a series circuit is equal to the sum of the individual p.d.s across each component
 - (d) the p.d. across an arrangement of parallel resistances is the same as the p.d. across one branch in the arrangement of the parallel resistances
- 2 Calculate the combined resistance of two or more resistors in series
- 3 Calculate the combined resistance of two resistors in parallel
- 4 Calculate current, voltage and resistance in parts of a circuit or in the whole circuit

4.3.3 Action and use of circuit components

- 1 Describe the action of negative temperature coefficient (NTC) thermistors and light-dependent resistors and explain their use as input sensors
- 2 Describe the action of a variable potential divider
- 3 Recall and use the equation for two resistors used as a potential divider

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

1. O/N/23/11/Q23

An aircraft flying through the air becomes electrically charged.

What causes this to happen?

- A** background radiation in the aircraft
- B** friction of the aircraft with the air
- C** heating of the aircraft surface
- D** radio waves emitted by the aircraft

2. O/N/23/11/Q24

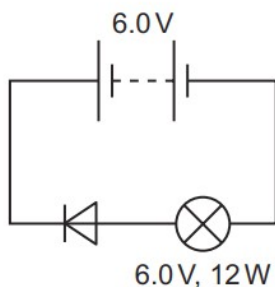
Two metal spheres are brought close to each other. The two spheres repel each other.

Which statement explains this?

- A** Both spheres are charged and both are positive.
- B** Both spheres are charged but one is positive and the other is negative.
- C** The spheres have the same number of electrons.
- D** The spheres have the same number of protons.

3. O/N/23/11/Q25

A student sets up the circuit shown. The lamp does not light.

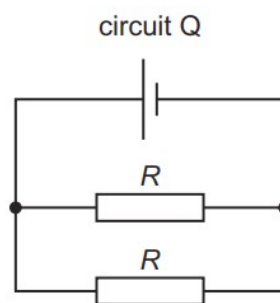
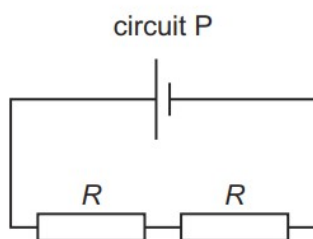


What makes the lamp light?

- A** changing the supply to a 12 V battery
- B** inserting a fuse
- C** reversing the battery connections
- D** reversing the connections to the lamp

4. O/N/23/11/Q26

Circuits P and Q each contain two identical resistors, each of resistance R .



What is the resistance of circuit P and what is the resistance of circuit Q?

	resistance of circuit P	resistance of circuit Q
A	R	$\frac{R}{2}$
B	R	$2R$
C	$2R$	$\frac{R}{2}$
D	$2R$	$2R$

5. O/N/23/11/Q27

The current in an electric motor is 2.0 A.

It transfers 20 J of energy in 5.0 s.

What is the potential difference (p.d.) across the motor?

- A** 0.50 V **B** 2.0 V **C** 8.0 V **D** 50 V

6. O/N/23/12/Q31

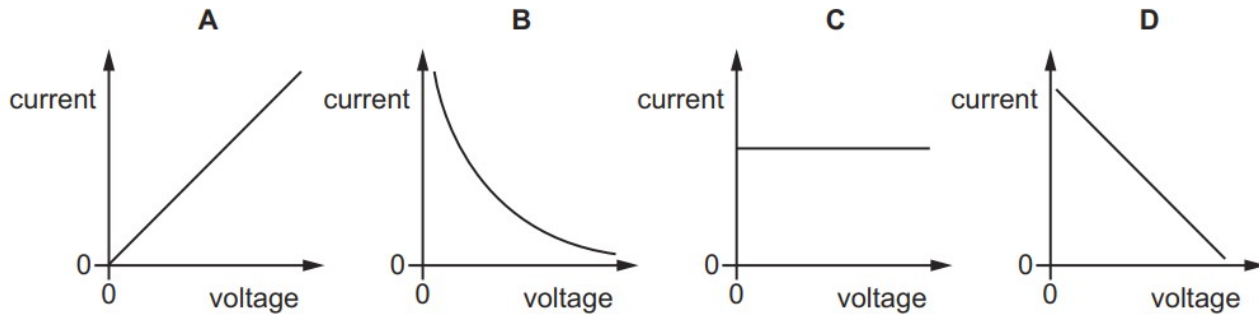
A student rubs different rods held in his hand.

Which rod gains a large electrostatic charge?

- A** a polythene rod rubbed with a steel magnet
B a polythene rod rubbed with a woollen duster
C an iron rod rubbed with a steel magnet
D an iron rod rubbed with a woollen duster

7. O/N/23/12/Q33

Which graph shows how the current changes when the voltage across a resistor of constant resistance is varied?

**8. O/N/23/12/Q34**

Two identical resistors are connected first in series and then in parallel.

The two resistors in series are connected to a power supply and the current is I_s . The parallel combination is connected to the same power supply and the current supplied is I_p .

What is the ratio $\frac{I_s}{I_p}$?

- A** 0.25 **B** 0.50 **C** 2.0 **D** 4.0

9. M/J/23/12/Q26

A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

- A** Negative electrons are transferred to the carpet.
B Negative electrons are transferred to the shoe.
C Positive electrons are transferred to the carpet.
D Positive electrons are transferred to the shoe.

10. M/J/23/12/Q27

A metal conductor is connected between the positive and negative terminals of a battery.

Which row gives the direction of movement of the particles that flow in the conductor and their name?

	direction of movement	name of particles
A	from positive terminal to negative terminal	protons
B	from positive terminal to negative terminal	electrons
C	from negative terminal to positive terminal	protons
D	from negative terminal to positive terminal	electrons

11. M/J/23/12/Q28

Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel rather than using a single cell?

- A** Each cell produces more energy.
- B** Each cell supplies more charge.
- C** Each cell takes longer to discharge.
- D** The e.m.f. is greater than that of a single cell.

12. M/J/23/12/Q29

A student has three $15\ \Omega$ resistors.

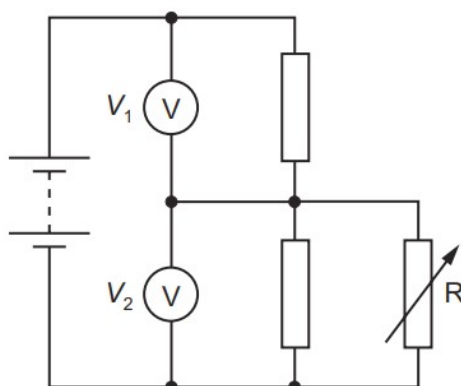
She connects the resistors together in different arrangements.

What is the minimum resistance and what is the maximum resistance of the arrangements that the student can produce?

	minimum resistance / Ω	maximum resistance / Ω
A	5.0	15
B	5.0	45
C	15	30
D	15	45

13. M/J/23/12/Q30

The circuit diagram shows a variable resistor R connected in parallel to the lower section of a potential divider.



The resistance of R increases.

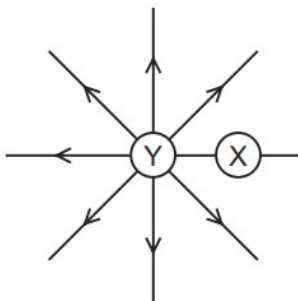
What happens to the two voltmeter readings V_1 and V_2 ?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

14. M/J/23/11/Q28

Object X is stationary and positively charged. It experiences a force due to the field produced by object Y .

The arrows show the direction of the field produced by Y .



Which statement about the direction of the force on X is correct?

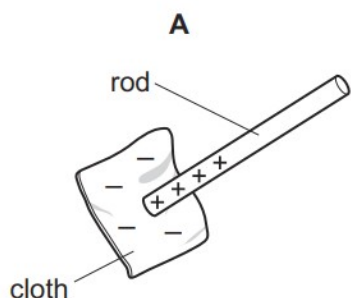
- A** It is towards the right because it is in an electric field.
- B** It is towards the left because it is in an electric field.
- C** It is towards the right because it is in a magnetic field.
- D** It is towards the left because it is in a magnetic field.

15. M/J/23/11/Q27

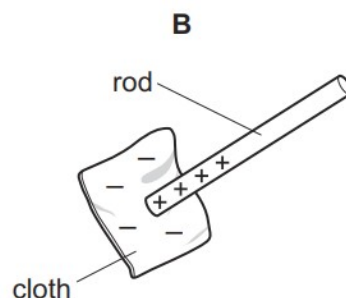
In an electrostatics experiment, a plastic rod is rubbed with a cloth.

The **cloth** becomes negatively charged.

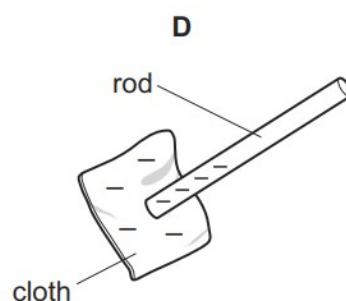
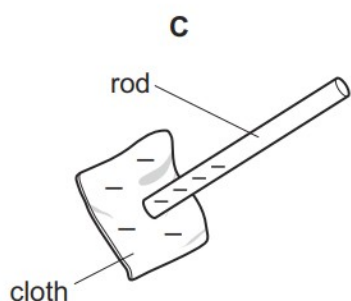
Which diagram shows the charge on the rod and describes the movement of charge?



electrons move from the rod on to the cloth

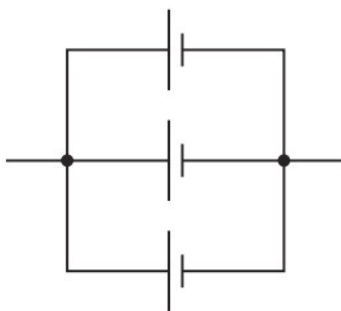


protons move from the cloth on to the rod



16. M/J/23/11/Q29

The diagram shows a battery of three 1.5 V cells.

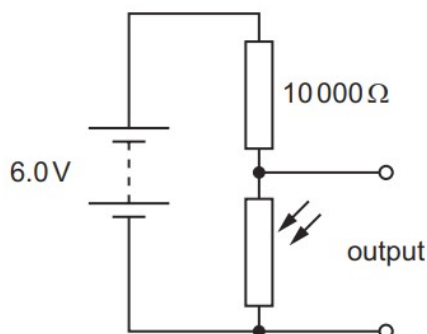


What is an advantage of this arrangement of cells?

- A** The battery can supply a current for a longer time than a single 1.5 V cell.
- B** The battery can supply any e.m.f. between 0 V and 4.5 V.
- C** The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.
- D** The e.m.f. of the battery is 4.5 V.

17. M/J/23/11/Q30

A potential divider consists of a light-dependent resistor (LDR) and a resistor connected to a 6.0 V battery.



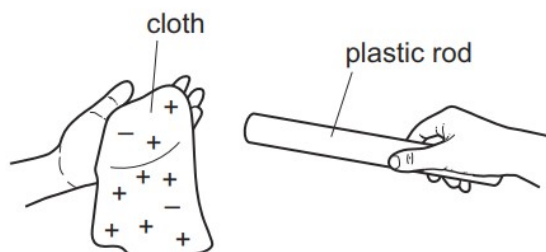
The output is 3.0 V.

What is the resistance of the LDR?

- A** 0
- B** between 0 and 10000Ω
- C** 10000Ω
- D** more than 10000Ω

18. O/N/22/11/Q27

A plastic rod and a cloth are initially uncharged. The rod is then rubbed with the cloth and the cloth becomes positively charged.



What is the sign of the charge on the plastic rod and why does it become charged?

	charge on rod	reason why rod becomes charged
A	negative	electrons move from the cloth to the rod
B	negative	protons move from the rod to the cloth
C	positive	electrons move from the rod to the cloth
D	positive	protons move from the cloth to the rod

19. O/N/22/11/Q28

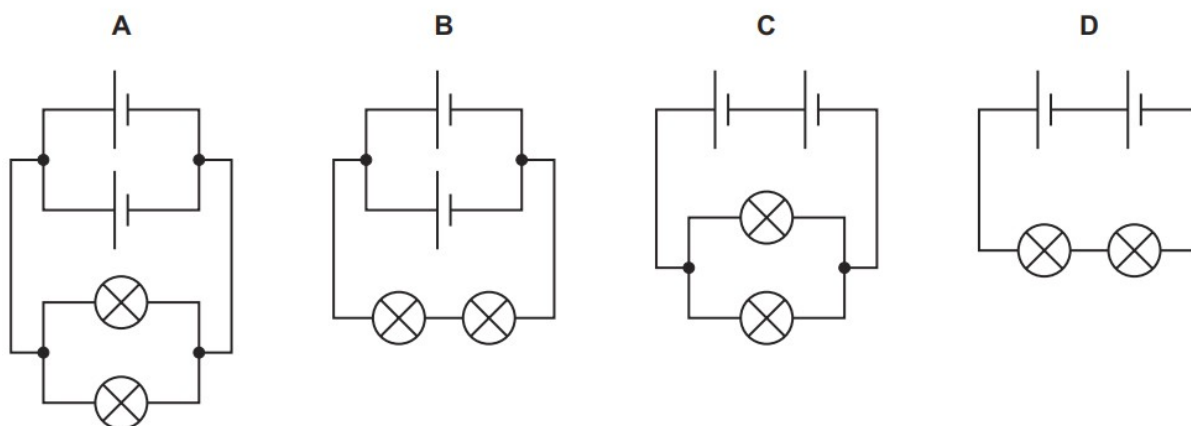
Which expression gives the electromotive force (e.m.f.) of a battery?

- A work done $\times$ time
- B work done $\times$ charge
- C $\frac{\text{work done}}{\text{time}}$
- D $\frac{\text{work done}}{\text{charge}}$

20. O/N/22/11/Q29

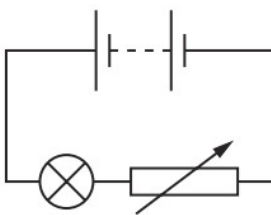
A circuit contains two cells and two lamps. The cells are new and are fully charged.

In which arrangement of the cells and lamps do the cells discharge the most quickly?



21. O/N/22/11/Q30

A lamp is connected in series with a battery and a variable resistor. The resistance of the variable resistor is decreased gradually.



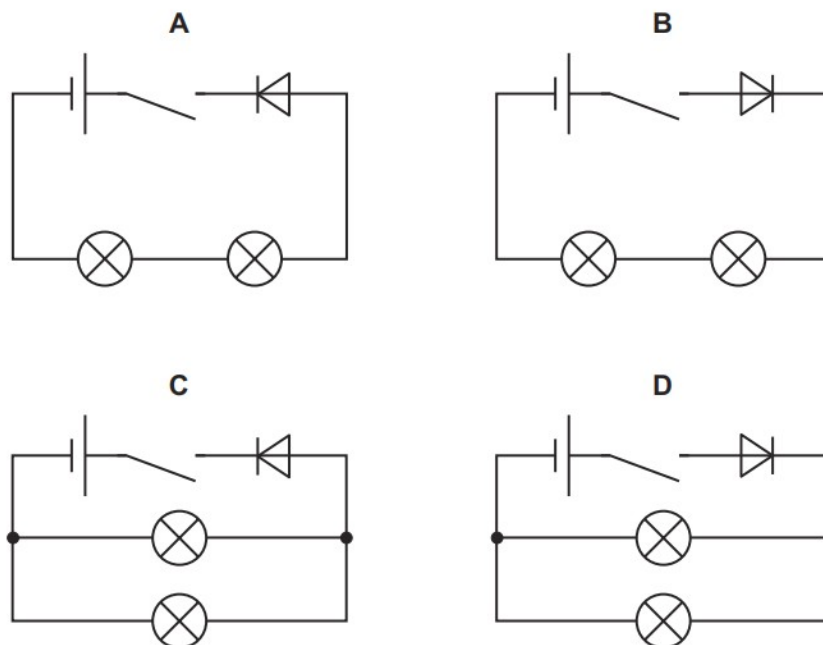
Which statement is correct?

- A The current in the lamp increases.
- B The current in the variable resistor decreases.
- C The e.m.f. of the battery decreases.
- D The total resistance in the circuit increases.

22. O/N/22/11/Q31

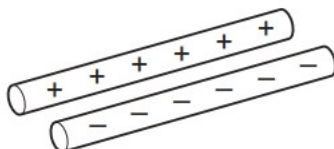
In the circuits shown all the cells are identical and all the lamps are identical. The switches are closed.

In which circuit are both lamps the brightest?



23. O/N/22/12/Q31

Two oppositely charged plastic rods are placed next to each other.



Which row describes and explains how the rods move?

	movement	explanation
A	away from each other	opposite charges attract
B	away from each other	opposite charges repel
C	towards each other	opposite charges attract
D	towards each other	opposite charges repel

24. O/N/22/12/Q32

A meter that measures the rate of flow of charge in a resistor is placed in a circuit.

What is the name of this meter and how is it connected to the resistor?

	name	connection
A	ammeter	in parallel
B	ammeter	in series
C	voltmeter	in parallel
D	voltmeter	in series

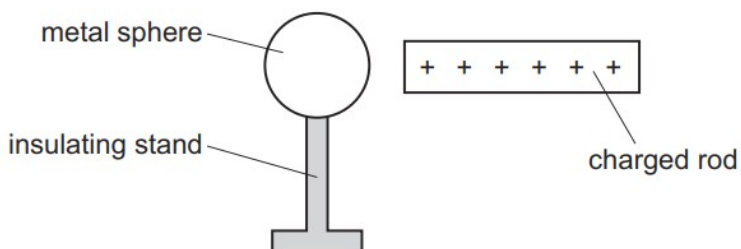
25. O/N/22/12/Q33

Which device is used to convert electrical energy to kinetic energy?

- A** iron
- B** kettle
- C** lamp
- D** motor

26. M/J/22/12/Q30

A positively charged rod is held near an uncharged metal sphere on an insulating stand.

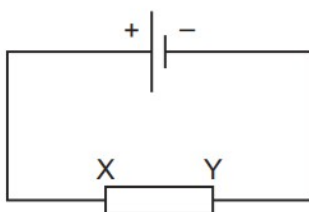


What is the distribution of charge on the sphere?



27. M/J/22/12/Q32

The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

28. M/J/22/12/Q33

Charge of 0.40 C passes through a resistor in 1.0 s.

In 2.0 s, 20 J of thermal energy is produced in the resistor.

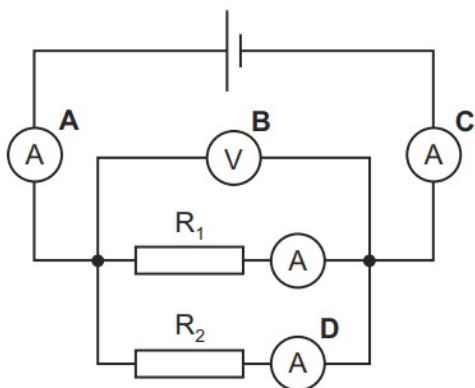
What is the potential difference across the resistor?

- A** 0.020 V **B** 0.040 V **C** 25 V **D** 50 V

29. M/J/22/12/Q34

The diagram shows a simple electric circuit.

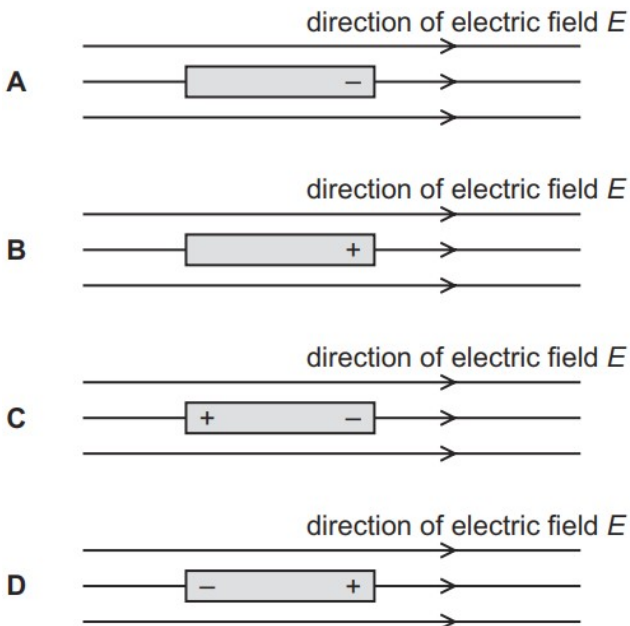
Which meter measures the current in resistor R_2 ?



30. M/J/22/11/Q30

An initially uncharged copper rod is placed in a uniform electric field E . The rod is parallel to the field.

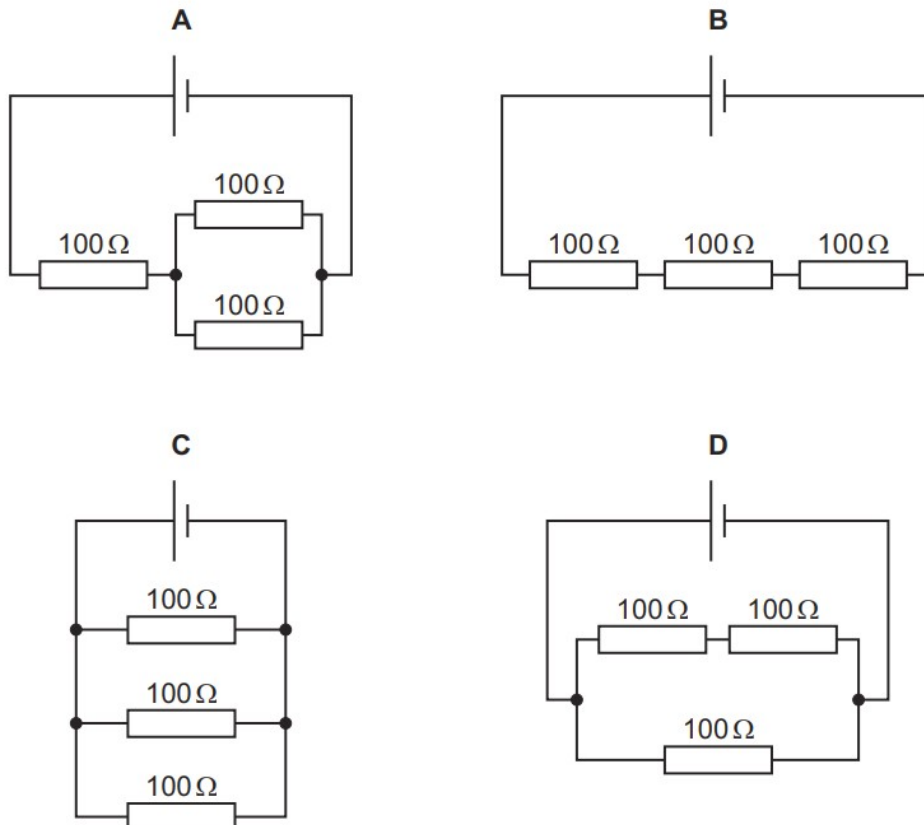
Which diagram shows the charges induced on the rod?



31. M/J/22/11/Q34

The diagrams show four circuits. Each circuit contains three $100\ \Omega$ resistors.

Which circuit has the smallest total resistance?



32. M/J/22/11/Q33

What is the unit of potential difference (p.d.) and which measuring instrument is used to measure p.d.?

	unit	instrument
A	ampere	ammeter
B	ampere	voltmeter
C	volt	ammeter
D	volt	voltmeter

33. O/N/21/11/Q29

A piece of paper torn out of an exercise book is shown.

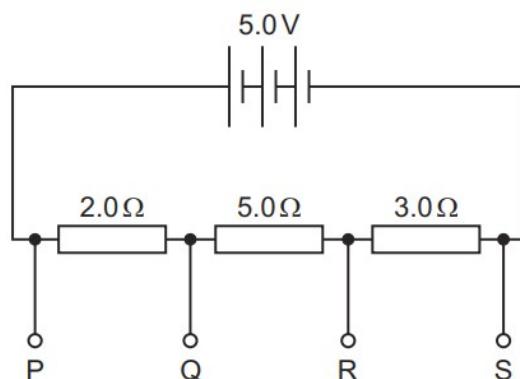
	is the work done (measured in joules)
	on unit charge (measured in coulombs) as it
	moves around the circuit.

What is being defined?

- A** electrical resistance
- B** electric current
- C** electrical power
- D** electromotive force

34. O/N/21/11/Q31

The diagram shows a circuit with a 5.0 V power supply, three resistors and four output terminals P, Q, R and S.

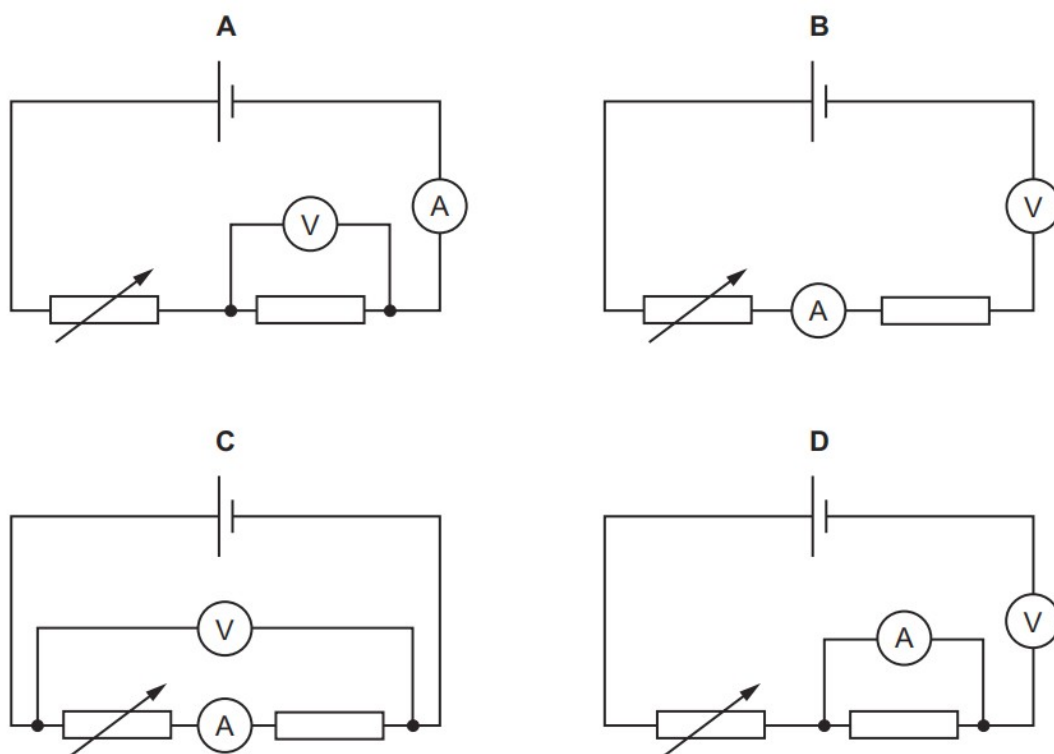


Between which pair of terminals is there a potential difference of 1.5 V?

- A** PQ
- B** PR
- C** QS
- D** RS

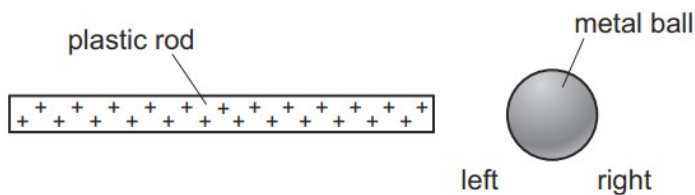
35. O/N/21/11/Q30

Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?



36. O/N/21/12/Q31

A positively charged plastic rod is held near but not touching an insulated and previously uncharged metal ball. No charge leaves the rod.



What are possible charges on the left-hand side and right-hand side of the ball?

	charge on left-hand side of ball	charge on right-hand side of ball
A	$-2.0 \times 10^{-9} \text{ A}$	$-1.0 \times 10^{-9} \text{ A}$
B	$-2.0 \times 10^{-9} \text{ A}$	$+2.0 \times 10^{-9} \text{ A}$
C	$-2.0 \times 10^{-9} \text{ C}$	$+2.0 \times 10^{-9} \text{ C}$
D	$+2.0 \times 10^{-9} \text{ C}$	$-2.0 \times 10^{-9} \text{ C}$

37. O/N/21/12/Q33

Quantity X and quantity Y have the same unit Z.

X is the energy dissipated by a battery in driving unit charge round the complete circuit.

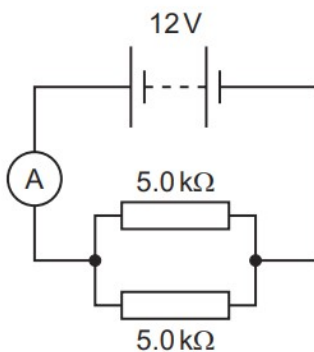
Y is the work done in a component in driving unit charge through the component.

What are X, Y and Z?

	X	Y	Z
A	electromotive force	potential difference	coulomb
B	electromotive force	potential difference	volt
C	potential difference	electromotive force	coulomb
D	potential difference	electromotive force	volt

38. M/J/21/12/Q32

The diagram shows a circuit.



The ammeter has four different range settings.

Which range setting gives a reading near the mid-point of the range?

- A** 0–1 mA **B** 0–10 mA **C** 0–1.0 A **D** 0–10 A

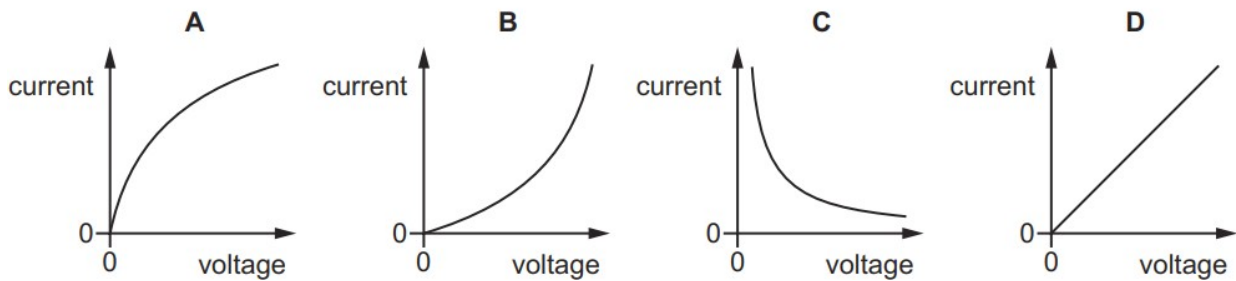
39. M/J/21/12/Q33

In which example is the current 1.0 A?

- A** a charge of 0.10 C passes through a resistor in 100 ms
B a 10 V lamp with input power 0.10 W
C a resistor of resistance 0.50Ω with a potential difference (p.d.) of 2.0 V across it
D two resistors with a current of 0.50 A connected in series

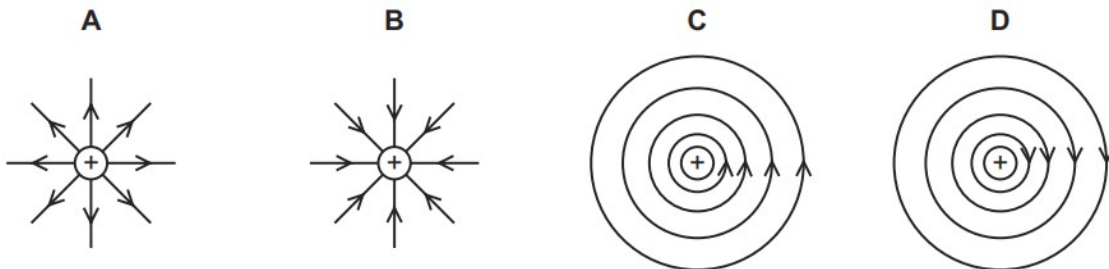
40. M/J/21/12/Q34

Which graph shows the relationship between current and voltage for a filament lamp?



41. M/J/21/11/Q27

Which diagram shows the pattern and direction of the electric field lines near a positive charge?



42. O/N/20/11/Q34

A small negative charge is placed at a point where an electric field is vertically upwards.

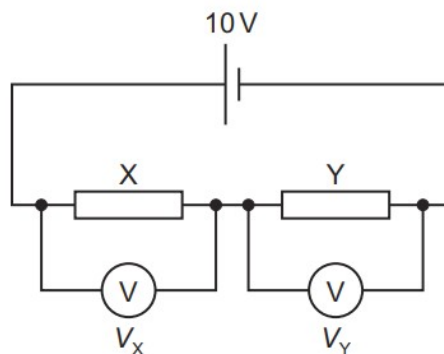
There is a force on the charge due to the field.

In which direction does it act?

- A** vertically upwards
- B** vertically downwards
- C** horizontally to the right
- D** horizontally to the left

43. O/N/20/11/Q35

In the circuit shown, the resistance of resistor Y is four times greater than the resistance of resistor X.



What is the difference $V_Y - V_X$ of the voltages shown on the voltmeters?

- A** 2.0V
- B** 5.0V
- C** 6.0V
- D** 8.0V

44. O/N/20/11/Q36

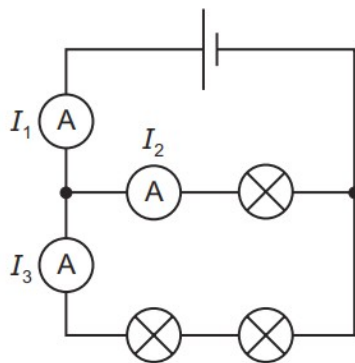
Two identical resistors connected in parallel have a total resistance of $4.0\ \Omega$.

What is the total resistance when the same two resistors are connected in series?

- A** $1.0\ \Omega$ **B** $4.0\ \Omega$ **C** $8.0\ \Omega$ **D** $16\ \Omega$

45. O/N/20/11/Q37

Three identical lamps and three ammeters are connected as shown.



The readings on the ammeters are I_1 , I_2 and I_3 .

How are the readings related?

- A** $I_1 = I_2 = I_3$
B $I_1 > I_2$ and $I_2 = I_3$
C $I_1 > I_3 > I_2$
D $I_1 > I_2 > I_3$

46. O/N/20/12/Q29

What is stored in a battery and what is its unit?

	quantity	unit
A	current	A
B	current	As
C	energy	J
D	energy	J/s

47. O/N/20/12/Q30

A car battery delivers $150\ \text{kC}$ to a circuit before it needs recharging.

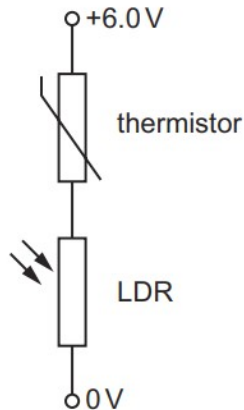
Two headlamps connected in parallel to the battery are switched on. There is a current of $4.0\ \text{A}$ in each lamp.

How much time passes before the battery needs recharging?

- A** 2.6 hours **B** 5.2 hours **C** 7.8 hours **D** 10.4 hours

48. O/N/20/11/Q38

A thermistor and a light-dependent resistor (LDR) are connected in series. A potential difference (p.d.) of 6.0 V is applied across them as shown.



The thermistor has a resistance of $6000\ \Omega$ in a cold room and $1000\ \Omega$ in a warm room. The LDR has a resistance of $2000\ \Omega$ in dim light and $500\ \Omega$ in bright light.

When is the p.d. across the LDR equal to 2.0 V?

- A** in a cold room with bright light
- B** in a cold room with dim light
- C** in a warm room with bright light
- D** in a warm room with dim light

49. M/J/20/12/Q36

A stationary negative charge in an electric field experiences an electric force in the direction shown.



What is the direction of the electric field?

- A** to the left
- B** to the right
- C** down the page
- D** up the page

50. M/J/20/12/Q37

A metal wire of length l and cross-sectional area A has resistance R .

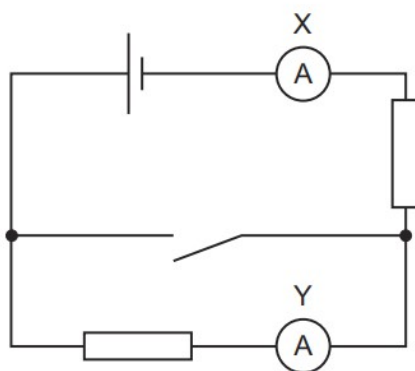
A second wire is made from the same metal. It has a length $2l$ and a cross-sectional area $4A$.

What is the resistance of the second wire?

- A** $8R$
- B** $2R$
- C** $\frac{R}{2}$
- D** $\frac{R}{8}$

51. M/J/20/12/Q38

The diagram shows a circuit.

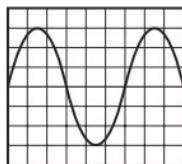


What effect does closing the switch have on the readings of ammeters X and Y?

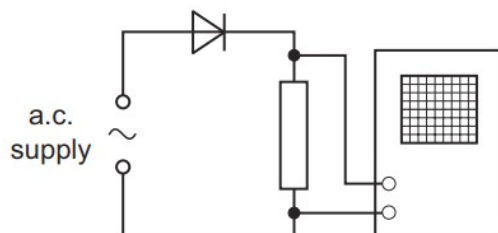
	reading on ammeter X	reading on ammeter Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

52. O/N/19/11/Q33

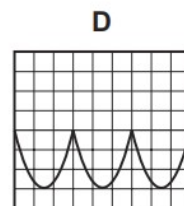
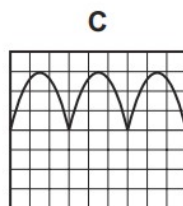
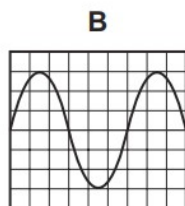
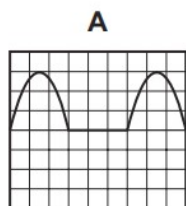
An a.c. supply connected to an oscilloscope produces the trace shown.



A diode is now connected in series to the circuit as shown.



What is the new trace on the screen?



53. O/N/19/11/Q34

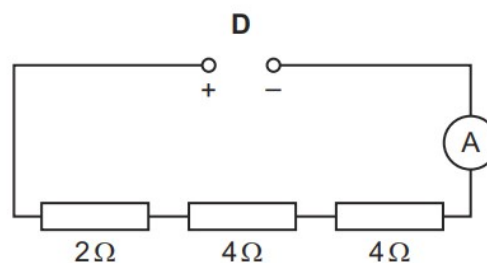
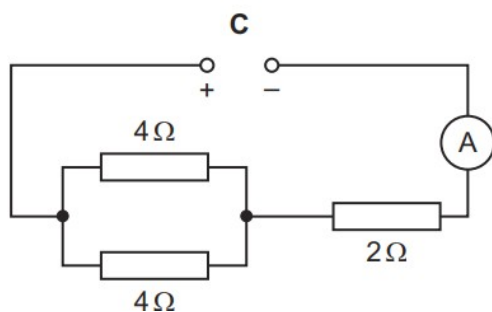
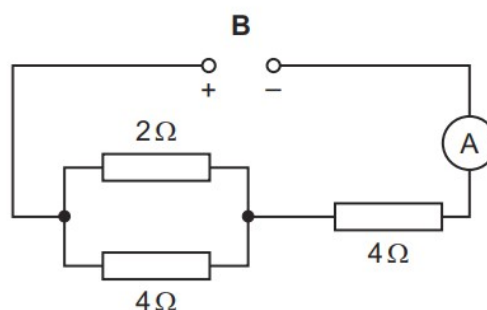
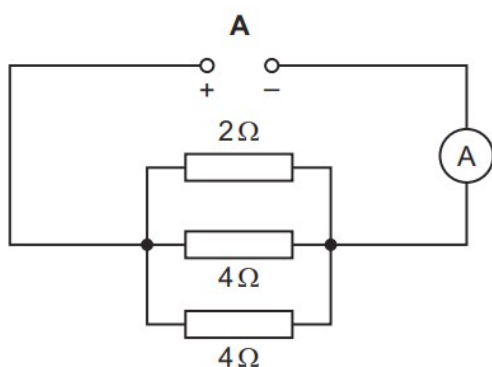
Resistors of different resistances are connected to a power supply in either a parallel circuit or a series circuit. Which statement is correct?

- A The current is the same in all resistors connected in parallel.
- B The current is the same in all resistors connected in series.
- C The voltage across each resistor is different for all resistors connected in parallel.
- D The voltage across each resistor is the same for all resistors connected in series.

54. O/N/19/11/Q35

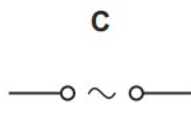
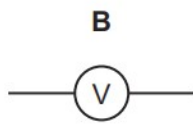
An ammeter is connected to three resistors and a power supply.

Which arrangement of resistors gives the greatest ammeter reading?



55. O/N/19/12/Q32

Which symbol does **not** represent a power source?



56. O/N/19/12/Q35

A hotplate on an electric cooker contains two identical resistors.

The switch has three positions.

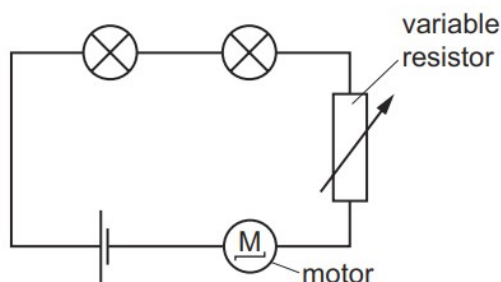
- position 1 The two resistors are connected in series to the mains supply.
position 2 The two resistors are connected in parallel to the mains supply.
position 3 Just one resistor is connected to the mains supply.

Which positions correspond to the low, the medium and the high power settings?

	low power	medium power	high power
A	1	2	3
B	1	3	2
C	3	1	2
D	3	2	1

57. M/J/19/12/Q33

The diagram shows a simple circuit.

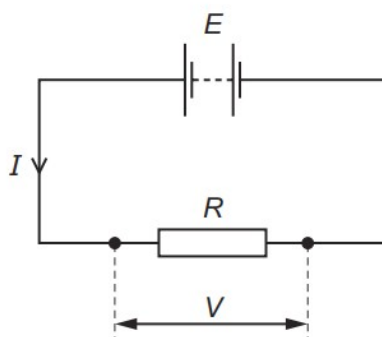


What happens when the resistance of the variable resistor is increased?

- A** The lamps are dimmer and the motor rotates more quickly.
B The lamps are dimmer and the motor rotates more slowly.
C The lamps have the same brightness and the motor rotates more quickly.
D The lamps have the same brightness and the motor rotates more slowly.

58. M/J/19/12/Q34

The diagram shows a simple electrical circuit.

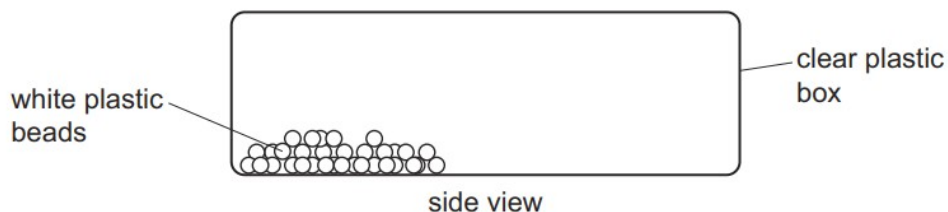


Which statement **cannot** be correct?

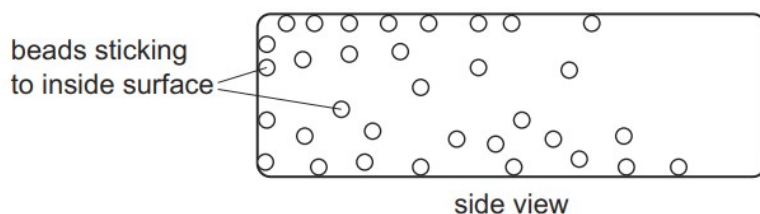
- A The current I is 2 A.
- B The e.m.f. E is 12 J.
- C The p.d. V (across the resistor) is 10 V.
- D The resistance R is 5Ω .

59. M/J/19/11/Q31

The diagram shows some white plastic beads in a clear plastic box.



The box is shaken and the beads rub against the box. The beads stick to the inside surface of the box.



Which row is a possible explanation for this?

	box	beads
A	gains electrons	no change of electrons
B	loses electrons	gain electrons
C	no change of electrons	lose electrons
D	no change of electrons	gain electrons

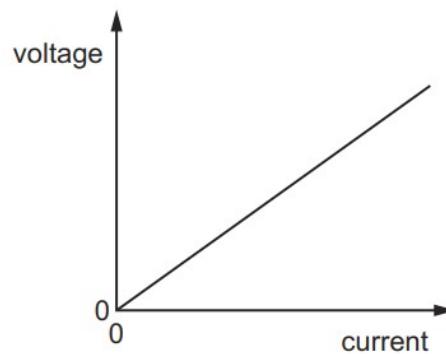
60. M/J/19/11/Q32

Which unit is the same as a volt?

- A ampere / ohm
- B ampere / watt
- C ohm / ampere
- D watt / ampere

61. M/J/19/11/Q33

The voltage / current graph for a metal wire is shown.



What does the gradient of this graph represent?

- A the charge passing through the wire
- B the e.m.f. of the battery connected to the wire
- C the energy produced in the wire
- D the resistance of the wire

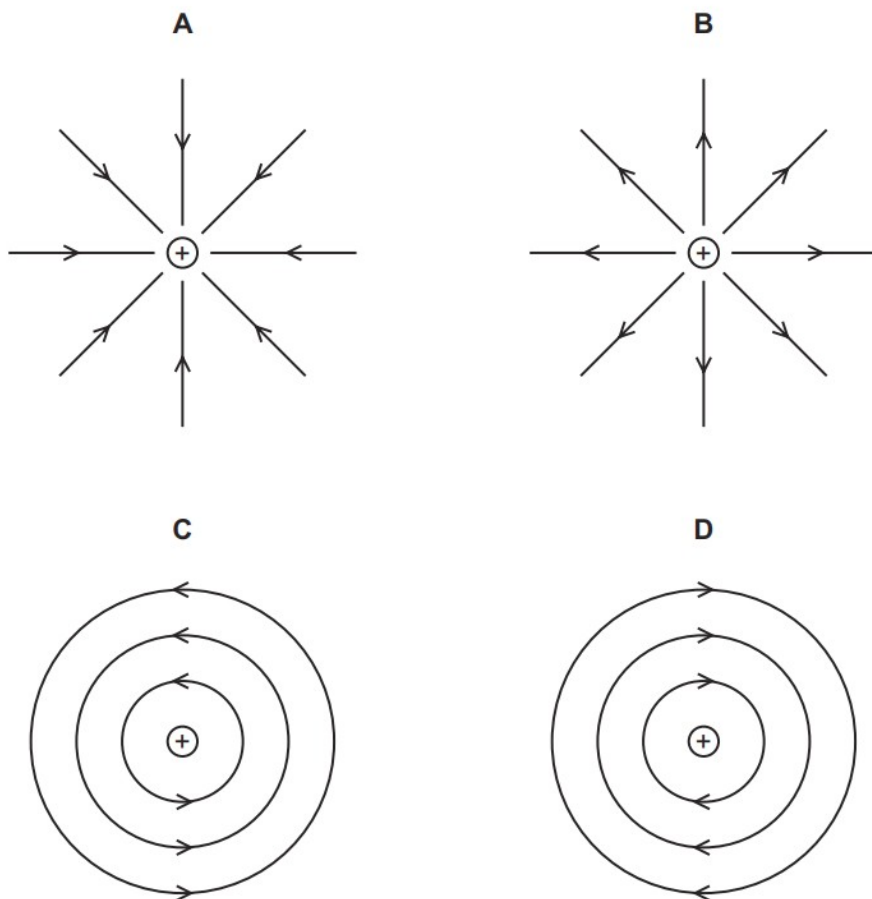
62. M/J/19/11/Q34

What is the symbol for a device that measures current?



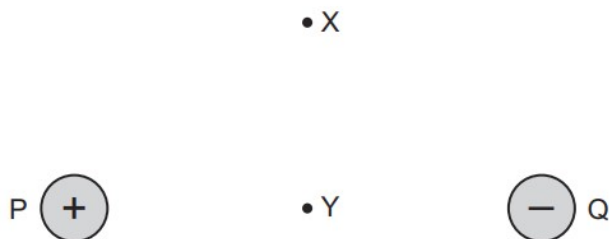
63. O/N/18/11/Q24

Which diagram shows the electric field around an isolated point positive charge?



64. O/N/18/11/Q25

An electric field is formed by two isolated, equal and opposite charges P and Q.



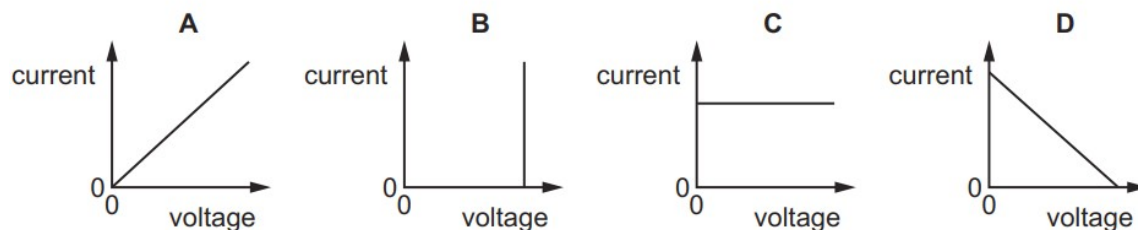
How does the field at X compare with the field at Y?

- A** It is stronger at X than at Y and is in the opposite direction.
- B** It is stronger at X than at Y and is in the same direction.
- C** It is weaker at X than at Y and is in the opposite direction.
- D** It is weaker at X than at Y and is in the same direction.

65. O/N/18/11/Q27

The voltage across a resistor of fixed resistance changes.

Which graph shows how the current in the resistor changes with the voltage?



66. O/N/18/11/Q28

A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.75Ω .

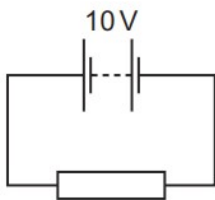
Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6} \text{ m}^2$.

What is the resistance of the longer wire?

- A** 0.094Ω **B** 0.38Ω **C** 1.5Ω **D** 6.0Ω

67. O/N/18/11/Q29

A battery of electromotive force (e.m.f.) 10 V is connected to a resistor.



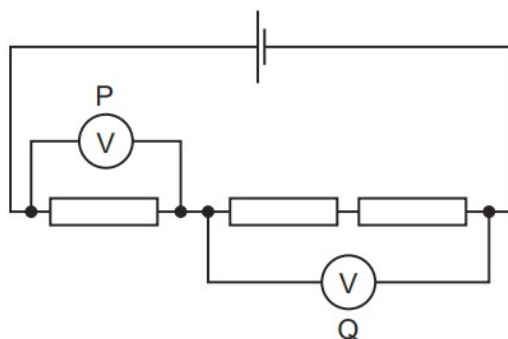
A charge of 2.0 coulombs passes through the resistor.

How much work is done as the charge passes through the resistor?

- A** 0.20 J **B** 2.0 J **C** 5.0 J **D** 20 J

68. O/N/18/11/Q30

Three identical resistors are joined in series to a cell.



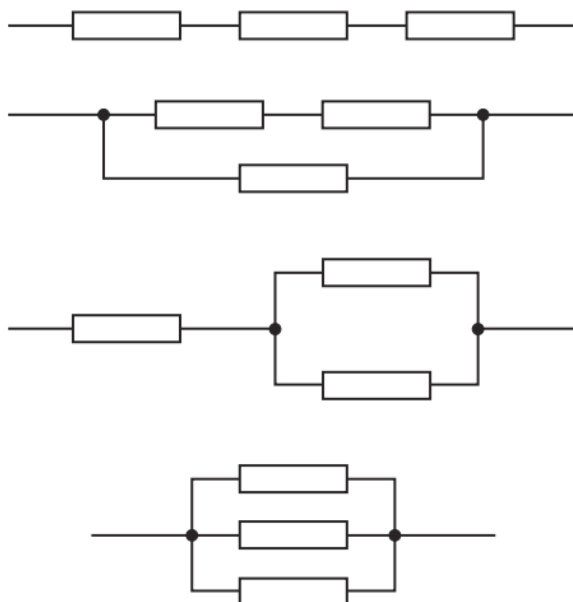
Voltmeter Q reads 8.0 V.

What is the reading on voltmeter P and what is the e.m.f. of the cell?

	reading on P/V	e.m.f. of cell/V
A	4.0	8.0
B	4.0	12
C	8.0	8.0
D	8.0	12

69. O/N/18/12/Q30

The diagram shows three 6.0Ω resistors connected to give various total resistances.



What is the difference between the smallest total resistance and the largest total resistance?

- A** 7.0Ω **B** 9.0Ω **C** 12Ω **D** 16Ω

70. O/N/18/12/Q29

Students are asked to describe an experiment to measure the resistance of a metallic conductor.

Which description is correct?

- A** Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{I}{V}$.
- B** Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{V}{I}$.
- C** Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{I}{V}$.
- D** Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{V}{I}$.

71. O/N/18/12/Q32

A defibrillator is a device that is used to give an electric shock to a patient's heart.

It supplies an electric shock with energy 240 J at an average voltage of 2000 V for 10 ms.

What is the average current it supplies?

- A** 0.012 A **B** 1.2 A **C** 12 A **D** 120 A

72. M/J/18/12/Q35

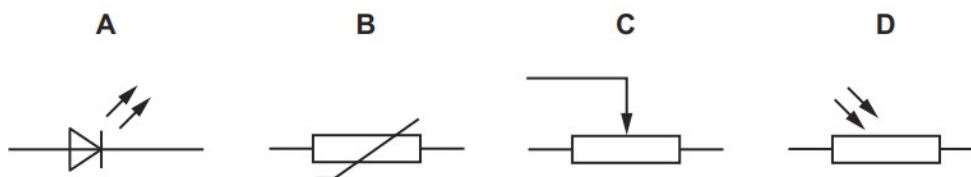
V is a potential difference, I is a current, R is a resistance, and t is a time.

Which expression has units of energy?

- A** It **B** I^2R **C** VIt **D** $\frac{V^2}{R}$

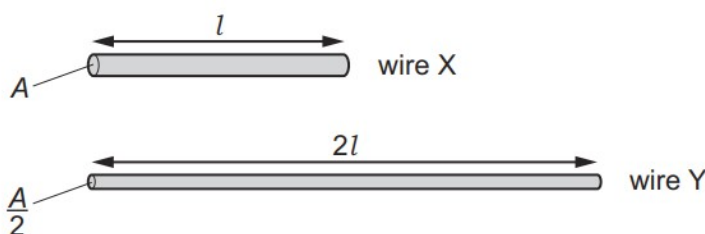
73. M/J/18/12/Q37

Which symbol is used for a light-dependent resistor?



74. M/J/18/11/Q29

A copper wire X has resistance R . Another copper wire Y has twice the length and half the cross-sectional area of X.

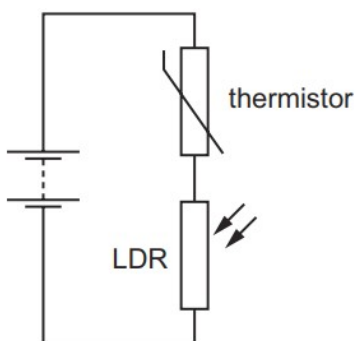


What is the resistance of Y?

- A** $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

75. M/J18/11/Q30

In the circuit shown, the temperature of the room and the amount of light affect the current.



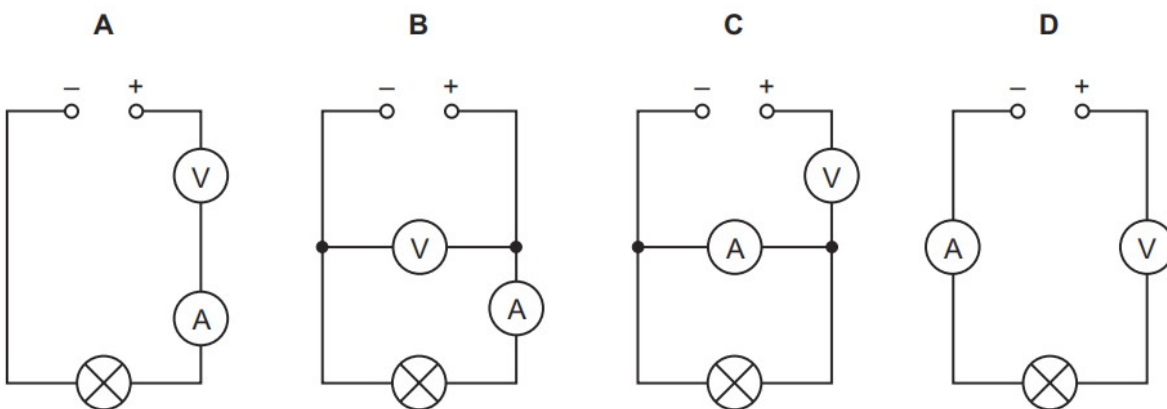
Under which conditions is the current in the circuit the largest?

	temperature	amount of light
A	high	in bright light
B	high	in the dark
C	low	in bright light
D	low	in the dark

76. O/N/17/11/Q30

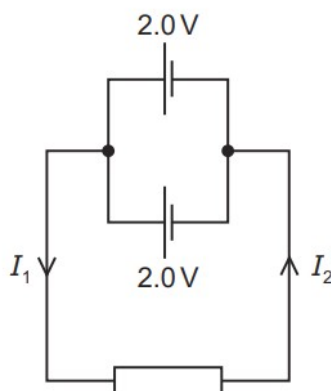
In a circuit, a voltmeter is used to measure the potential difference across a lamp. An ammeter is used to measure the current in the lamp.

Which diagram shows the circuit?



77. O/N/17/11/Q31

The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f. / V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

78. O/N/17/12/Q30

A piece of wire has a resistance of 16Ω .

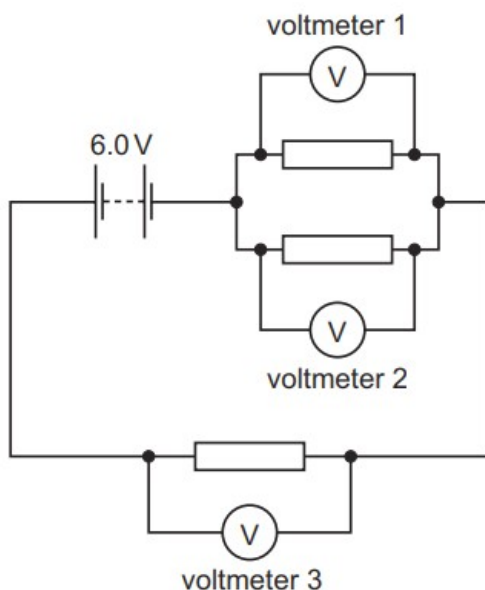
The wire is 20 cm long and has a cross-sectional area of 2.0 mm^2 .

Which wire of the same material has a resistance of 8.0Ω ?

	length / cm	cross-sectional area / mm^2
A	10	1.0
B	10	4.0
C	20	1.0
D	20	4.0

79. O/N/17/12/Q29

A 6.0 V battery, three resistors and three voltmeters are connected in the circuit shown.



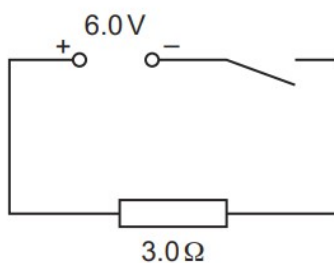
The reading on voltmeter 1 is 2.0 V.

What is the reading on voltmeter 2 and what is the reading on voltmeter 3?

	reading on voltmeter 2 /V	reading on voltmeter 3 /V
A	2.0	4.0
B	2.0	8.0
C	4.0	4.0
D	4.0	8.0

80. M/J/17/12/Q33

The circuit shown is switched on for 1.0 minute.



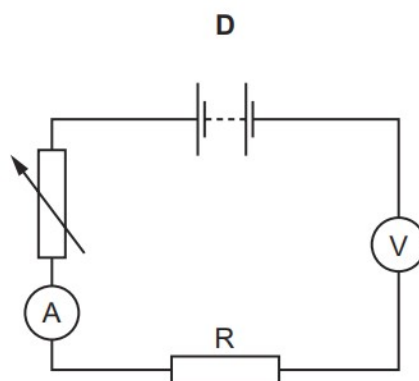
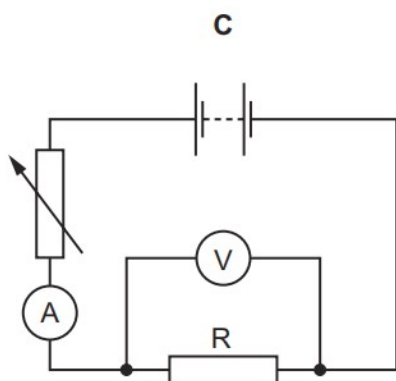
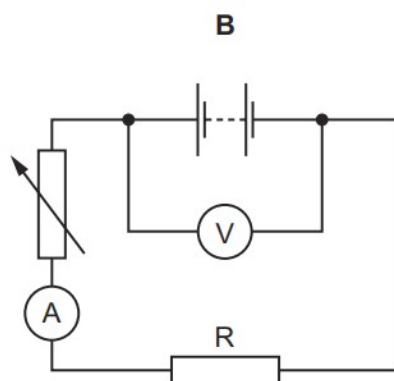
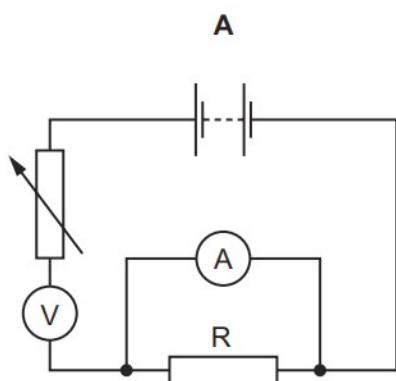
How much charge passes through the 3.0 Ω resistor?

- A** 2.0 C **B** 12 C **C** 120 C **D** 720 C

81. M/J/17/12/Q34

The readings of an ammeter and a voltmeter are used to calculate the resistance of resistor R.

Which circuit shows the ammeter and voltmeter in the correct positions?



82. M/J/17/11/Q36

During a thunderstorm, a lightning flash occurs when charge flows between a cloud and the Earth. On average, 20 C of charge flows during each flash and there are three flashes every minute.

What is the average current between the cloud and the Earth during the thunderstorm?

- A** 1.0 A **B** 36 A **C** 60 A **D** 1800 A

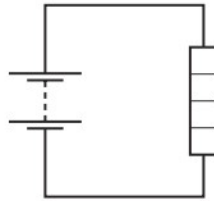
83. M/J/17/11/Q37

Why are batteries sometimes made from several identical cells arranged in parallel?

- A** The battery has a greater e.m.f. than a single cell on its own.
- B** The battery has a lower e.m.f. than a single cell on its own.
- C** The battery lasts longer than a single cell on its own.
- D** The battery supplies a smaller current than a single cell on its own.

84. O/N/16/12/Q33

An ammeter and a voltmeter are connected into the circuit shown when determining the power produced by a heater.



How are these meters connected, in relation to the heater?

	ammeter	voltmeter
A	in parallel	in parallel
B	in parallel	in series
C	in series	in parallel
D	in series	in series

85. M/J/16/12/Q25

A charge of 45 C flows through an electric appliance in 3.0 minutes.

What is the average current in the appliance?

- A** 0.25 A **B** 4.0 A **C** 15 A **D** 135 A

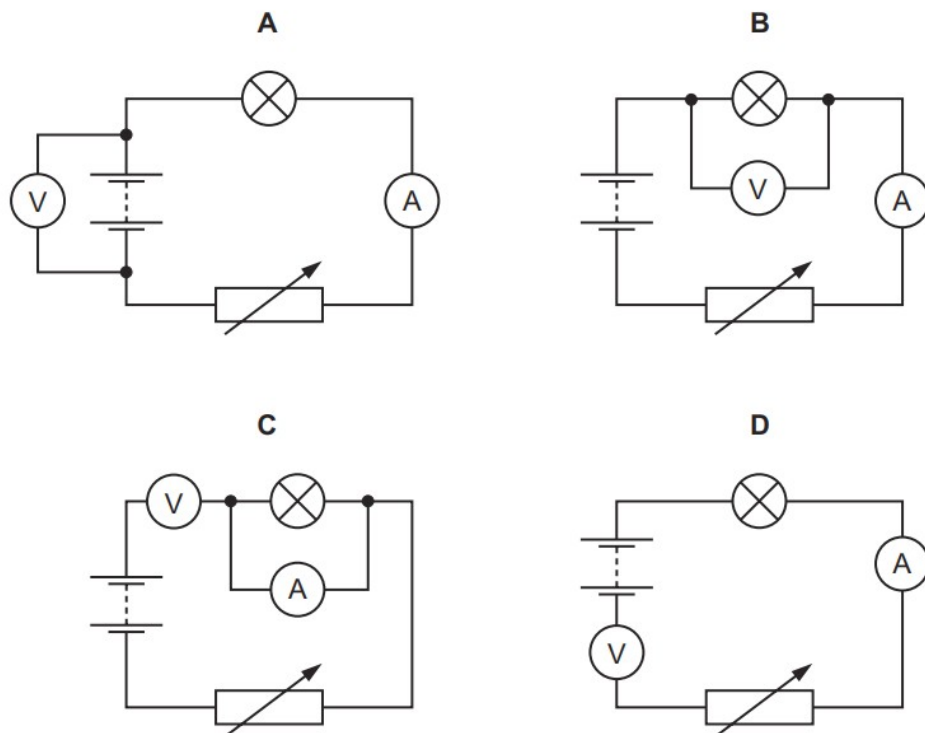
86. M/J/16/12/Q26

How can *one volt* also be expressed?

- A** one coulomb per ampere
B one coulomb per joule
C one joule per ampere
D one joule per coulomb

87. M/J/16/12/Q28

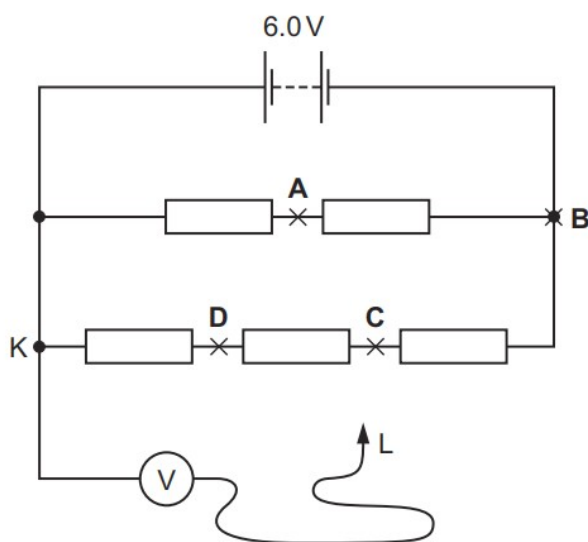
Which diagram shows the arrangement of the ammeter and voltmeter to obtain readings to find the power of a lamp?



88. M/J/16/12/Q29

A 6.0V battery is connected to a network containing five identical resistors. A voltmeter has one lead connected to point K as shown.

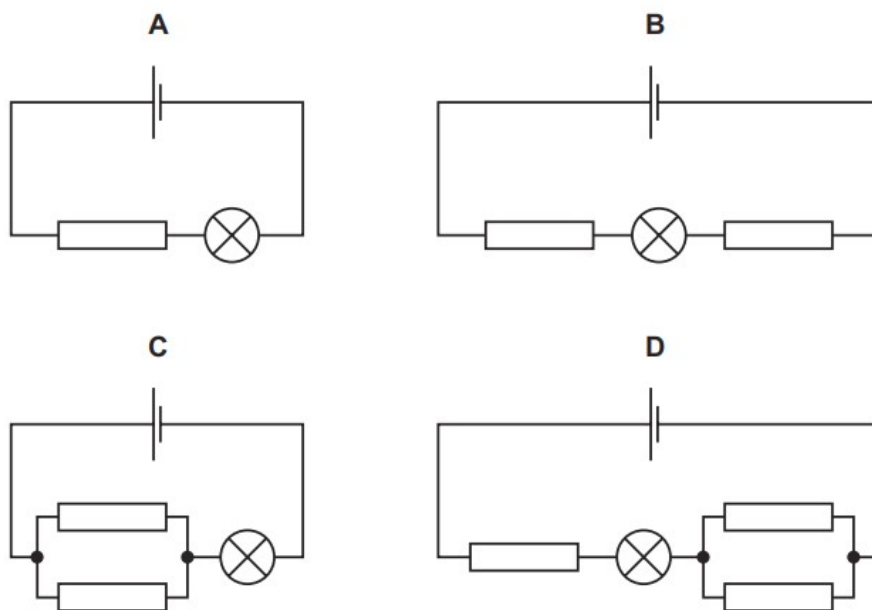
At which point should lead L be connected so that the voltmeter reads 3.0V?



89. M/J/16/11/Q31

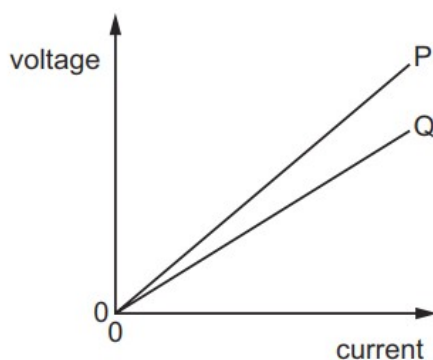
The cells, lamps and resistors in the circuits are identical.

In which circuit is the lamp the brightest?



90. M/J/16/11/Q32

The graph is the voltage-current graph for two resistance wires P and Q.



The wires are made from the same material and have equal lengths.

The resistances of the wires and their cross-sectional areas are different.

Which wire has the greater resistance and which wire has the larger cross-sectional area?

	greater resistance	larger cross-sectional area
A	P	P
B	P	Q
C	Q	P
D	Q	Q

91. O/N/15/11/Q24

A polythene rod becomes charged when rubbed with a dry woollen cloth.

The charge caused is a result of

- A** friction producing a movement of electrons.
- B** friction producing a movement of protons.
- C** magnetism producing a movement of electrons.
- D** magnetism producing a movement of protons.

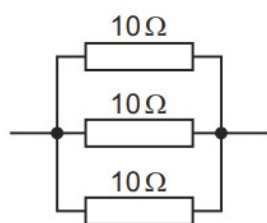
92. O/N/15/11/Q27

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?

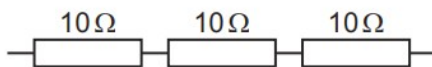
- A** current
- B** electromotive force (e.m.f.)
- C** energy
- D** power

93. O/N/15/11/Q28

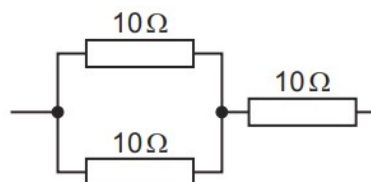
Three 10Ω resistors are connected together to form networks X, Y and Z.



network X



network Y



network Z

What is the order of resistance of the networks going from the smallest total resistance to the largest total resistance?

- A** $X \rightarrow Y \rightarrow Z$ **B** $X \rightarrow Z \rightarrow Y$ **C** $Z \rightarrow X \rightarrow Y$ **D** $Z \rightarrow Y \rightarrow X$

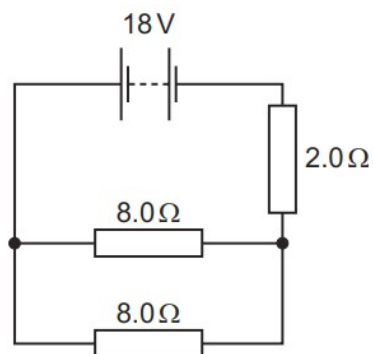
94. O/N/15/12/Q29

Which row correctly names an electrical conductor and an insulator?

	electrical conductor	insulator
A	aluminium	iron
B	iron	rubber
C	plastic	aluminium
D	rubber	plastic

95. O/N/15/12/Q33

A power supply of 18 V is connected to three resistors, as shown.



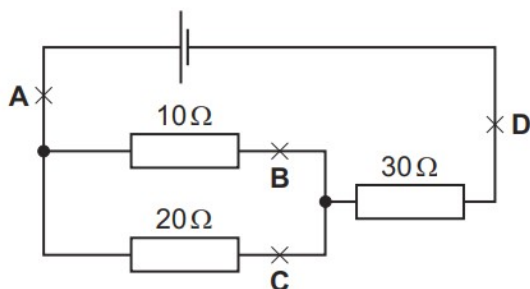
What is the potential difference across the 2.0Ω resistor?

- A** 2.0 V **B** 3.6 V **C** 6.0 V **D** 12 V

96. M/J/15/12/Q33

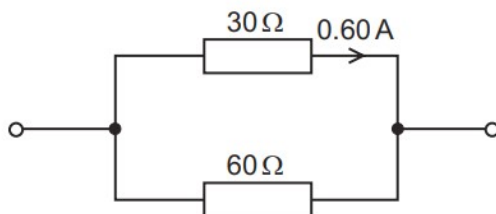
The diagram shows a circuit.

Where must an ammeter be connected to measure the smallest current?



97. M/J/15/12/Q34

Two resistors of resistances 30Ω and 60Ω are arranged in parallel. The current in the 30Ω resistor is 0.60 A.

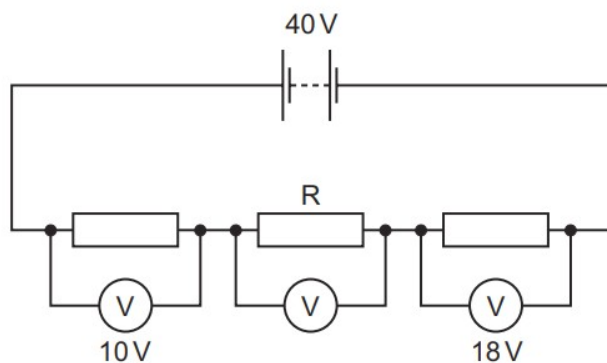


What is the potential difference across the 60Ω resistor?

- A** 9.0 V **B** 18 V **C** 36 V **D** 54 V

98. M/J/15/12/Q35

The circuit shows three resistors in series connected to a battery. Each resistor has a voltmeter across it and two of the voltages are shown.

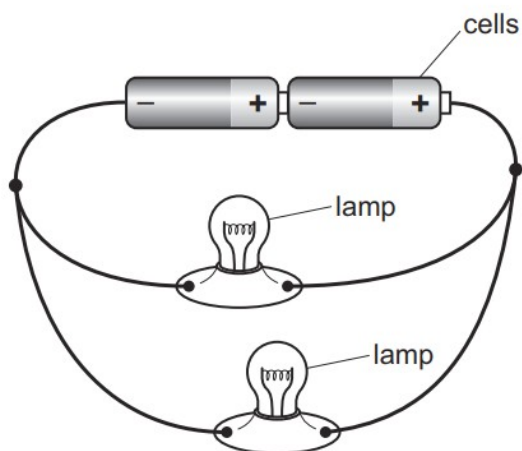


What is the potential difference (p.d.) across the resistor R?

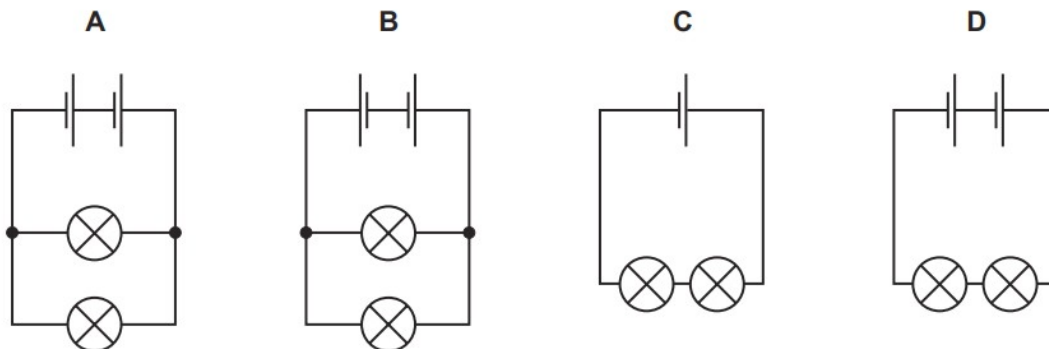
- A** 12V **B** 22V **C** 30V **D** 68V

99. M/J/15/11/Q28

The diagram shows a circuit.



Which circuit diagram shows this circuit?



100. M/J/15/11/Q29

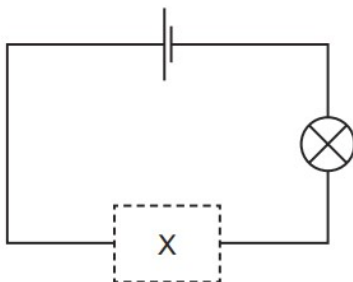
A charge of 7.5 C flows through a resistor in 5.0 s . A student has ammeters with different ranges that he can use to measure the current in the resistor.

Which ammeter range is the most appropriate?

- A** 0-1 A **B** 0-2 A **C** 0-5 A **D** 0-40 A

101. O/N/14/11/Q28

With component X placed in the circuit below, the brightness of the lamp can be gradually reduced until it goes out.



What is component X?

- A** a reed relay
B a switch
C a transformer
D a variable resistor

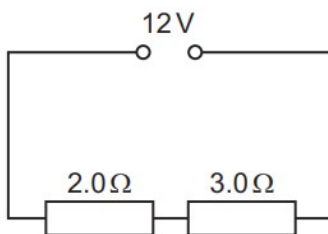
102. O/N/14/12/Q24

Which list contains only electrical insulators?

- A** glass, plastic, rubber
B glass, plastic, steel
C glass, rubber, steel
D plastic, rubber, steel

103. O/N/14/12/Q26

The diagram shows a circuit that has two resistors in series with a 12 V supply.

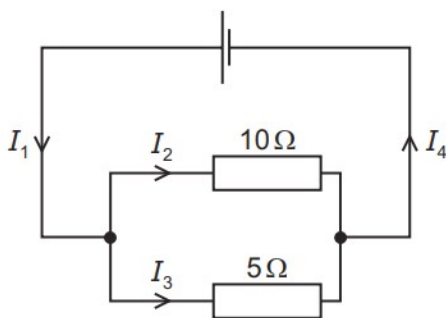


What is the current in the circuit?

- A** 2.4 A **B** 10 A **C** 14 A **D** 60 A

104. O/N/14/12/Q25

The currents in different parts of the circuit are I_1 , I_2 , I_3 and I_4 .



Which statement is correct?

- A** $I_1 = I_4$ and I_2 is greater than I_3 .
- B** $I_1 = I_4$ and I_3 is greater than I_2 .
- C** I_2 is greater than I_1 and less than I_3 .
- D** I_2 is greater than I_1 and greater than I_3 .

105. M/J/14/11/Q30

A piece of wire has a resistance of 16Ω .

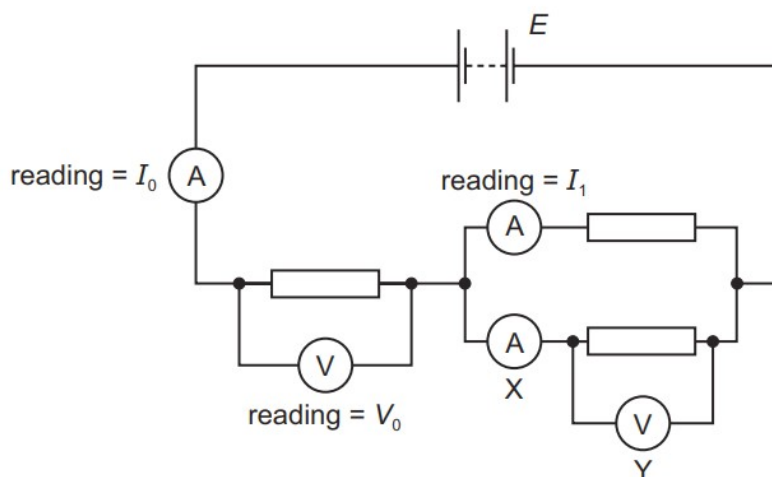
Another wire made from the same metal has four times the length and twice the cross-sectional area.

What is the resistance of the wire?

- A** 8Ω
- B** 32Ω
- C** 96Ω
- D** 128Ω

106. M/J/14/11/Q31

A battery of e.m.f. E is connected in a circuit containing three resistors.



The readings of two ammeters and a voltmeter are shown.

Which readings are shown on ammeter X and on voltmeter Y?

	X	Y
A	$I_0 - I_1$	$E - V_0$
B	$I_0 - I_1$	$E + V_0$
C	$I_0 + I_1$	$E + V_0$
D	$I_0 + I_1$	$E - V_0$

107. O/N/13/11/Q27

Which row shows a conductor and an insulator?

	conductor	insulator
A	aluminium	copper
B	aluminium	glass
C	plastic	copper
D	plastic	glass

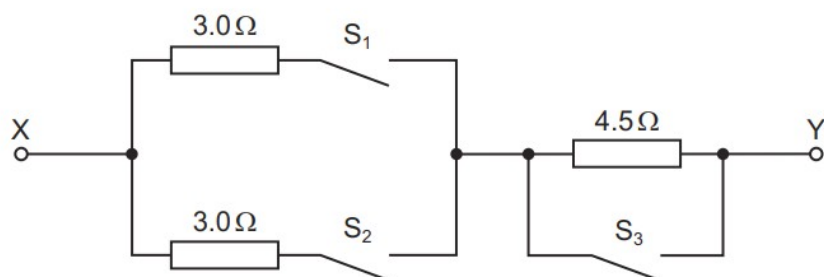
108. O/N/13/11/Q28

Which of the following is equivalent to one coulomb?

- A** one ampere second
- B** one ampere per volt
- C** one volt ampere
- D** one volt per ampere

109. O/N/13/11/Q29

The diagram shows a circuit in which all the switches are open.

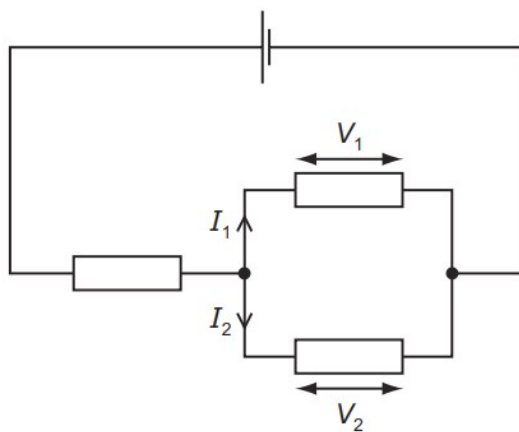


Which switch positions give a resistance of 6.0Ω between X and Y?

	S_1	S_2	S_3
A	closed	closed	closed
B	closed	closed	open
C	closed	open	closed
D	closed	open	open

110. O/N/13/11/Q30

The circuit shows three resistors connected to a cell.



The resistors have different values of resistance.

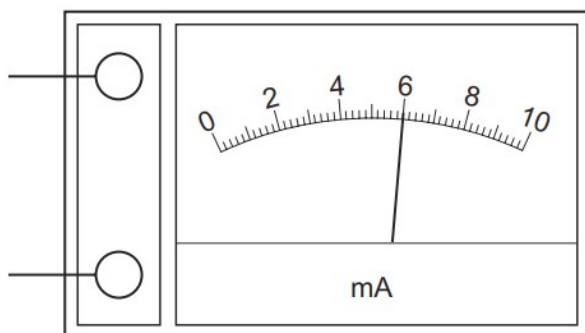
V_1 and V_2 are potential differences and I_1 and I_2 are currents as shown.

Which pair of relationships is correct?

- A** $I_1 = I_2$ and $V_1 = V_2$
- B** $I_1 = I_2$ and $V_1 \neq V_2$
- C** $I_1 \neq I_2$ and $V_1 = V_2$
- D** $I_1 \neq I_2$ and $V_1 \neq V_2$

111. O/N/13/12/Q28

An ammeter is used to measure the current in a $300\ \Omega$ resistor. The ammeter is shown below.

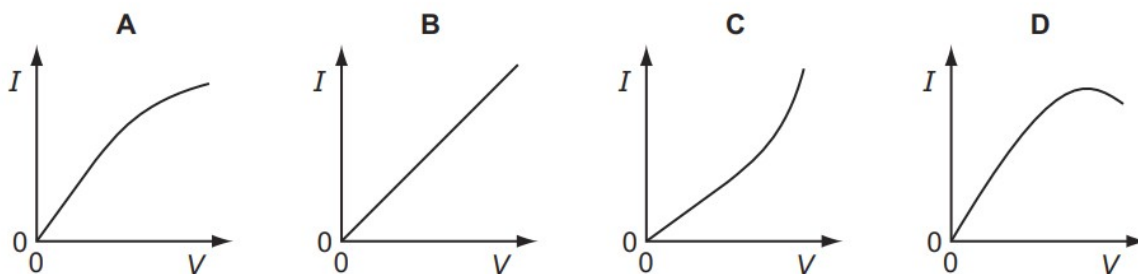


What is the potential difference across the resistor?

- A** 0.050 V **B** 1.8 V **C** 50 V **D** 1800 V

112. O/N/13/12/Q30

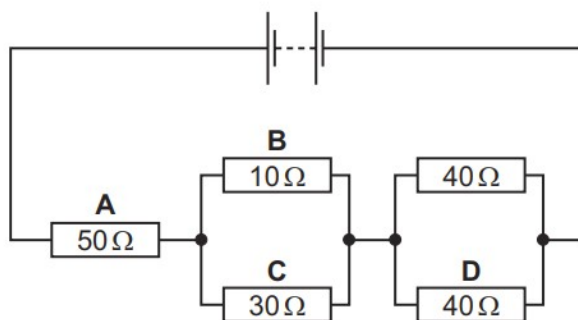
Which is the current/voltage (I/V) graph of a filament lamp?



113. O/N/13/12/Q31

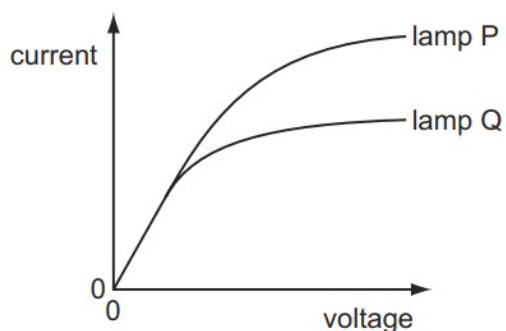
The diagram shows a circuit containing five resistors connected to a battery.

In which resistor is the current the smallest?



114. M/J/13/12/Q33

The diagram shows the current-voltage graph of two filament lamps.

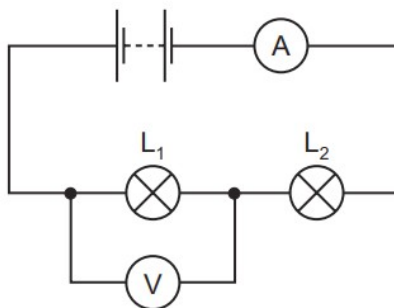


Which statement is correct?

- A** Lamp P has a lower resistance than lamp Q at all currents.
- B** Lamp P has the same resistance as lamp Q at low currents.
- C** Lamp P has the same resistance as lamp Q at high currents.
- D** Lamp P has a higher resistance than lamp Q at all currents.

115. M/J/13/12/Q34

A circuit containing two lamps L_1 and L_2 is connected as shown.



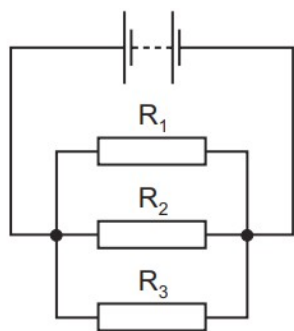
A voltmeter measures the potential difference across the lamp L_1 .

The filament of lamp L_1 breaks. What happens to the readings of the ammeter and of the voltmeter?

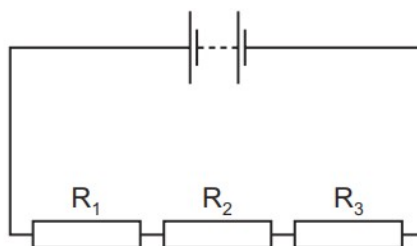
	reading on the ammeter	reading on the voltmeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

116. M/J/13/12/Q35

In the circuits P and Q below, resistors R_1 , R_2 and R_3 have different resistances.



circuit P



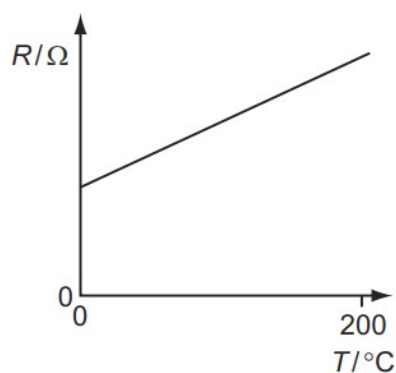
circuit Q

In which circuit are the currents in the resistors equal and in which circuit are the potential differences across the resistors equal?

	currents equal	potential differences equal
A	P	P
B	P	Q
C	Q	P
D	Q	Q

117. M/J/13/11/Q31

The electrical resistance of a metal wire varies with temperature as shown.



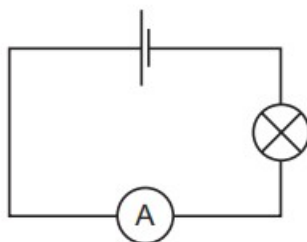
The resistance of the wire at 0°C is $1.00\ \Omega$ and at 100°C is $1.40\ \Omega$.

What is the resistance of the wire at 200°C ?

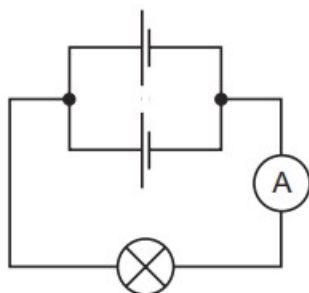
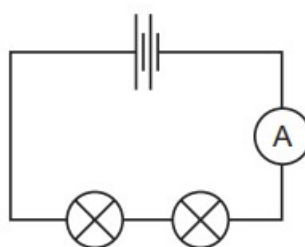
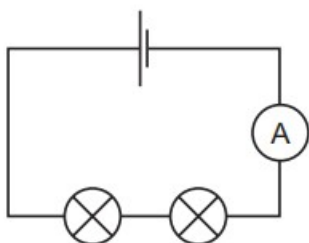
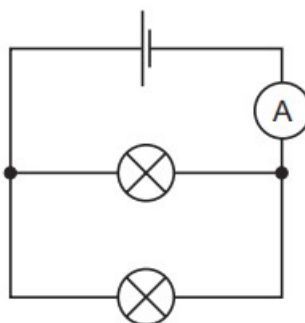
- A** $1.80\ \Omega$ **B** $1.96\ \Omega$ **C** $2.00\ \Omega$ **D** $2.80\ \Omega$

118. M/J/13/11/Q32

A cell is connected in series with an ammeter and a lamp. The reading on the ammeter is 1 A.



In which circuit, using identical lamps, ammeters and cells, is the reading on the ammeter 2 A?

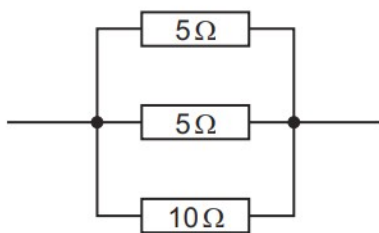
A**B****C****D****119. M/J/13/11/Q30**

Which of the following is equivalent to 1 V?

- A** 1 J/A
- B** 1 J/C
- C** 1 J/s
- D** 1 J/W

120. O/N/12/11/Q29

The diagram shows three resistors in parallel.

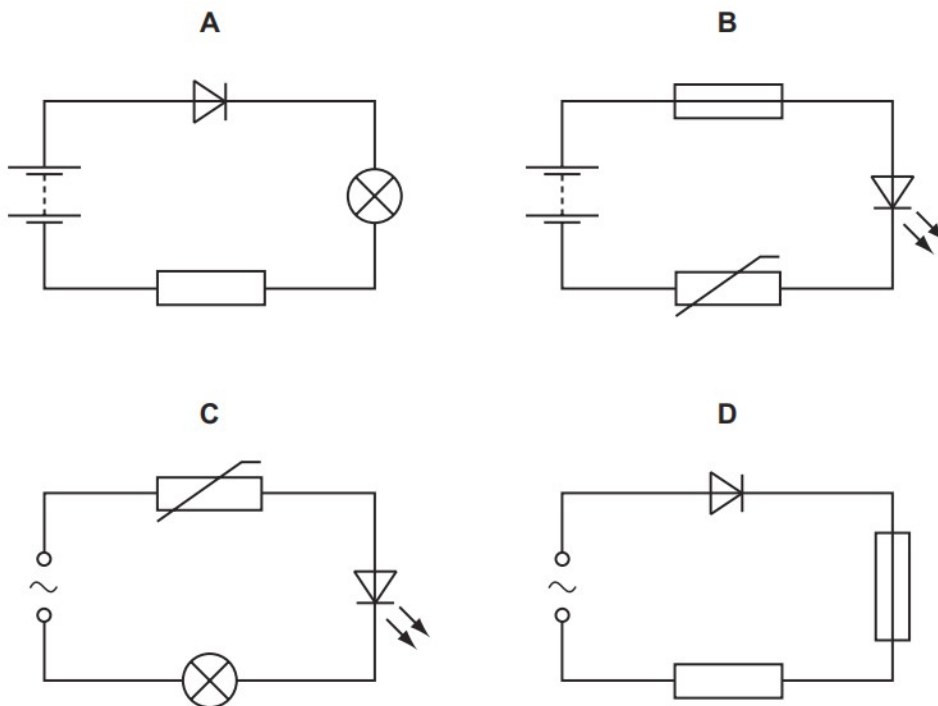


What is the combined resistance?

- A** $0.50\ \Omega$ **B** $2.0\ \Omega$ **C** $20\ \Omega$ **D** $250\ \Omega$

121. O/N/12/11/Q30

Which circuit contains a fuse and a rectifying diode?



122. O/N/12/12/Q28

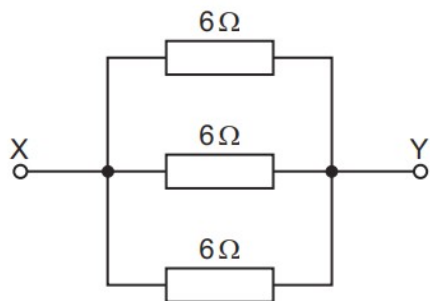
A metal wire has length l and cross-sectional area A .

What is the resistance proportional to?

- A** $A + l$ **B** $\frac{A}{l}$ **C** $\frac{l}{A}$ **D** $l \times A$

123. O/N/12/12/Q29

A student joins three $6\ \Omega$ resistors as shown in the diagram.

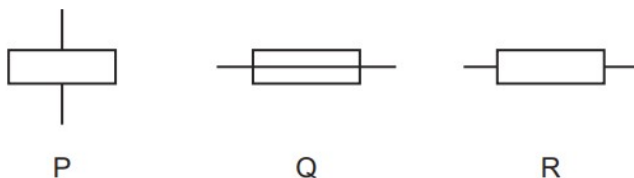


What is the total resistance between points X and Y?

- A** $0.5\ \Omega$ **B** $2\ \Omega$ **C** $6\ \Omega$ **D** $18\ \Omega$

124. O/N/12/12/Q30

P, Q and R are electrical symbols.

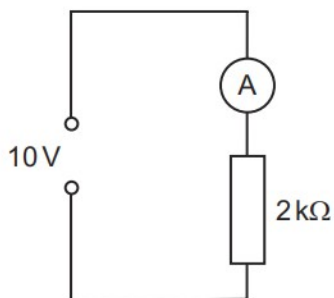


What do these symbols represent?

	P	Q	R
A	fuse	resistor	relay coil
B	relay coil	fuse	resistor
C	relay coil	resistor	fuse
D	resistor	fuse	relay coil

125. M/J/12/12/Q29

The diagram shows an ammeter connected in a circuit.

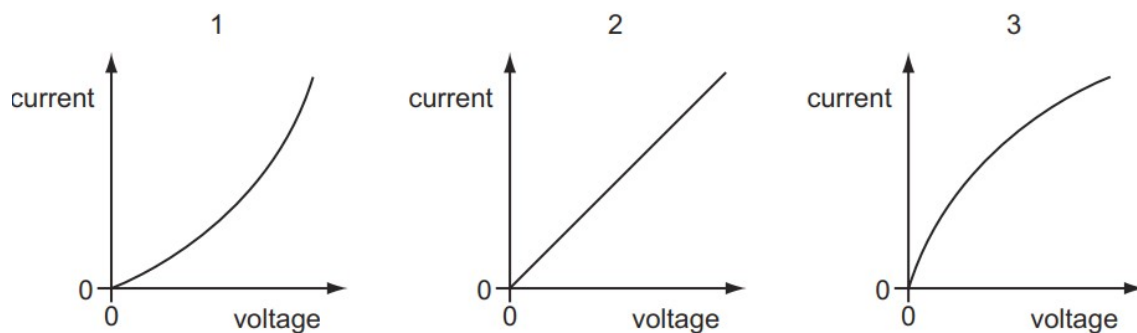


What is the current in the ammeter?

- A** 5 mA **B** 20 mA **C** 0.2 A **D** 5 A

126. M/J/12/12/Q28

The current/voltage graphs are for different electrical components.

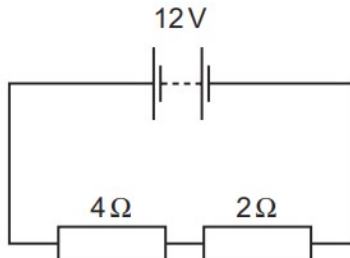


Which graph is for a resistor at constant temperature and which is for a filament lamp?

	resistor	lamp
A	1	2
B	2	1
C	2	3
D	3	2

127. M/J/12/12/Q30

In the circuit shown, the potential difference (p.d.) across the 4Ω resistor is 8 V .

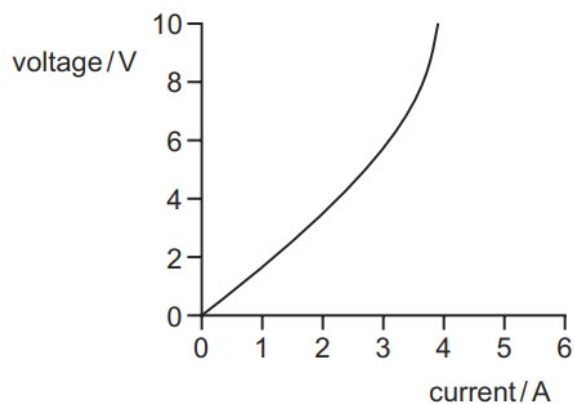


What is the p.d. across the 2Ω resistor?

- A** 4 V **B** 6 V **C** 8 V **D** 16 V

128. M/J/12/11/Q28

The voltage / current graph for a filament lamp is shown.

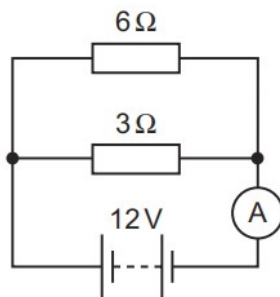


The voltage across the lamp increases. What happens to the temperature of the lamp and the resistance of the lamp?

	temperature	resistance
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

129. M/J/12/11/Q29

A 12 V battery is connected across a parallel arrangement of two resistors.

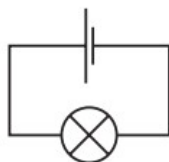


What is the reading on the ammeter?

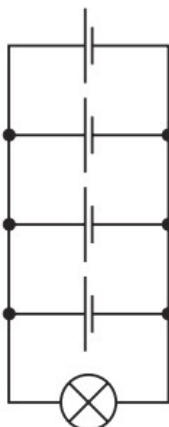
- A** 1.3 A **B** 2.0 A **C** 4.0 A **D** 6.0 A

132. O/N/11/11/Q29

A lamp, designed to work at 1.5 V, is connected to a cell of electromotive force (e.m.f.) 1.5 V. The lamp lights at normal brightness.



The lamp is now connected to four similar cells, each of e.m.f. 1.5 V, arranged in parallel.

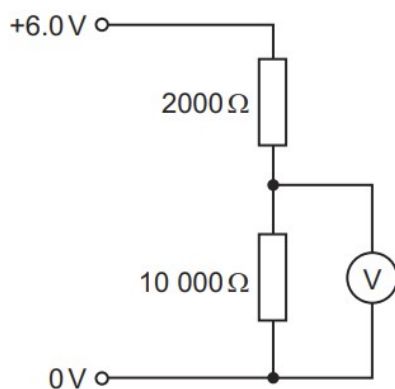


What is the effect of connecting the extra cells in this way?

- A** The lamp burns out.
- B** The lamp is dimmer.
- C** The lamp produces light for a longer time.
- D** The lamp produces light for a shorter time.

133. O/N/11/12/Q28

The diagram shows a potential divider system of two resistors connected to a 6.0 V power supply.

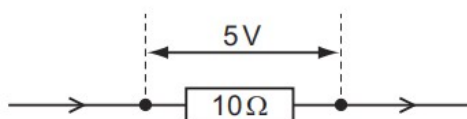


What is the reading on the voltmeter?

- A** 1.0 V
- B** 1.2 V
- C** 3.0 V
- D** 5.0 V

134. M/J/11/12/Q27

The potential difference (p.d.) across a 10Ω resistor is 5 V .



How much charge passes through the 10Ω resistor in 30 seconds?

- A** 2 C **B** 15 C **C** 60 C **D** 1500 C

135. M/J/11/12/Q28

Which changes both cause a decrease in the resistance of a copper wire?

	size of wire	temperature of wire
A	decrease in length	lower
B	increase in length	lower
C	decrease in thickness	higher
D	increase in thickness	higher

136. M/J/11/12/Q30

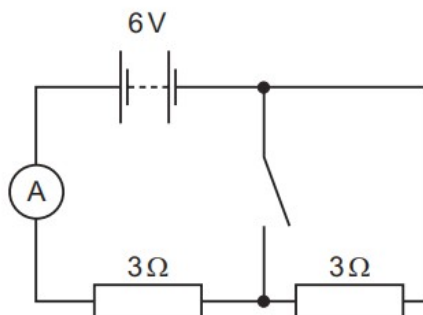
A lamp rated 6 V , 2 A is switched on for 60 s.

How much energy is used?

- A** 0.2 J **B** 20 J **C** 180 J **D** 720 J

137. M/J/11/12/Q31

The diagram shows a circuit.

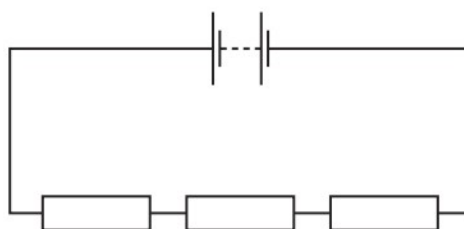


What is the reading on the ammeter when the switch is open, and the reading when it is closed?

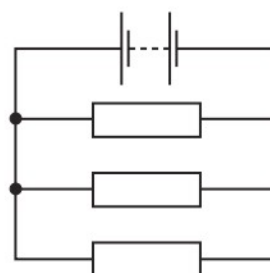
	ammeter reading when open / A	ammeter reading when closed / A
A	1	1
B	1	2
C	2	1
D	2	2

138. O/N/10/11/Q30

When three identical resistors are connected in series, their combined resistance is $6\ \Omega$.



series



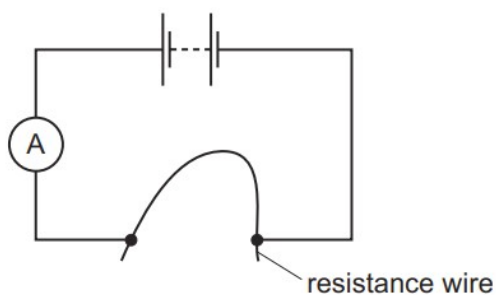
parallel

What is their combined resistance when they are connected in parallel?

- A** $\frac{1}{6}\ \Omega$ **B** $\frac{2}{3}\ \Omega$ **C** $\frac{3}{2}\ \Omega$ **D** $6\ \Omega$

139. M/J/10/12/Q27

A length of resistance wire is used as a resistor in a simple circuit.



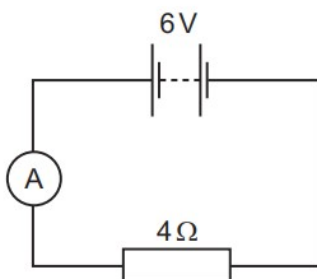
Four separate changes are made to the wire.

Which change will **not** reduce the value of the resistance of the wire?

- A** It is covered in an insulating sleeve.
B Its cross-sectional area is increased.
C Its length is decreased.
D Its temperature is decreased.

140. M/J/10/12/Q31

A 6 V supply is connected in series with an ammeter and a 4Ω resistor.



What is the reading on the ammeter?

- A** 0.67 A **B** 1.5 A **C** 10 A **D** 24 A

141. M/J/10/12/Q30

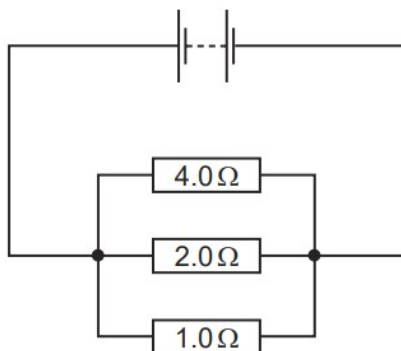
Sets of voltage-current readings are obtained for different electrical components.

Which set of readings is for a 100Ω resistor?

A	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-30	-15	-5	0	+5	+15	+30
B	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-30	-20	-10	0	+10	+20	+30
C	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-60	-40	-20	0	+20	+40	+60
D	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-60	-45	-30	0	+30	+45	+60

142. O/N/09Q26

The circuit diagram shows three resistors in parallel with a battery.



What is the effective resistance of these three resistors?

- A** 0.57Ω **B** 0.86Ω **C** 1.75Ω **D** 7.00Ω

143. O/N/09/Q27

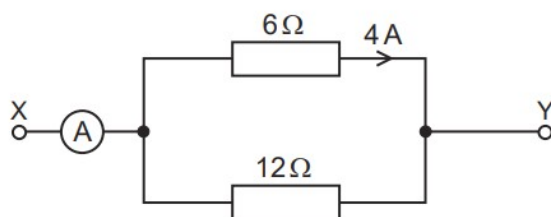
Ohm's law states that the current in a conductor is proportional to the potential difference across it, provided that a certain quantity remains constant.

What is this quantity?

- A length
- B pressure
- C temperature
- D thickness

144. O/N/09/Q28

Two resistors of $6\ \Omega$ and $12\ \Omega$ are arranged in parallel. A potential difference is connected across the terminals X and Y. The current in the $6\ \Omega$ resistor is $4\ \text{A}$.



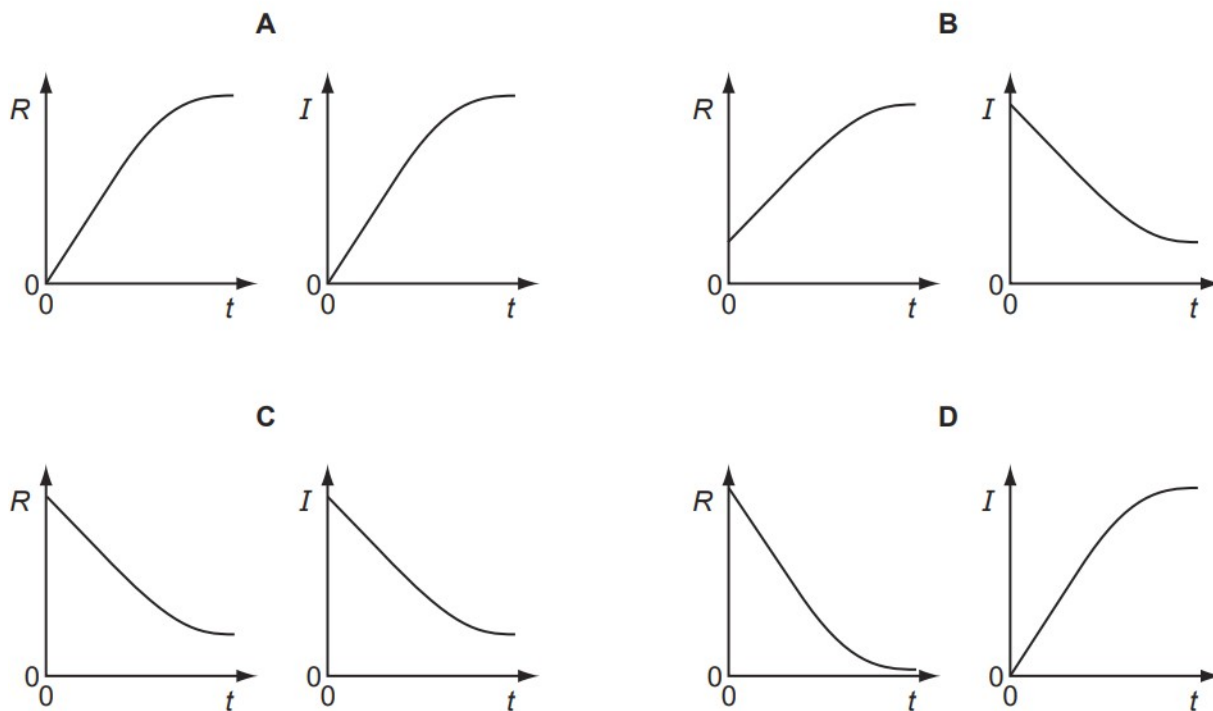
What is the current in the ammeter?

- A 4 A B 6 A C 8 A D 12 A

145. M/J/09/Q26

When a filament lamp is switched on, there is a current in the lamp. As the temperature of the filament rises, its resistance changes.

Which pair of graphs shows how the resistance R of the filament and the current I vary with time after the lamp is switched on?



146. O/N/08/Q27

A piece of polythene is rubbed with a cloth duster. The polythene becomes negatively charged and the cloth becomes positively charged.

What happens to the polythene and to the cloth to cause this?

	polythene	cloth
A	gains electrons	gains protons
B	gains electrons	loses electrons
C	loses protons	gains protons
D	loses protons	loses electrons

147. O/N/08/Q28

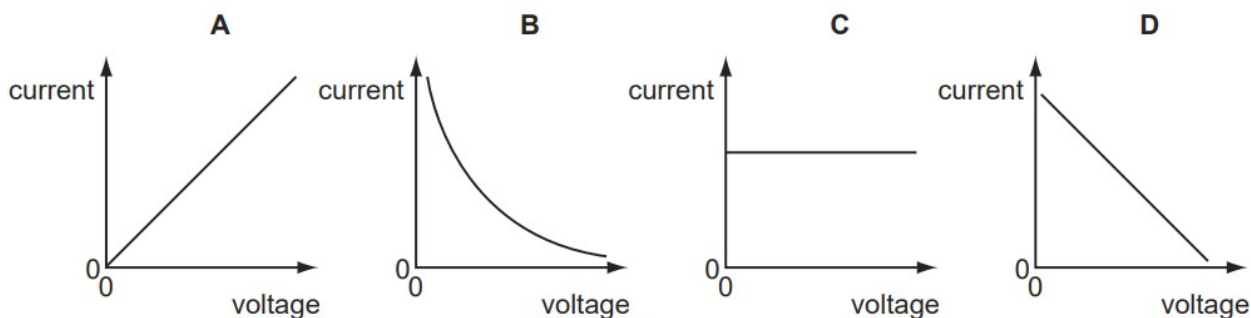
The current in an electric heater is 10 A. It is switched on for five minutes.

How much charge flows through the heater?

- A** 0.5 C **B** 2 C **C** 50 C **D** 3000 C

148. O/N/08/Q29

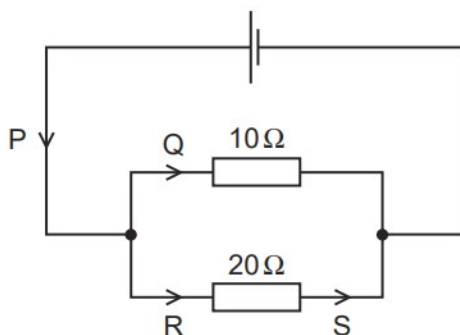
Which graph shows how the current changes when the voltage across a fixed resistance is varied?



149. O/N/08/Q30

The circuit diagram shows a parallel arrangement of resistors.

P, Q, R and S represent the current at the points shown.



Which statement is correct?

- A** P is greater than Q.
- B** Q is equal to R.
- C** R is greater than S.
- D** S is equal to P.

150. M/J/08/Q28

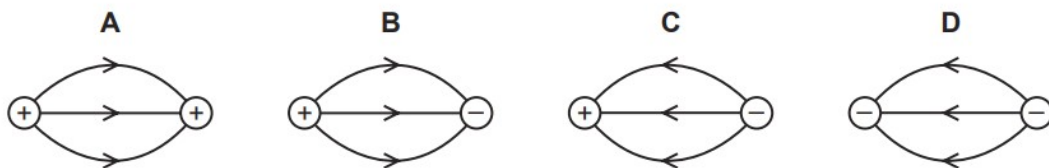
A perspex rod can be charged positively by rubbing it with a woollen cloth.

How does the rod gain its charge?

- A** The rod gains electrons.
- B** The rod gains protons.
- C** The rod loses electrons.
- D** The rod loses protons.

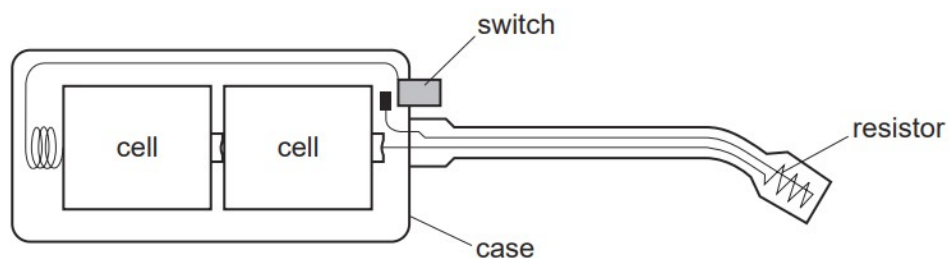
151. M/J/08/Q29

Which diagram correctly shows the electric field lines between two point charges?

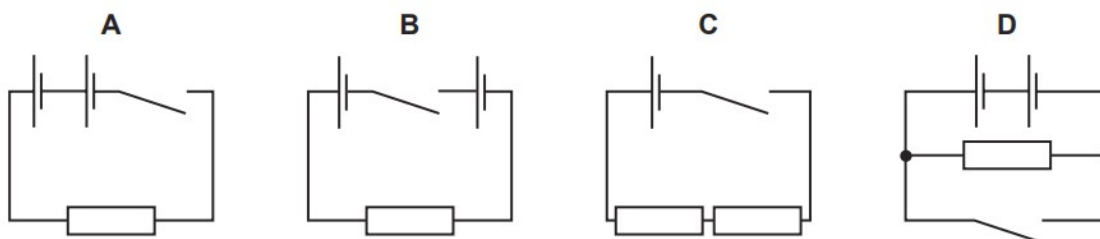


152. M/J/08/Q31

The diagram shows the components of a lighter for a gas cooker.



Which circuit diagram is correct for this lighter?



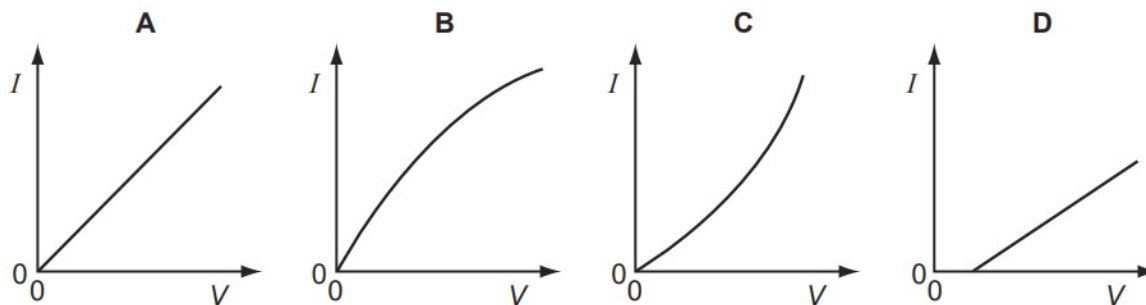
153. M/J/07/Q27

Which factors will both increase the resistance of a wire in a circuit?

	size of wire	temperature of wire
A	longer	lower
B	shorter	lower
C	thicker	higher
D	thinner	higher

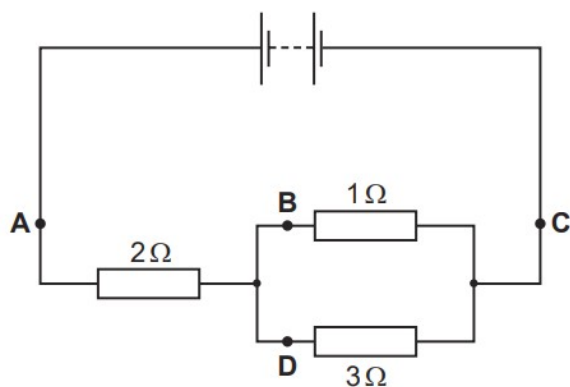
154. M/J/07/Q28

Which graph best represents how current I varies with voltage V in a component in which the resistance **increases** as the current **increases**.



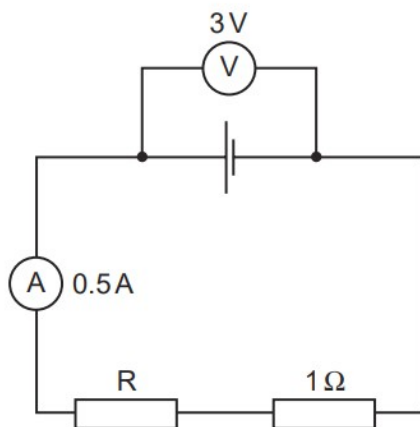
155. M/J/07/Q29

At which point in the circuit is the current the smallest?



156. M/J/07/Q30

The diagram shows a circuit.



The ammeter has negligible resistance.

What is the resistance of the resistor R?

- A** 0.5Ω **B** 1.5Ω **C** 5Ω **D** 6Ω

157. O/N/06/Q27

A wire has a resistance of 8Ω . A second wire, made of the same material, has half the length and twice the cross-sectional area.

What is the resistance of the second wire?

- A** 1Ω **B** 2Ω **C** 8Ω **D** 16Ω

158. M/J/06/Q30

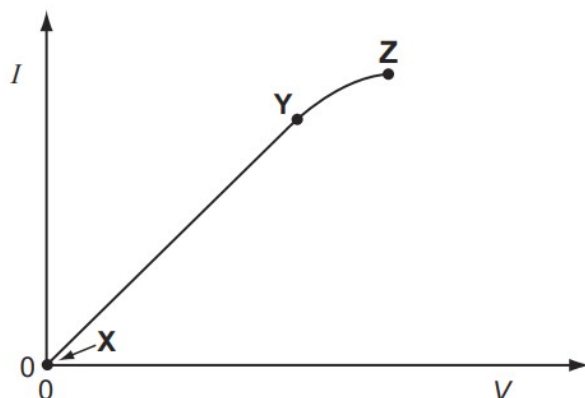
An electrical quantity is defined as 'the energy converted by a source in driving a unit charge round a complete circuit.'

What is this quantity called?

- A current
- B electromotive force
- C potential difference
- D power

159. M/J/06/Q31

The diagram shows the current I /voltage V graph for a length of resistance wire.



Where can Ohm's law be applied to the wire?

- A at Y only
- B at Z only
- C from X to Y
- D from X to Z

160. M/J/06/Q33

How much energy is converted in a resistor of 5.0Ω carrying a current of 2.0 A for 10 seconds?

- A 4.0 J
- B 25 J
- C 100 J
- D 200 J

161. M/J/06/Q32

Diagram 1 shows a resistor connected to a battery, an ammeter and a voltmeter.

The ammeter reading is 0.5 A and the voltmeter reading is 3.0 V.

A second identical resistor is now connected in parallel with the first resistor, as shown in diagram 2.

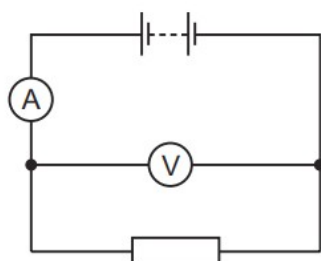


diagram 1

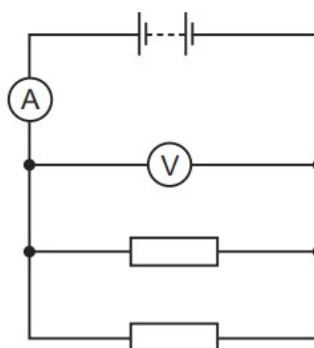


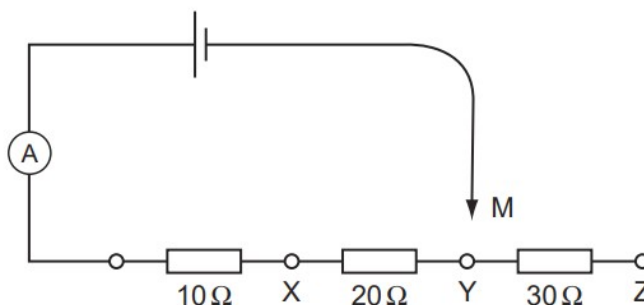
diagram 2

What are the ammeter and voltmeter readings in the circuit shown in diagram 2?

	ammeter reading / A	voltmeter reading / V
A	1.0	3.0
B	1.0	1.5
C	0.5	6.0
D	0.5	3.0

162. O/N/05/Q28

The diagram shows a cell connected in series with an ammeter and three resistors ($10\ \Omega$, $20\ \Omega$, $30\ \Omega$). The circuit can be completed by a moveable contact M.



When M is connected to X, the ammeter reads 0.6 A.

What is the ammeter reading when M is connected to Y?

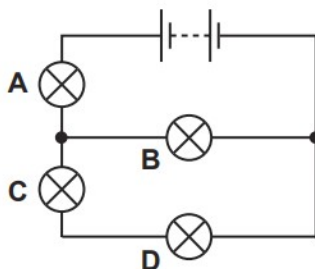
- A** 0.1 A **B** 0.2 A **C** 0.3 A **D** 0.6 A

163. O/N/05/Q29

In the circuit shown, the battery lights up all four lamps.

When one of the lamp filaments melts, the other three lamps stay on.

Which lamp filament melts?



164. M/J/05/Q27

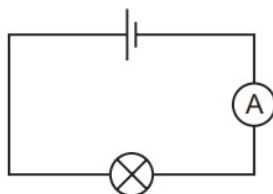
A cell is connected to a resistor.

What is the e.m.f. of the cell equal to?

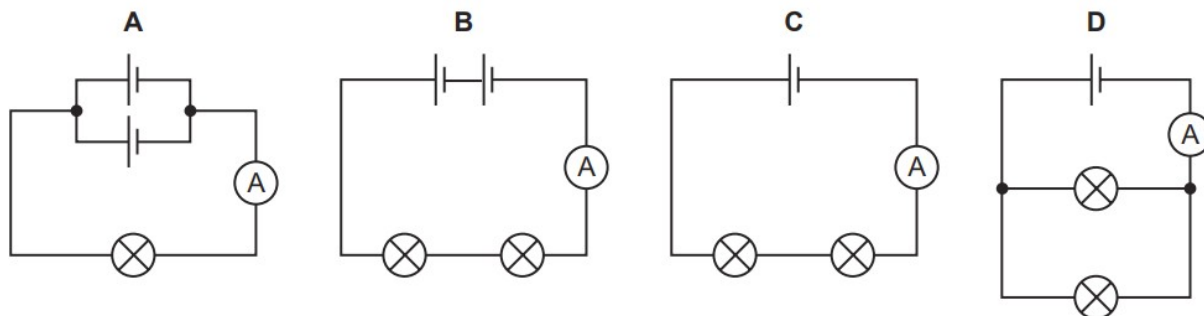
- A** The potential difference across the resistor for each unit of current.
- B** The power produced in the circuit for each unit of charge that passes.
- C** The work done in the circuit for each unit of charge that passes.
- D** The work done in the circuit for each unit of current.

165. M/J/05/Q28

A cell is connected in series with an ammeter and a lamp. The current is 1 A.

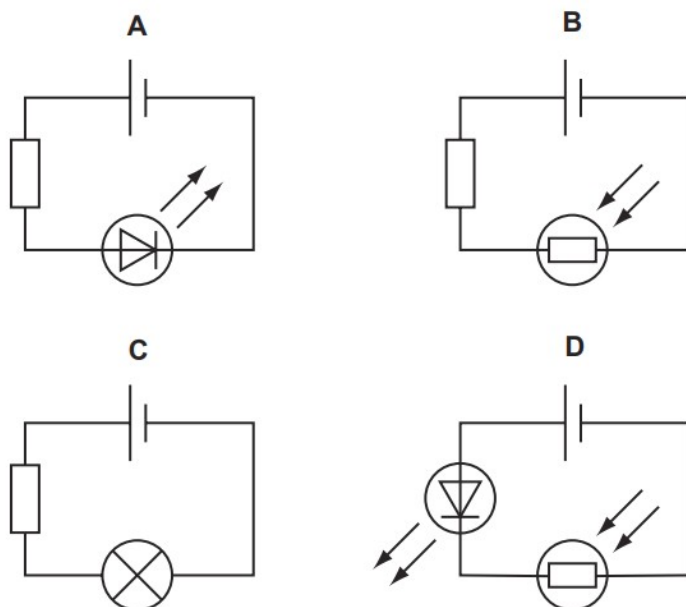


In which circuit, using identical cells, lamps and ammeters, is the current reading 2 A?



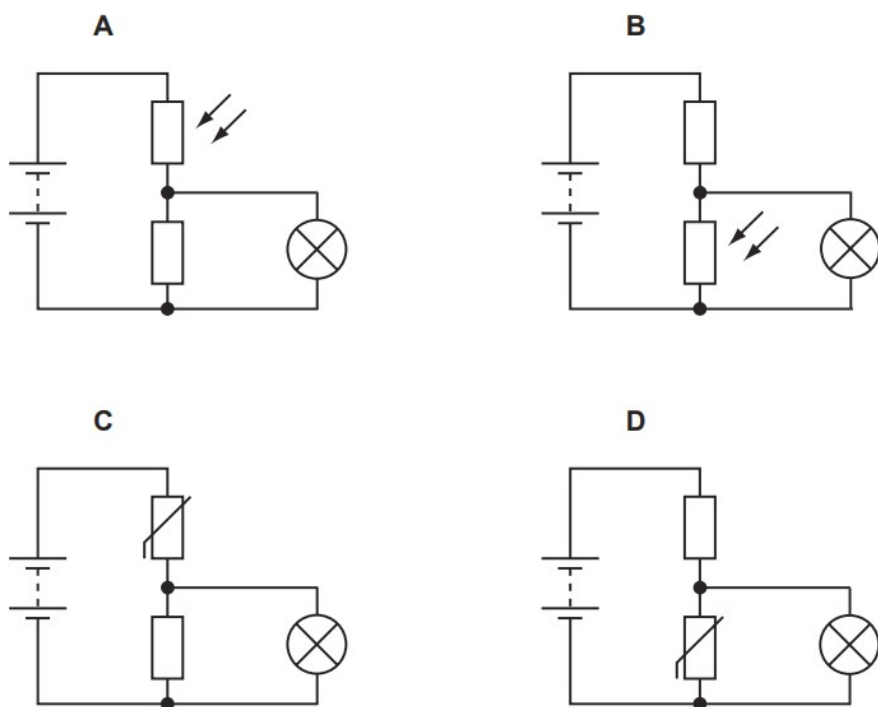
166. M/J/05/Q29

Which circuit contains a cell, a light-emitting diode and a fixed resistor?



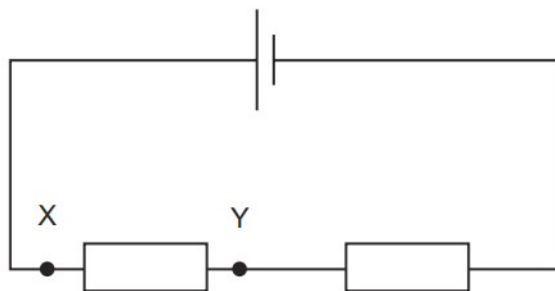
167. O/N/04/Q36

In which circuit will a lamp glow more brightly when less light shines on a light-dependent resistor?



168. M/J/04/Q28

The diagram shows two resistors connected in series with a cell.

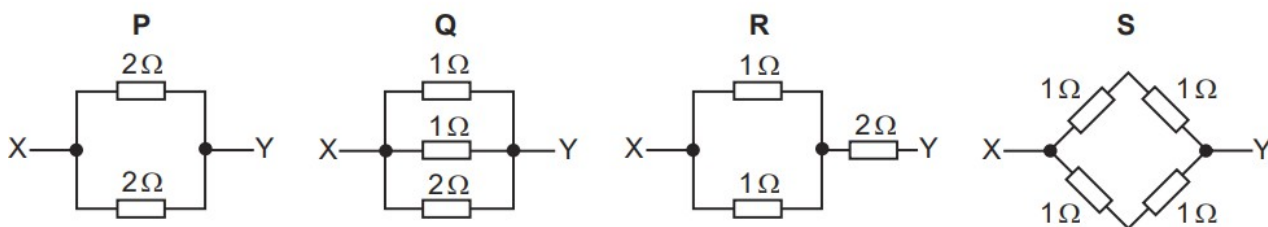


Which statement defines the potential difference across XY?

- A** the power needed to drive a unit charge through the cell
- B** the power needed to drive a unit charge between X and Y
- C** the work done in driving a unit charge through the cell
- D** the work done in driving a unit charge between X and Y

169. M/J/04/Q29

Which two resistor combinations have the same resistance between X and Y?



- A** P and Q
- B** P and S
- C** Q and R
- D** R and S

ANSWERS

1. B	48. C	97. B	146. B
2. A	49. A	98. A	147. D
3. C	50. C	99. A	148. A
4. C	51. C	100. B	149. A
5. B	52. A	101. D	150. C
6. B	53. B	102. A	151. B
7. A	54. A	103. A	152. A
8. A	55. B	104. B	153. D
9. A	56. B	105. B	154. B
10. D	57. B	106. A	155. D
11. C	58. B	107. B	156. C
12. B	59. B	108. A	157. B
13. B	60. D	109. B	158. B
14. A	61. D	110. C	159. C
15. A	62. A	111. B	160. D
16. A	63. B	112. A	161. A
17. C	64. D	113. C	162. B
18. A	65. A	114. B	163. B
19. D	66. D	115. B	164. C
20. C	67. D	116. C	165. D
21. A	68. B	117. A	166. A
22. D	69. D	118. D	167. B
23. C	70. D	119. B	168. D
24. B	71. C	120. B	169. B
25. D	72. C	121. D	
26. C	73. D	122. C	
27. B	74. D	123. B	
28. C	75. A	124. B	
29. D	76. B	125. A	
30. D	77. C	126. C	
31. C	78. D	127. A	
32. D	79. A	128. D	
33. D	80. C	129. D	
34. D	81. C	130. B	
35. A	82. A	131. B	
36. C	83. C	132. C	
37. B	84. C	133. D	
38. B	85. A	134. B	
39. A	86. D	135. A	
40. A	87. B	136. D	
41. A	88. A	137. B	
42. B	89. C	138. B	
43. C	90. B	139. A	
44. D	91. A	140. B	
45. D	92. C	141. B	
46. C	93. B	142. A	
47. B	94. B	143. C	
	95. C	144. D	
	96. C	145. B	

1. O/N/23/22/Q7

An electric circuit contains a battery, a fixed resistor R , a voltmeter and a thermistor.

Fig. 7.1 is an incomplete circuit diagram of the arrangement.

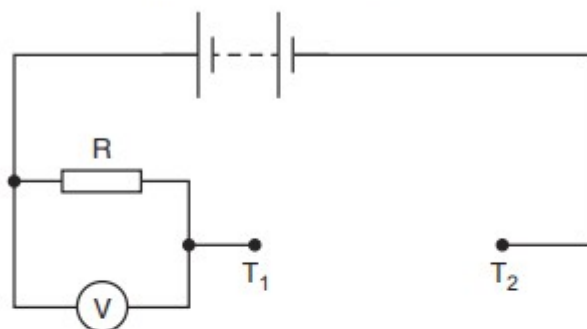


Fig. 7.1

The thermistor is connected between terminals T_1 and T_2 .

- (a) Complete Fig. 7.1 by drawing the circuit symbol for a thermistor connected between T_1 and T_2 . [1]

- (b) The resistance of R is 2.5Ω . The reading on the voltmeter is 0.40 V .

Calculate:

- (i) the current in R

current = A [2]

- (ii) the power transferred in R .

power = W [2]

- (c) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....

 [2]

2. O/N/23/21/Q6

Fig. 6.1 shows a circuit that contains two identical cells, each of e.m.f. 1.5V, that are connected in parallel.

The circuit also contains two ammeters, a thermistor T and two resistors R and S.

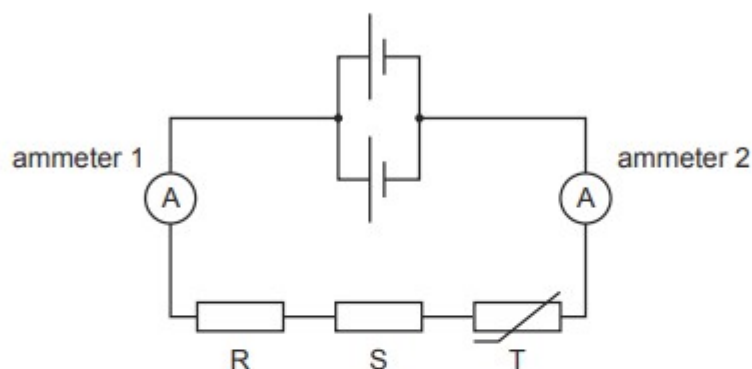


Fig. 6.1

The reading on ammeter 1 is 0.20A.

- (a) State why the reading on ammeter 2 is also 0.20A.

.....
 [1]

- (b) The resistance of R is 1.5Ω and the resistance of S is 3.5Ω .

- (i) Calculate the resistance of T.

resistance = Ω [3]

- (ii) The temperature of T increases.

Explain why the potential difference (p.d.) across T decreases.

.....

 [3]

3. M/J/23/22/Q6

Fig. 6.1 shows an electric circuit containing a filament lamp, a resistor R, a 12 V battery and five meters.

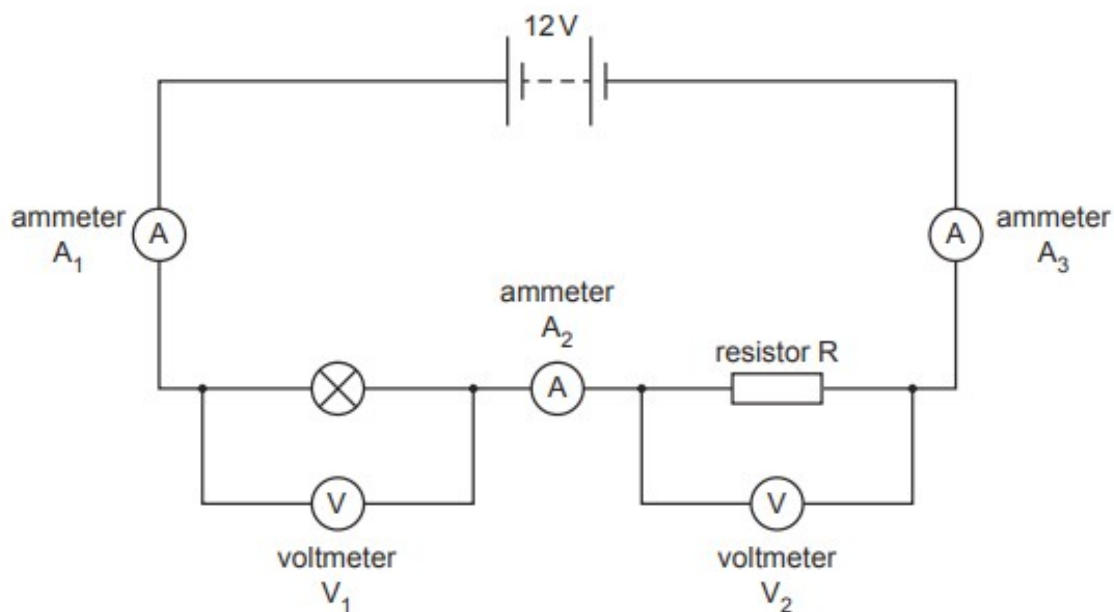


Fig. 6.1

- (a) (i) The reading on ammeter A_1 is 0.25 A.

The reading on voltmeter V_1 is 3.0 V.

Determine the readings on the other meters.

reading on ammeter A_2 = A

reading on ammeter A_3 = A

reading on voltmeter V_2 = V
[2]

- (ii) Calculate the resistance of resistor R.

resistance of resistor R = Ω [2]

(iii) The resistor obeys Ohm's law.

State Ohm's law.

.....
.....
..... [2]

(b) Fig. 6.2 shows the current–voltage graph for the filament lamp.

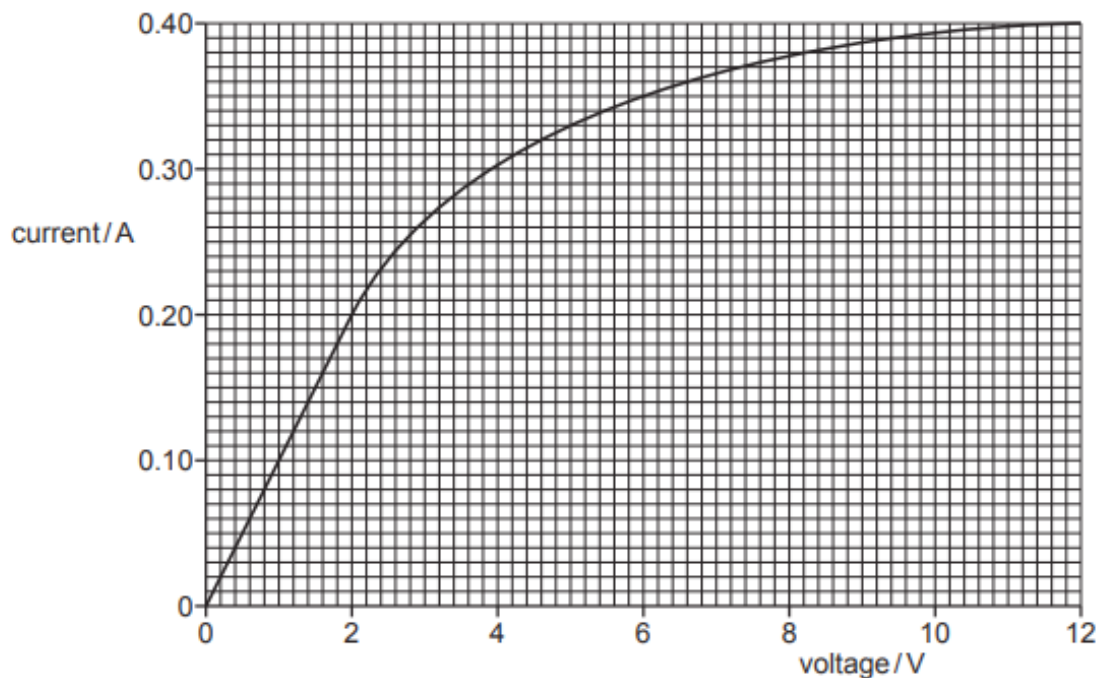


Fig. 6.2

The battery in Fig. 6.1 is replaced with a different battery which has a different e.m.f. (electromotive force).

The voltage across the lamp increases to 6.0 V.

Use data from the graph to determine the e.m.f. of the second battery.

Show your working.

e.m.f. = V [3]

4. M/J/23/21/Q3

Fig. 3.1 shows a circuit containing three resistors, a 12 V power supply and an ammeter.

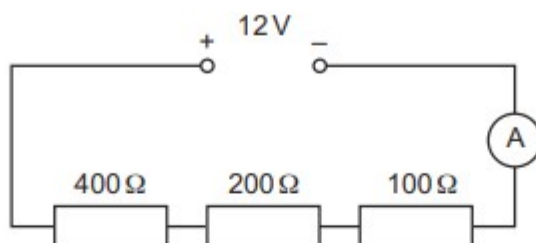


Fig. 3.1

- (a)** Calculate the current in the 400 Ω resistor.

current = A [2]

- (b)** Calculate the power produced in the 400 Ω resistor.

power = W [2]

- (c)** A student uses all of the components shown in Fig. 3.1 in another circuit.

She connects them so that there is the largest possible current in each resistor.

The current in one of the resistors is larger than the current in the other two resistors. The student connects the ammeter into the circuit to measure the current in this resistor.

Draw the circuit diagram of the arrangement. Label each resistor with the value of its resistance.

5. O/N/22/22/Q5

The power supply in a circuit is made from several cells, each with the same electromotive force (e.m.f.). The cells are connected in parallel.

- (a) Explain what is meant by 'electromotive force'.

.....

 [2]

- (b) State **one** advantage of using several cells in parallel rather than a single cell as the power supply.

.....
 [1]

- (c) The power supply is connected in series to a resistor of resistance $4000\ \Omega$ and a thermistor. There is a voltmeter in parallel with the resistor.

Fig. 5.1 is the circuit diagram.

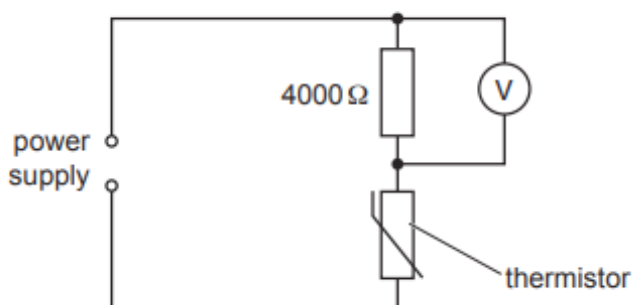


Fig. 5.1

- (i) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....

 [3]

- (ii) The e.m.f. of the power supply is 1.5 V.

Calculate the reading on the voltmeter when the resistance of the thermistor is $8000\ \Omega$.

reading = [2]

6. O/N/22/22/Q8

An uncharged plastic rod is rubbed with an uncharged woollen cloth.

As a result, the rod becomes negatively charged.

- (a) Explain, in terms of particles, how the rod becomes charged and state what happens to the cloth.

.....

.....

.....

..... [2]

- (b) Fig. 8.1 shows a metal sphere X on a plastic stand.

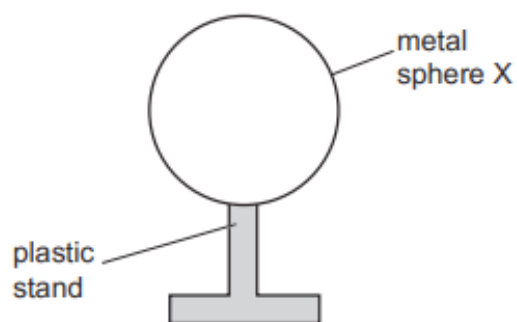


Fig. 8.1

Explain why metals are electrical conductors and why plastics are electrical insulators.

.....

.....

.....

..... [2]

- (d) A second metal sphere Y is negatively charged.

There is an electric field surrounding Y.

- (i) State what is meant by 'electric field'.

.....
..... [1]

- (ii) Fig. 8.4 shows the negatively charged sphere Y. The negative charges are not shown.



Fig. 8.4

On Fig. 8.4, draw the pattern and mark the direction of the electric field lines surrounding sphere Y. [3]

7. M/J/22/22/Q5

(a) A student connects a battery, thermistor X and resistor Y in parallel, as shown in Fig. 5.1.

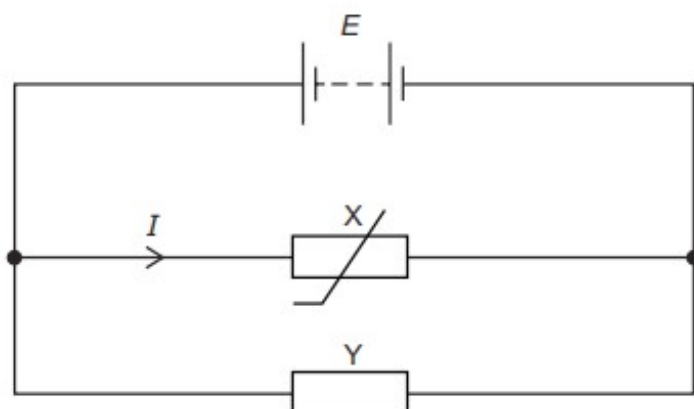


Fig. 5.1

- The electromotive force (e.m.f.) of the battery is E .
- The current in X is I .
- The resistance of Y is greater than the resistance of X .

Tick **two** boxes, one to show the correct statement about the potential difference across Y and the other to show the correct statement about the current in Y .

The potential difference across Y :

is less than E

☐

is equal to E

☐

is greater than E .

☐

The current in Y :

is less than I

☐

is equal to I

☐

is greater than I .

☐

[1]

- (b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0V, as shown in Fig. 5.2.

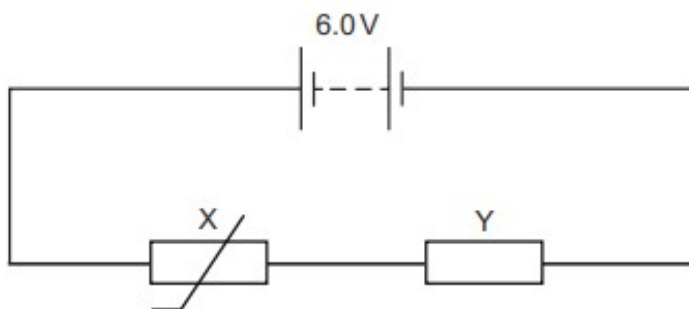


Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is 600Ω and the current in thermistor X is 0.0020 A.

- (i) Calculate the power produced in the thermistor.

power = [2]

- (ii) Calculate the resistance of Y.

resistance = [3]

- (iii) The thermistor is cooled.

Explain why this causes the potential difference across Y to decrease.

.....

 [2]

[Total: 8]

8. M/J/22/21/Q6

- (a) A light-dependent resistor (LDR) is used to sense the amount of light in a room.

Fig. 6.1 shows part of the circuit used. The LDR is not shown.

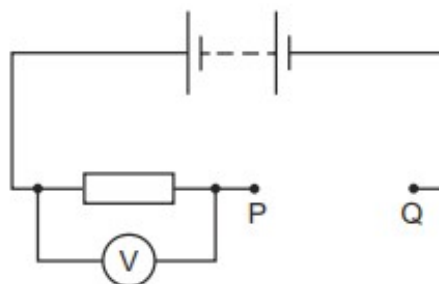


Fig. 6.1

- (i) Complete Fig. 6.1 by drawing the circuit symbol for an LDR between points P and Q. [1]
- (ii) Explain how the voltmeter reading changes as the amount of light falling on the LDR increases.

.....

.....

..... [2]

- (b) Fig. 6.2 shows part of a circuit containing three resistors.

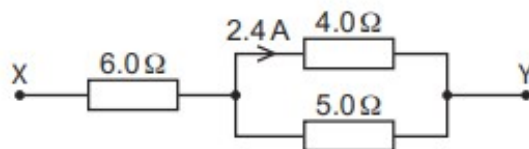


Fig. 6.2

The current in the $4.0\ \Omega$ resistor is $2.4\ \text{A}$.

- (i) Show that the current in the $5.0\ \Omega$ resistor is $1.9\ \text{A}$.

- (ii) Calculate the potential difference (p.d.) between points X and Y.

[2]

p.d. = [2]

9. O/N/21/22/Q5

Fig. 5.1 shows a circuit that contains a direct current (d.c.) power supply, a light-dependent resistor (LDR), two fixed resistors, M and N, and two ammeters.

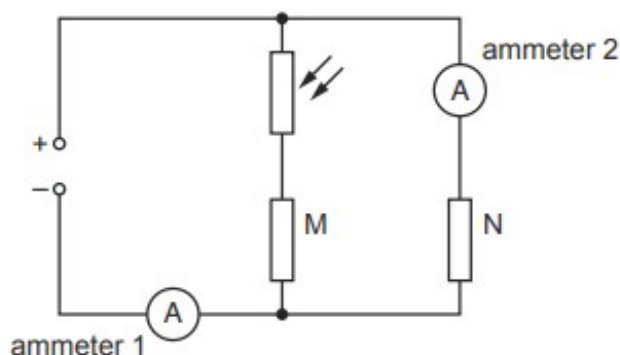


Fig. 5.1

- (a) The reading on ammeter 1 is 1.6 mA.

Calculate the time taken for a charge of 7.2 C to pass through ammeter 1.

time = [2]

- (b) Some time later, the brightness of the light incident on the LDR decreases.

Explain what happens to:

- (i) the reading on ammeter 2

.....
 [1]

- (ii) the time taken for a charge of 7.2 C to pass through ammeter 1.

.....

 [2]

[Total: 5]

10. O/N/21/21/Q6

A circuit contains a battery of electromotive force (e.m.f.) 4.5V, a 28Ω resistor and a variable resistor. Fig. 6.1 is the circuit diagram.

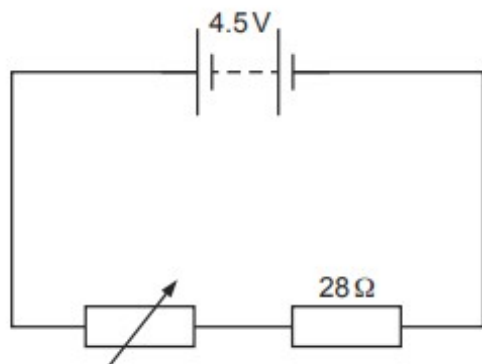


Fig. 6.1

- (a) The battery consists of three identical cells in series.

Determine the e.m.f. of each cell.

e.m.f. of each cell = [1]

- (b) The potential difference (p.d.) across the variable resistor is 2.5V.

- (i) State what is meant by *potential difference*.

.....
.....
.....
..... [2]

- (ii) Calculate the resistance of the variable resistor.

resistance = [3]

[Total: 6]

11. M/J/21/22/Q6

The power input to a filament lamp when connected to a 12 V supply is 20 W.

- (a) State what is meant by *power*.

.....
..... [1]

- (b) Calculate:

- (i) the current in the filament

current = [2]

- (ii) the resistance of the filament.

resistance = [1]

- (c) The filament in another lamp is made from thinner wire of the same length and made from the same material. Both lamps are connected to a 12 V supply.

State and explain how using thinner wire in the filament affects the power input to the lamp.

.....
.....
..... [2]

[Total: 6]

12. M/J/21/22/Q8

Fig. 8.1 shows a potential divider circuit containing two resistors R_1 and R_2 .

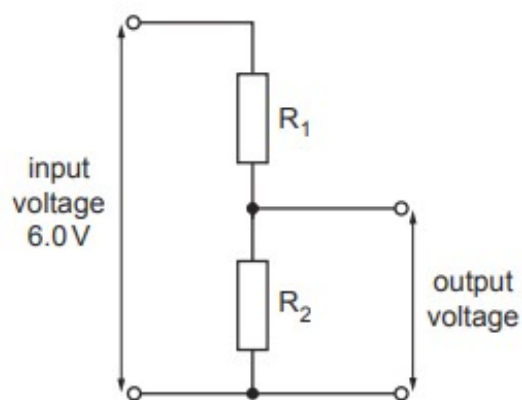


Fig. 8.1

- (a) There is an input voltage of 6.0 V.

Complete Table 8.1 to show the output voltage for different values of the resistances of the two resistors.

Table 8.1

resistance of R_1 / Ω	resistance of R_2 / Ω	output voltage / V
8.0	8.0	
800	1200	

[3]

(b) In the circuit shown in Fig. 8.1, resistor R_1 is replaced by a light dependent resistor (LDR).

(i) In the space below, draw the circuit symbol for an LDR.

[1]

(ii) The LDR is moved from darkness into daylight.

State what happens to the resistance of the LDR and the output voltage.

resistance

output voltage

[1]

(iii) A potential divider circuit can also be used to produce an output voltage that changes with temperature.

State the name of the component that is used instead of the LDR.

..... [1]

13. M/J/21/21/Q7

(a) Fig. 7.1 shows two circuit components.



Fig. 7.1

For each of these components state what causes an **increase** in its resistance.

thermistor

.....

LDR

.....

[2]

(b) Fig. 7.2 shows a battery connected to a diode, an ammeter and a resistor.

Fig. 7.3 shows the current–voltage graph for the diode.

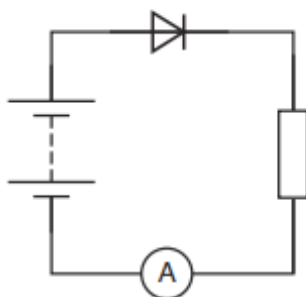


Fig. 7.2

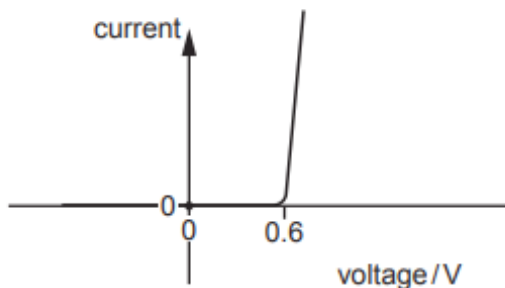


Fig. 7.3

The current in the resistor is 2.0 A and the potential difference across the resistor is 5.4 V.

(i) State what is meant by the *electromotive force* (e.m.f.) of the battery.

.....

..... [1]

(ii) Explain why the e.m.f. of the battery must be greater than 5.4 V.

.....

.....

..... [2]

- (iii) Calculate the electrical energy input to the resistor in 5.0 minutes.

energy input = [2]

- (iv) The circuit is reconnected with the diode in the opposite direction, as shown in Fig. 7.4.

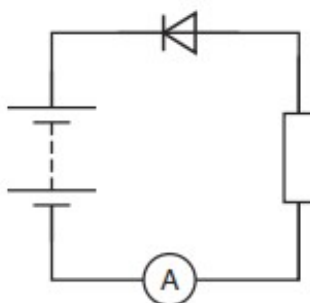


Fig. 7.4

Explain what is observed.

.....
.....
..... [1]

[Total: 8]

14. M/J/21/21/Q9

Fig. 9.1 is the current–voltage graph for a filament lamp.

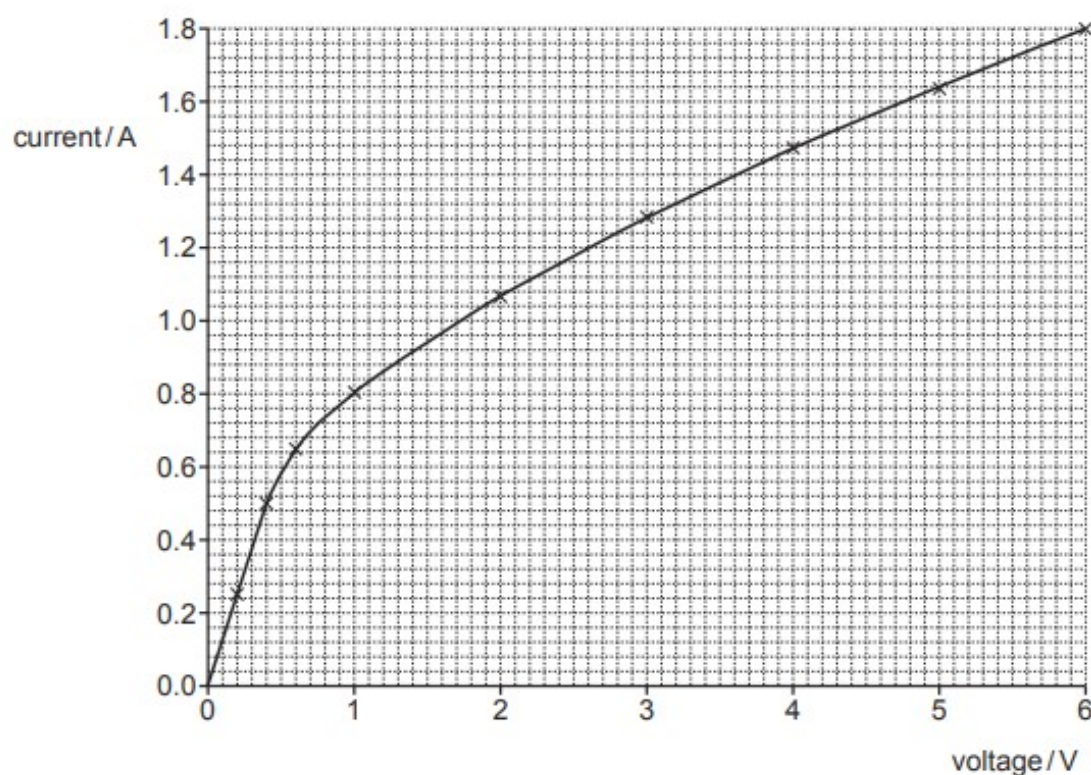


Fig. 9.1

- (a) Fig. 9.2 shows an incomplete circuit diagram of the circuit that a student uses to obtain the readings for the graph.

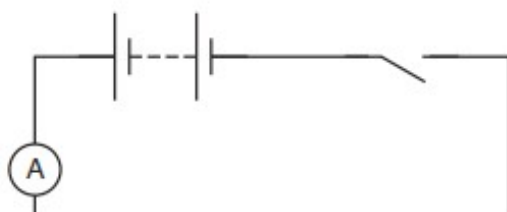


Fig. 9.2

Complete the circuit diagram using suitable circuit symbols for the lamp, a voltmeter and a variable resistor. [3]

- (b) The ammeter used by the student has ranges 0–10A, 0–1A, 0–100mA and 0–10mA.

Describe how the student uses the different ranges to obtain readings which allow all nine points to be plotted precisely on the graph while using the ammeter safely.

.....
.....
.....
..... [2]

- (c) State how Fig. 9.1 shows that the lamp does **not** obey Ohm's law.

.....
..... [1]

- (d) Using Fig. 9.1, determine the resistance of the lamp when:

- (i) the voltmeter reading is 0.40 V

resistance =

- (ii) the voltmeter reading is 6.0 V.

resistance =
[3]

- (e) (i) Explain why the resistance of the lamp changes as the voltage changes.

.....
.....
..... [2]

- (ii) State how the resistance of a wire depends upon its length and how it depends on its cross-sectional area.

length
.....
cross-sectional area
..... [2]

- (iii) The student obtains a long sample of the same type of wire as the filament in the lamp.

The wire he obtains has the same cross-sectional area as the filament.

He cuts a 1.0 m length of the wire and measures its resistance at room temperature.

He obtains the value $16\ \Omega$.

Using the appropriate resistance value from (d), estimate the length of the filament wire in the lamp.

length = [2]

[Total: 15]

15. O/N/20/21/Q10

Fig. 10.1 shows a wire of length l and cross-sectional area A .

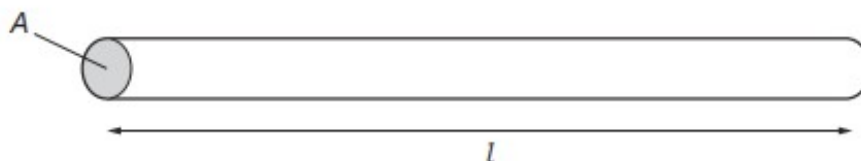


Fig. 10.1

(a) State how the resistance of the wire in Fig. 10.1 depends on:

(i) l

..... [1]

(ii) A .

..... [1]

(b) The cross-sectional area of a piece of metal wire is $7.5 \times 10^{-4} \text{ cm}^2$. The resistance of a 1.0 m length of the same wire is 6.4Ω .

The wire is made from metal W.

Fig. 10.2 shows a solid cube of side 1.0 cm. It is also made from metal W.

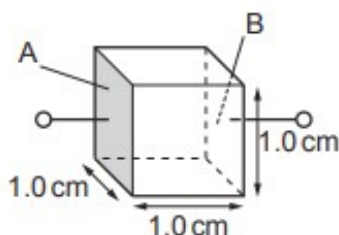


Fig. 10.2

Calculate the resistance between the two opposite faces A and B of the cube.

resistance = [2]

- (c) The wire in part (b) is taped to a metre rule.

Fig. 10.3 shows that a 1.0 m length of the wire (resistance $6.4\ \Omega$) is connected in series with a switch, a cell of electromotive force (e.m.f.) 1.2 V and a resistor of resistance $9.6\ \Omega$.

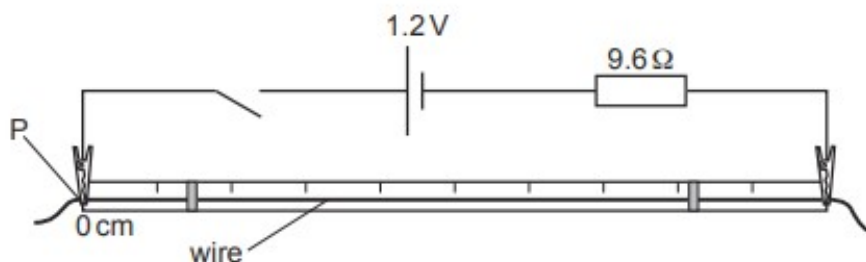


Fig. 10.3

The switch is closed.

- (i) Explain what is meant by *electromotive force (e.m.f.)*.

.....

 [2]

- (ii) Calculate the potential difference (p.d.) across the 1.0 m length of the wire.

p.d. = [3]

- (iii) One input terminal of an oscilloscope is connected to the wire at point P, the 0 cm mark of the metre rule.

The other terminal of the oscilloscope is connected to a sliding contact. Initially, this contact touches the wire at point P.

The Y-gain setting on the oscilloscope is 0.20 V/cm .

Fig. 10.4 shows the screen of the oscilloscope with a horizontal trace across the middle of the screen.

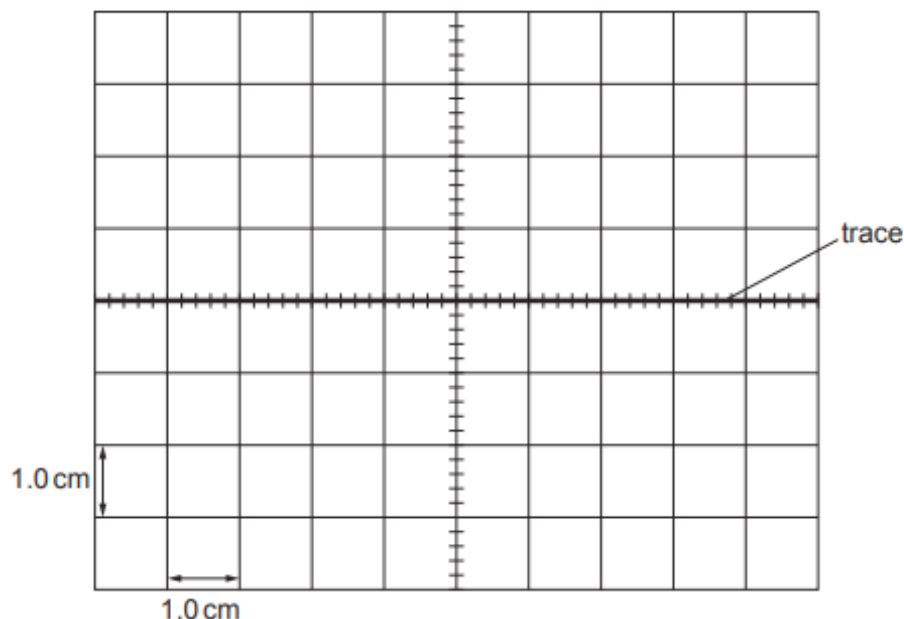


Fig. 10.4

The sliding contact is slowly moved along the wire until it reaches the other end of the metre rule.

Describe and explain what happens to the trace on the screen.

.....

.....

.....

..... [3]

(d) A second, identical 1.2 V cell is connected in parallel with the cell in the circuit in Fig. 10.3.

(i) State **one** advantage of using two cells in parallel rather than a single cell.

.....

..... [1]

(ii) State and explain the effect on the trace in (c)(iii) of adding the second cell in parallel.

.....

.....

..... [2]

[Total: 15]

16. M/J/20/22/Q8

Fig. 8.1 shows the circuit diagram of a temperature gauge. It contains an ammeter, a thermistor, a fixed resistor R and a battery.

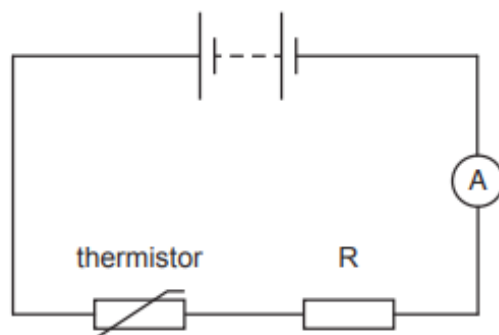


Fig. 8.1

The current is measured at different temperatures and a graph of the results is shown in Fig. 8.2.

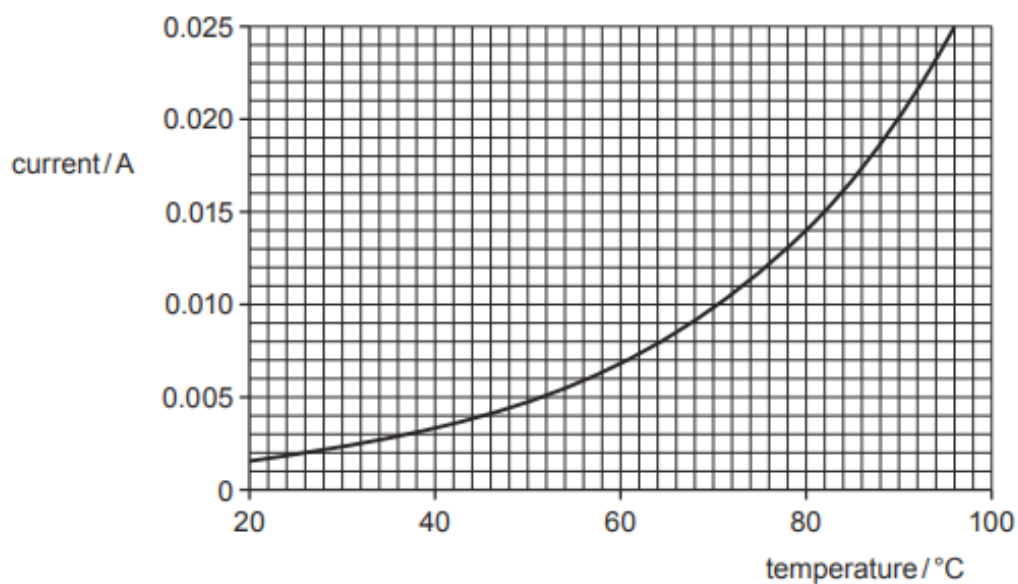


Fig. 8.2

- (a) (i) State how the resistance of the thermistor changes with temperature and explain how Fig. 8.2 shows this change.

.....

.....

..... [2]

- (ii) Resistance and current are used for the measurement of temperature.

State **one** other physical property that is used for the measurement of temperature.

.....
..... [1]

- (b) A temperature scale involves the use of fixed points.

State what is meant by:

- (i) *the ice point*

.....
..... [1]

- (ii) *the steam point.*

.....
..... [1]

- (c) At 80°C , the potential difference (p.d.) across the thermistor is 3.6 V.

- (i) State what is meant by *potential difference*.

.....
..... [1]

- (ii) Calculate the resistance of the thermistor when the temperature is 80°C .

resistance = [3]

- (iii) The electromotive force (e.m.f.) of the battery is 25 V.

Calculate the resistance of the resistor R.

resistance = [2]

17. M/J/20/21/Q8

Fig. 8.1 shows a lamp from a car. It contains two metal filaments.

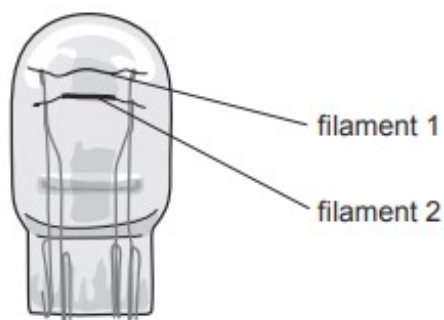


Fig. 8.1

- (a) (i) Complete the boxes to describe the transfer of energy that takes place when the lamp is switched on.

..... energy	→	 and energy
--------------	---	------------------------

[3]

- (ii) The efficiency of the metal filament lamp is less than 10%.

State what is meant by *efficiency*.

.....

..... [2]

- (b) The two filaments are usually connected in parallel to a car battery.

A student investigates what happens when the filaments are connected in series, rather than in parallel. He uses the same battery for the investigation.

State whether the current, the voltage across each filament and the total power produced *increases, decreases or stays the same* when the two filaments are connected in series.

current

voltage

power

[2]

(c) Fig. 8.2 shows the current–voltage graph for the two filaments.

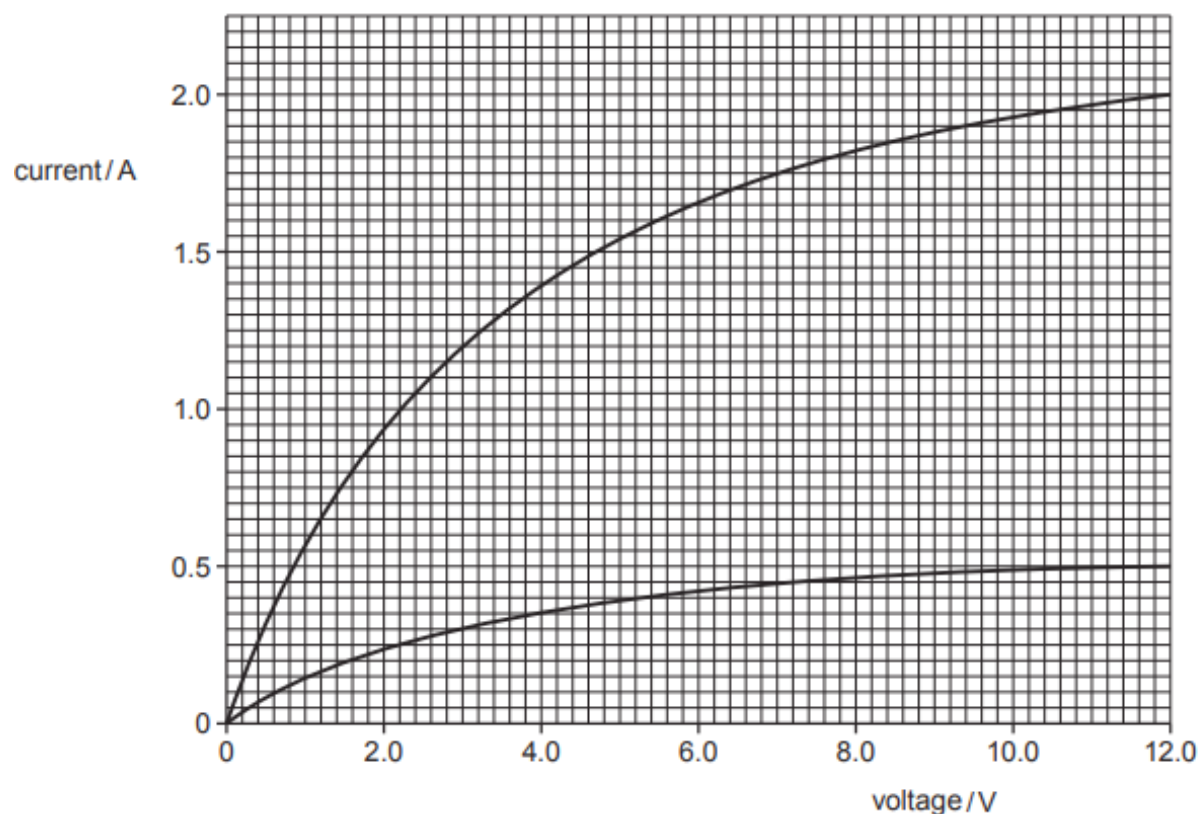


Fig. 8.2

- (i) Calculate the total resistance of the two filaments when they are connected in parallel to a voltage of 12 V.

resistance = [3]

- (ii) The two filaments are made from the same type of metal and have the same length, when uncoiled. They both operate at the same temperature.

Suggest why one filament has a resistance that is greater than that of the other filament.

.....

..... [1]

18. O/N/19/22/Q6

An electric circuit contains a $700\ \Omega$ resistor and a light-dependent resistor (LDR). Fig. 6.1 is the circuit diagram.

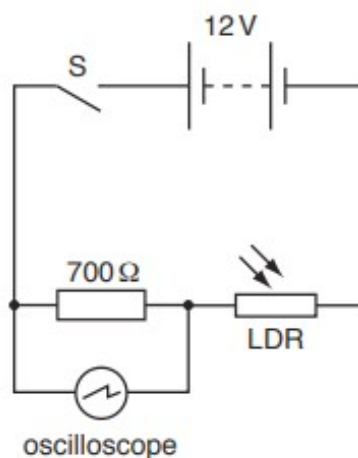


Fig. 6.1

The electromotive force (e.m.f.) of the battery is 12 V.

An oscilloscope is connected across the fixed resistor. Fig. 6.2 shows the oscilloscope, including the settings of the timebase and the Y-gain controls. Line Q shows the position of the trace on the oscilloscope when the switch S is open.

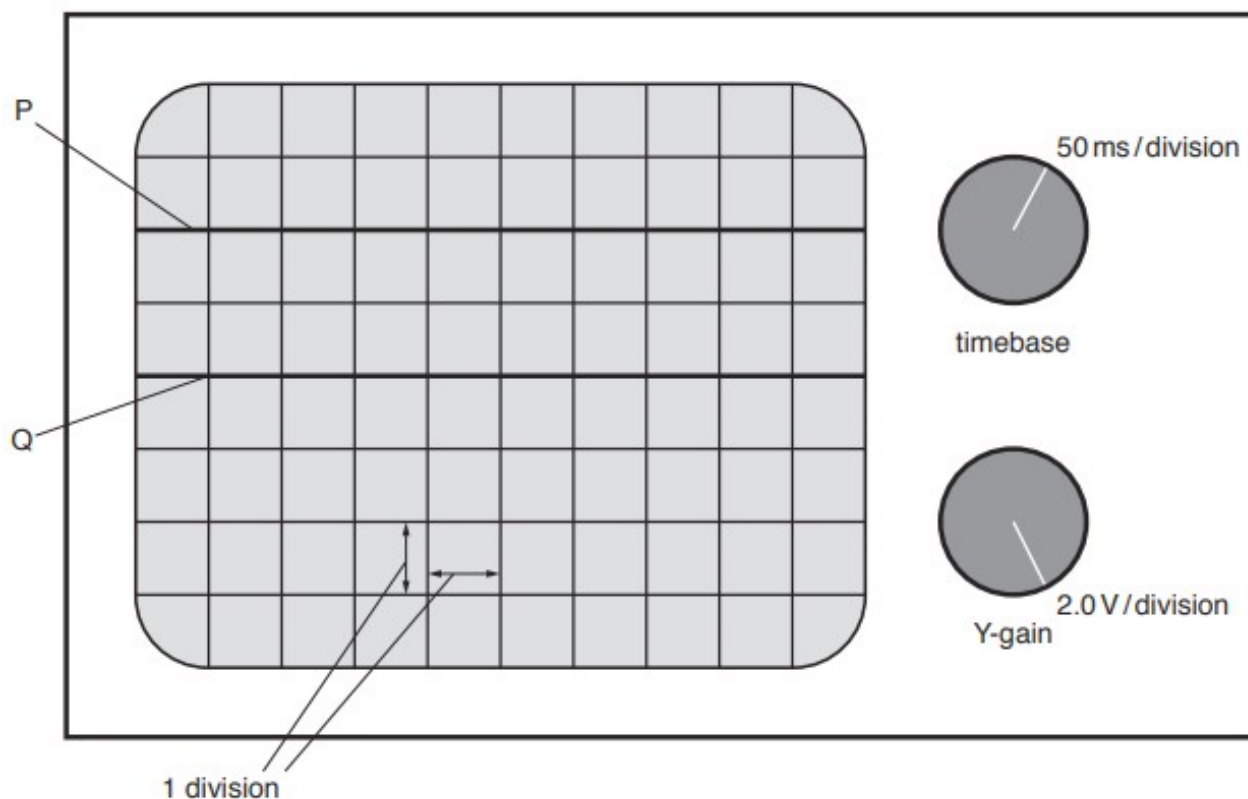


Fig. 6.2

The switch S is closed and the trace on the oscilloscope moves to the position shown by line P in Fig. 6.2.

- (a) (i) Determine the potential difference (p.d.) across the $700\,\Omega$ resistor.

p.d. = [1]

- (ii) Determine the resistance of the LDR.

resistance = [3]

- (b) The intensity of the light incident on the LDR gradually increases.

State and explain how the trace on the oscilloscope screen moves.

.....
.....
.....
..... [3]

[Total: 7]

19. O/N/19/21/Q6

A 60 W filament lamp is powered by the 240 V mains electricity supply.

The lamp is switched on.

(a) Calculate:

(i) the current in the lamp

current = [2]

(ii) the resistance of the lamp.

resistance = [2]

(b) The lamp is now unplugged from the mains supply and connected in series with two 1.5 V cells. Fig. 6.1 is the circuit diagram.

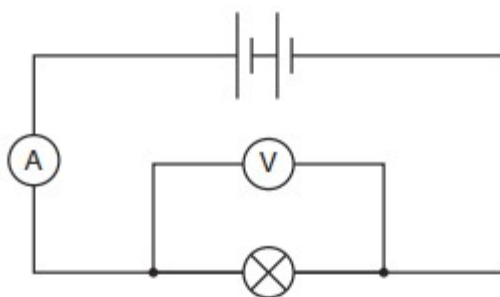


Fig. 6.1

(i) State the reading on the voltmeter.

reading = [1]

(ii) State and explain how the resistance of the lamp now differs from the value calculated in **(a)(ii)**.

.....

 [2]

(iii) When a filament lamp blows, it very often does so immediately after being switched on.

Suggest why this is the case.

.....
 [1]

20. M/J/19/22/Q6

Two lamps, P and Q, are connected to a battery of electromotive force (e.m.f.) 6.0V and an ammeter, as shown in Fig. 6.1.

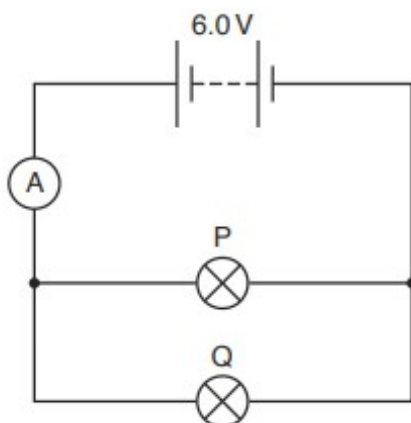


Fig. 6.1

Lamp P has a resistance of 15Ω . The ammeter reading is 0.65A.

(a) Calculate the current in lamp P.

current = [2]

(b) Calculate the resistance of lamp Q.

resistance = [2]

(c) The two lamps are now connected in series to the ammeter and the same battery.

(i) In the space below draw the circuit diagram.

[1]

(ii) Explain why the ammeter reading is less than 0.65A when the lamps are connected in series.

.....
 [1]

21. M/J/19/21/Q5

A student uses the circuit shown in Fig. 5.1 to investigate a resistor R.

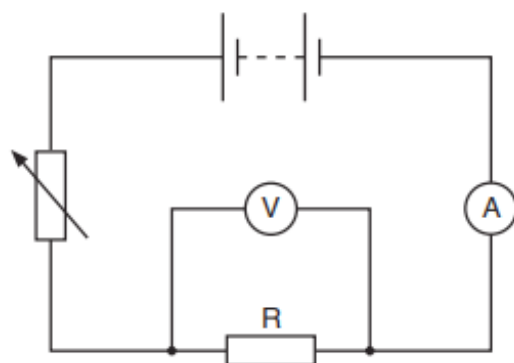


Fig. 5.1

- (a) Describe how the student uses the apparatus in Fig. 5.1 to obtain a range of ammeter and voltmeter readings.

.....
 [1]

- (b) The readings obtained are shown in Table 5.1.

Table 5.1

voltmeter reading / V	ammeter reading / mA
7.6	320
5.2	220
2.4	100

- (i) State Ohm's law.

.....

 [2]

- (ii) Using the results in Table 5.1, show that the resistor obeys Ohm's law.

.....

 [1]

(iii) The student has available an ammeter with five ranges marked:

- 0 – 10A
- 0 – 200mA
- 0 – 20mA
- 0 – 2mA
- 0 – 0.2mA

Suggest how the student makes best use of the different ranges during the investigation.

.....

.....

.....

..... [1]

[Total: 5]

22. O/N/18/22/Q10

A student sets up the circuit shown in Fig. 10.1 in a laboratory at room temperature.

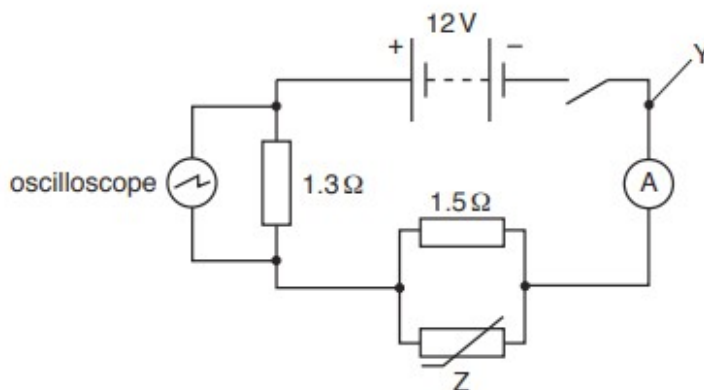


Fig. 10.1

The electromotive force (e.m.f.) of the battery is 12 V.

The switch is closed.

- (a) The connecting wires in the circuit are made from copper covered by insulating plastic.

State the name of the particles that flow in the copper and state the direction in which they are flowing at point Y in the circuit.

.....

 [2]

- (b) At room temperature, the resistance of component Z is $6.0\ \Omega$.

- (i) State the name of component Z.

..... [1]

- (ii) Calculate the resistance of the whole circuit.

resistance = [3]

- (iii) Calculate the current measured by the ammeter.

current = [2]

- (iv) The current in the ammeter is I_A , the current in the 1.5Ω resistor is I_R and the current in component Z is I_Z .

Write down an equation that relates I_A , I_R and I_Z .

..... [1]

- (c) Fig. 10.2 shows the screen of the oscilloscope.

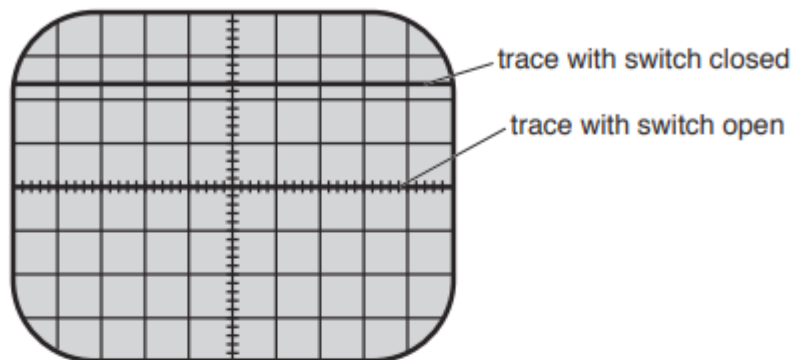


Fig. 10.2

Before the switch is closed, the trace is a horizontal line across the middle of the screen, as shown in Fig. 10.2. When the switch is closed, the trace remains horizontal and moves up the screen.

Component Z is heated.

State and explain what is observed on the oscilloscope screen as the temperature of Z increases.

.....

 [4]

- (d) The 12 V battery is made from cells of electromotive force (e.m.f.) 1.5 V which are all in series.

- (i) Calculate the number of 1.5 V cells that make up the battery.

number = [1]

- (ii) State the e.m.f. of a battery made by connecting all of these cells in parallel.

e.m.f. = [1]

23. O/N/18/21/Q7

A student sets up the circuit shown in Fig. 7.1.

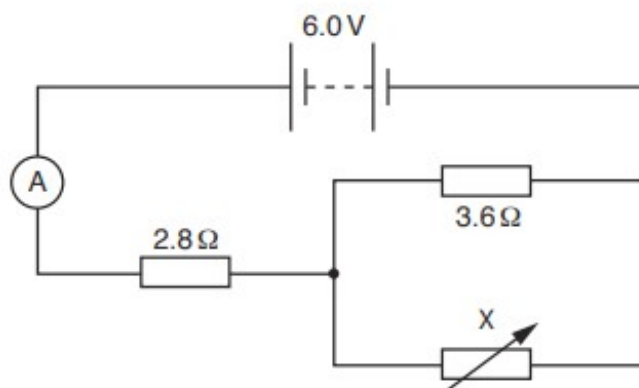


Fig. 7.1

The electromotive force (e.m.f.) of the battery is 6.0 V.

- (a)** The resistance of the variable resistor X is set to 1.8Ω .

Determine

- (i)** the total resistance of the circuit,

resistance = [3]

- (ii)** the current measured by the ammeter.

current = [2]

- (b) (i)** State what is meant by the potential difference (p.d.) across a component in a circuit.

.....
 [1]

- (ii)** Add to Fig. 7.1 the symbol for a component that is measuring the p.d. across the 2.8Ω resistor. [1]

(iii) The resistance of X is increased.

Explain why the p.d. across the $2.8\ \Omega$ resistor decreases.

.....

.....

..... [2]

24. M/J/18/22/Q10

A student connects a battery to two resistors. The circuit diagram is shown in Fig. 10.1.

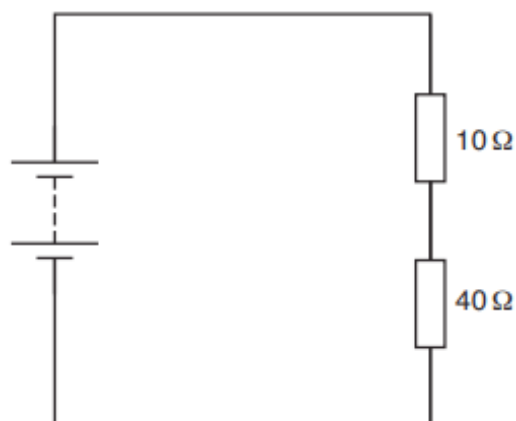


Fig. 10.1

The potential difference (p.d.) across the $40\ \Omega$ resistor is $9.6\ \text{V}$.

- (a) State what is meant by the *potential difference* across a resistor.

.....
[2]

- (b) (i) Calculate the current in the $40\ \Omega$ resistor.

current =[2]

- (ii) Calculate the electromotive force (e.m.f.) of the battery.

e.m.f. =[2]

- (c) The student has three different voltmeters to measure the p.d. across the $40\ \Omega$ resistor. These are labelled $0\text{--}2\ \text{V}$, $0\text{--}20\ \text{V}$ and $0\text{--}200\ \text{V}$. Each has a pointer that shows the p.d. on a scale.

State and explain which of the three voltmeters is best to measure this p.d.

.....

[2]

- (d) (i) Calculate the power P produced in the 10Ω resistor.

$$P = \dots\dots\dots [2]$$

- (ii) The student has available two 10Ω resistors, with power ratings of $\frac{1}{2}P$ and $2P$.

Suggest why a resistor with a power rating of $\frac{1}{2}P$ is not suitable for the circuit in Fig. 10.1.

.....
 [1]

- (e) The student adds a resistor R to the circuit, to make the circuit shown in Fig. 10.2.

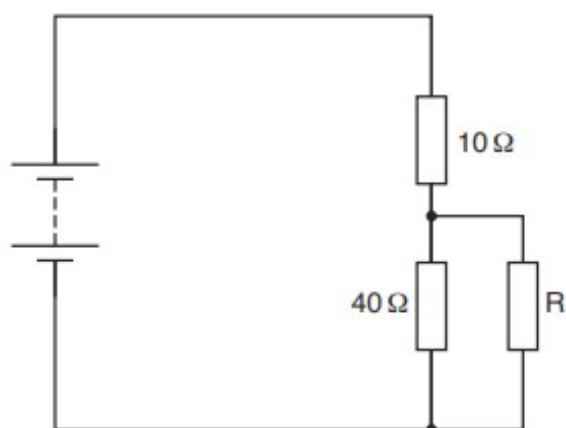


Fig. 10.2

Complete the table in Fig. 10.3 to show what happens as resistor R is connected.

You should state whether each quantity increases, decreases or stays the same and give a brief explanation of why any change occurs. Calculations are not required.

quantity	increases, decreases or stays the same as resistor R is added	brief explanation of why the change occurs
current in 10Ω resistor	increases	
p.d. across 10Ω resistor		
p.d. across 40Ω resistor		

[4]

Fig. 10.3

25. M/J/18/21/Q5

A student has three resistors; each resistor is labelled 3.9Ω , 3 W.

She checks the value of the resistance of **one** of the resistors.

- (a) In the space below, draw a circuit diagram of a circuit that she can use to find this resistance. Include a battery, a variable resistor, a voltmeter and an ammeter in the circuit.

[2]

- (b) When the circuit is turned on, the potential difference (p.d.) across the resistor is 4.2 V and the current in the resistor is 1.2 A.

- (i) Calculate the resistance of the resistor.

resistance =[2]

- (ii) Shortly after the circuit is turned on, the resistor overheats and gives off smoke.

Explain why it overheats. Include a calculation in your answer.

.....
.....
.....
.....
.....[2]

- (iii) The other two resistors are also labelled 3.9Ω , 3 W. The first resistor is now removed and the student connects these other two resistors in series to the p.d. of 4.2 V.

State and explain whether these resistors overheat.

.....
.....
.....
.....[1]

26. O/N/17/22/Q7

A 4.5 V battery is connected in a circuit with an ammeter, a light-dependent resistor (LDR) and a $1800\ \Omega$ fixed resistor. Fig. 7.1 is the circuit diagram.

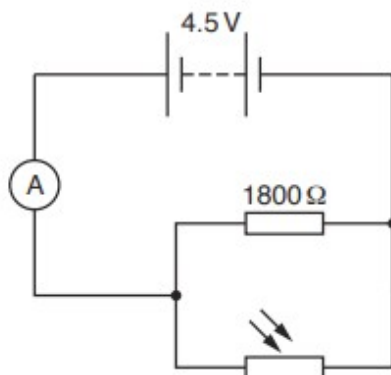


Fig. 7.1

- (a)** The resistance of the LDR is $9000\ \Omega$.

Calculate

- (i)** the total resistance of the circuit,

resistance =[2]

- (ii)** the reading on the ammeter.

reading =[2]

- (b)** A very bright lamp is switched on and the light intensity on the LDR increases.

State and explain what happens to

- (i)** the current in the LDR,

.....
[1]

- (ii)** the current in the $1800\ \Omega$ resistor.

.....
[1]

27. O/N/17/21/Q11(b)

(b) A circuit contains a 7.5 V d.c. power supply and three resistors.

Fig. 11.2 is the circuit diagram.

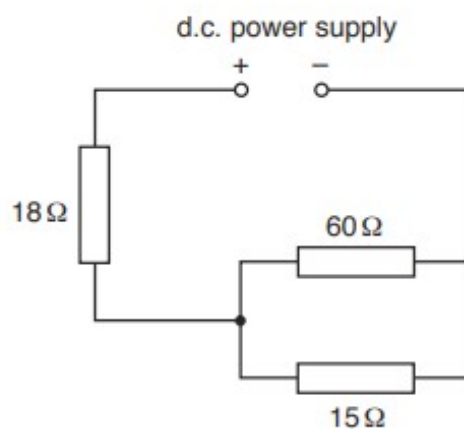


Fig. 11.2

(i) Determine the total resistance of the circuit.

resistance = [3]

(ii) Calculate the current in the power supply.

current = [2]

(iii) There is a different current in each of the three resistors in the circuit in Fig. 11.2.

Underline the resistor in which the current is

1. the largest,

15 Ω resistor

18 Ω resistor

60 Ω resistor

2. the smallest.

15 Ω resistor

18 Ω resistor

60 Ω resistor

[1]

- (iv) The 7.5 V d.c. power supply is made from five 1.5 V cells.

In the space below, draw a diagram that shows how the cells are arranged.

[1]

- (v) The $18\ \Omega$ resistor is made of metal.

State and explain what happens to the current in this resistor as its temperature increases.

.....

.....

..... [1]

28. M/J/17/22/Q8

Fig. 8.1 shows charges on a rod and on a cloth.

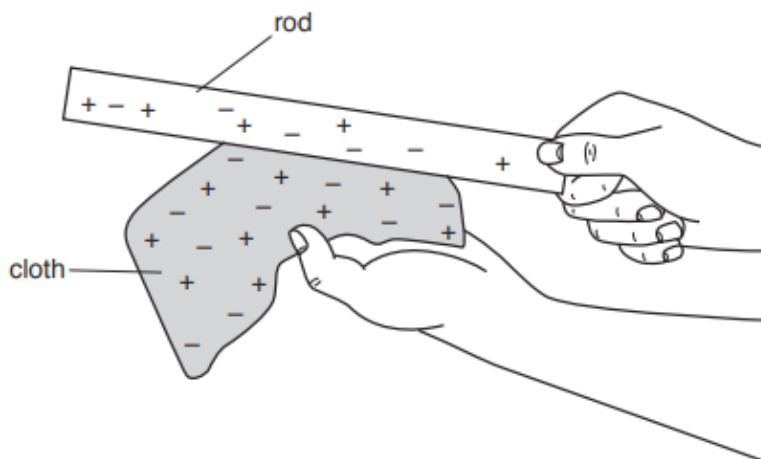


Fig. 8.1

- (a) State how Fig. 8.1 shows that the rod and the cloth are uncharged.

[1]

- (b)** The plastic rod becomes negatively charged when rubbed by the cloth.

- (i) Describe what happens to the charges on the rod and on the cloth during the charging process.

.....

.....

.....

.....

[2]

- (ii) Describe a simple experiment to show that the plastic rod is charged.

You may draw a diagram if you wish.

[2]

29. M/J/17/22/Q10

- (a) Measurements are made of the current in a metal wire and the potential difference (p.d.) across it. Fig. 10.1 shows a graph of the current against the p.d. for the wire.

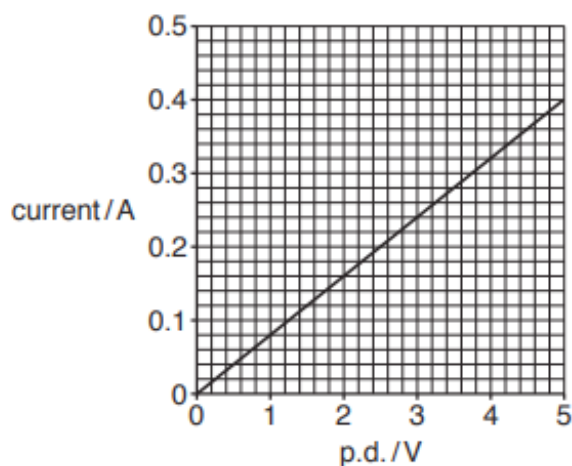


Fig. 10.1

- (i) State the relationship between the current and the p.d. for the metal wire.

.....
[1]

- (ii) State how the graph shows that the temperature of the wire does not change in the experiment.

.....
[1]

- (iii) Another wire of the same material has the same length as the original wire but has only half the cross-sectional area.

1. Compare the resistance of the new wire and the original wire.

.....
[1]

2. On Fig. 10.1, draw the graph for the new wire. [1]

- (b) Three resistors are connected to a d.c. power supply, as shown in Fig. 10.2. The electromotive force (e.m.f.) of the power supply is 6.0 V.

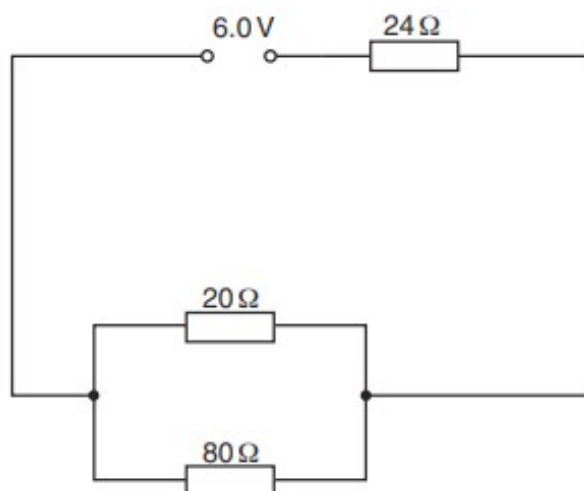


Fig. 10.2

- (i) Calculate

1. the total resistance of the circuit,

resistance =[2]

2. the current supplied by the power supply,

current =[2]

3. the p.d. across the 20 Ω resistor.

p.d. =[2]

- (ii) The d.c. power supply is a battery that contains a number of cells. Each cell has an electromotive force (e.m.f.) of 1.5 V.

1. State what is meant by the *electromotive force (e.m.f.)* of a cell.

.....
.....
.....[1]

2. In the space below, draw a diagram to show how four of these cells are connected to make a battery of e.m.f. 6.0 V. Mark the positive terminal of the battery.

[2]

3. In another arrangement to produce a battery of e.m.f. 6.0 V, eight cells of e.m.f. 1.5 V are connected. In this arrangement, some cells are connected in series and some in parallel.

In the space below, draw a diagram to show how the eight cells are connected to make the battery.

[1]

- (c) State one advantage of using a battery made from cells in parallel rather than a single cell.

.....
.....[1]

30. M/J/17/21/Q5

Fig. 5.1 shows an arrangement of three resistors.

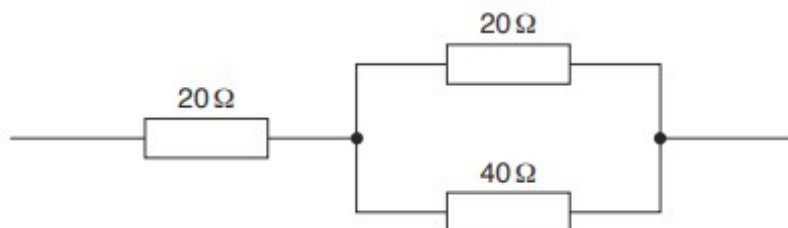


Fig. 5.1

- (a) Calculate the total resistance of this arrangement.

resistance =[3]

- (b) The arrangement is connected to a d.c. power supply.

Voltmeters are placed across the resistors, as shown in Fig. 5.2. There is a current in each resistor.

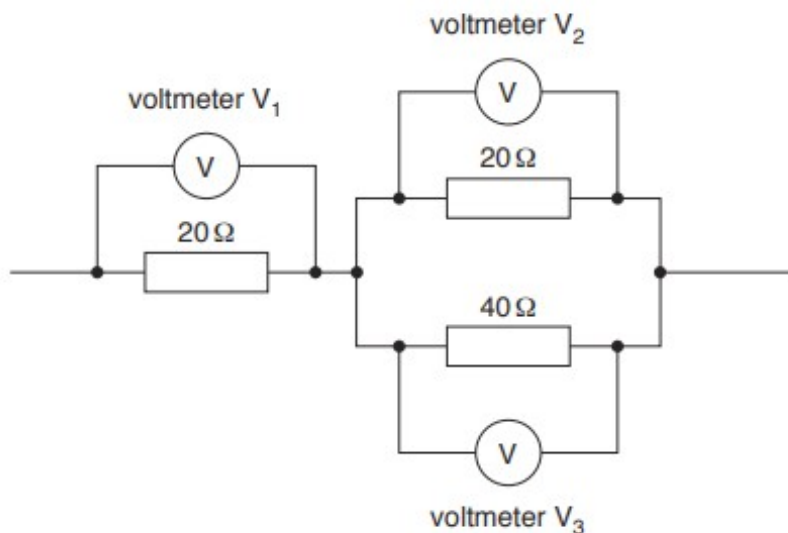


Fig. 5.2

Compare the readings on the three voltmeters. There is no need for any calculation.

.....

[2]

31. M/J/17/21/Q11

A student connects a thermistor in series with a battery and an ammeter. He uses the resistance of the thermistor to determine temperature.

- (a) In the space below, draw a circuit diagram showing the thermistor, the battery and the ammeter.

[3]

- (b) The student places the thermistor in a water bath at various temperatures and records the reading on the ammeter.

Fig. 11.1 shows how the current varies with temperature.

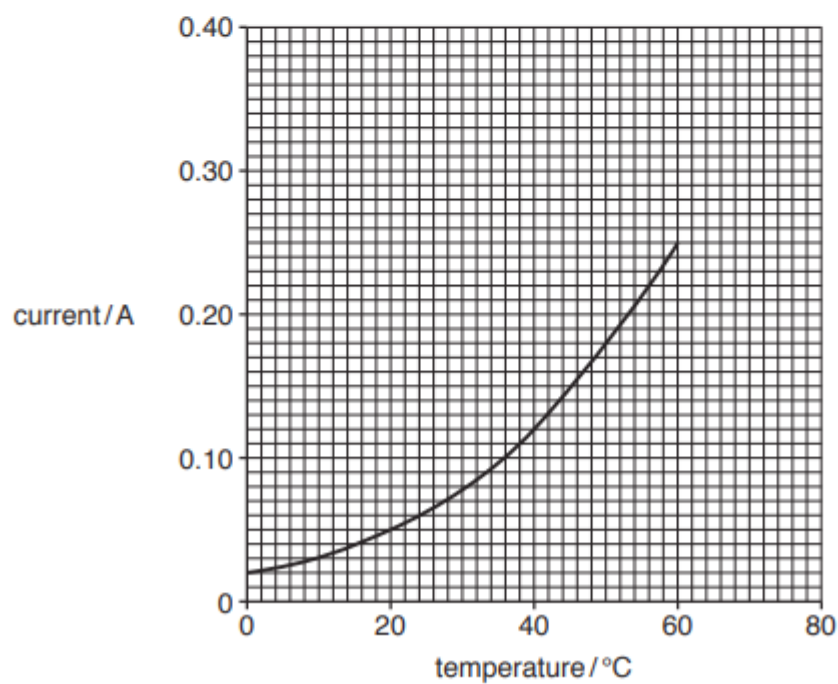


Fig. 11.1

The potential difference (p.d.) across the thermistor **remains constant**.

The resistance of the thermistor is $240\ \Omega$ at a temperature of 20°C .

(i) Using Fig. 11.1, calculate

1. the potential difference (p.d.) across the thermistor,

p.d. =[3]

2. the resistance of the thermistor at a temperature of 40°C .

resistance =[2]

- (ii) The resistance of the thermistor is $240\ \Omega$ at a temperature of 20°C and $48\ \Omega$ at a temperature of 60°C .

The resistance of the thermistor does not change linearly with temperature.

Describe how the resistances at 20°C , 40°C and 60°C show this is true.

.....
.....
.....
.....
.....[2]

32. M/J/16/22/Q9

Fig. 9.1 is a circuit diagram.

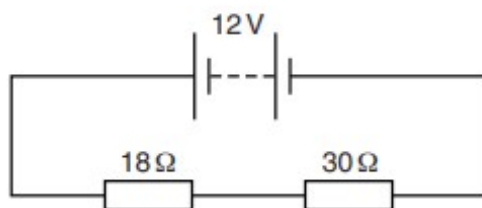


Fig. 9.1

The electromotive force (e.m.f.) of the battery is 12V.

(a) Explain what is meant by *electromotive force*.

.....
[2]

(b) An 18Ω resistor and a 30Ω resistor are connected in series with the battery.

(i) 1. State the relationship between the current I_B in the battery, the current I_1 in the 18Ω resistor and the current I_2 in the 30Ω resistor.

.....[1]

2. State an equation that relates the e.m.f. E of the battery, the potential difference V_1 across the 18Ω resistor and the potential difference V_2 across the 30Ω resistor.

.....[1]

(ii) Calculate the current in the battery.

current =[2]

(iii) Calculate the potential difference (p.d.) across the 18Ω resistor.

p.d. =[1]

- (iv) Calculate the power produced in the 18Ω resistor.

power =[2]

- (c) The resistors obey Ohm's law.

State Ohm's law and describe one limitation on this law.

.....
.....
.....[2]

- (d) The resistors are made from the same material, but have different cross-sectional areas and different lengths.

State the relationship between

- (i) the resistance R and the length l of a wire of constant cross-sectional area,

.....

- (ii) the resistance R and the cross-sectional area A of a wire of constant length.

.....
[2]

33. M/J/16/21/Q8

Fig. 8.1 shows a circuit that contains a battery, a fixed resistor of resistance $1000\ \Omega$ and a variable resistor R .

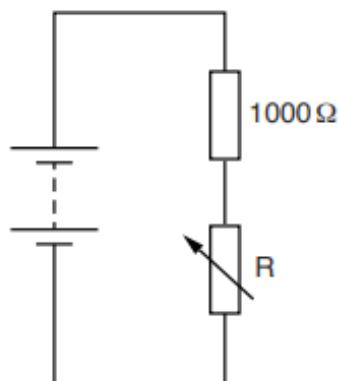


Fig. 8.1

- (a) (i) State the relationship between the current I_B in the battery, the current I_1 in the fixed resistor and the current I_2 in the variable resistor.

.....
[1]

- (ii) State an equation that relates the potential difference (p.d.) V_B across the battery, the p.d. V_1 across the fixed resistor and the p.d. V_2 across the variable resistor.

.....
[1]

- (b) The p.d. across the battery is 9.0V. The resistance of R is set at $500\ \Omega$.

Calculate the current in R .

current =[2]

34. O/N/15/21/Q8

A potential divider is made from a light-dependent resistor (LDR) and a $6.0\text{ k}\Omega$ fixed resistor. The potential divider is connected in series with a 12 V d.c. power supply and a voltmeter is connected across the $6.0\text{ k}\Omega$ resistor. Fig. 8.1 is the circuit diagram.

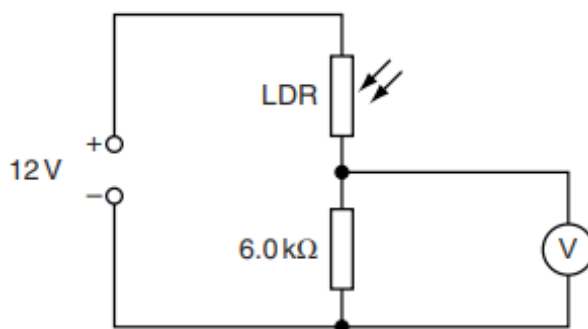


Fig. 8.1

A light shines on the LDR. The resistance of the LDR is $2.0\text{ k}\Omega$.

(a) Calculate

(i) the current in the LDR,

current = [2]

(ii) the reading on the voltmeter.

reading = [1]

(b) The brightness of the light on the LDR is gradually increased.

State and explain what happens to the reading on the voltmeter.

.....

 [2]

(c) Suggest a use for a potential divider made from an LDR and a fixed resistor.

.....
 [1]

35. M/J/15/21/Q6

Fig. 6.1 shows a circuit that contains a resistor connected to a power supply of 6.0V and a lamp L.

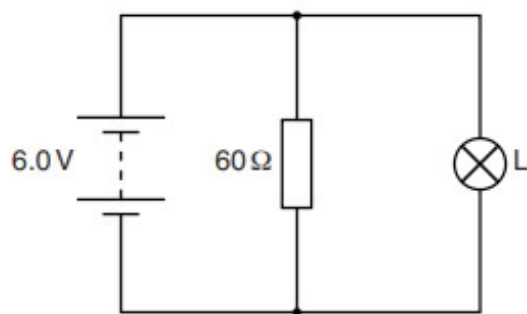


Fig. 6.1

The resistor has a resistance of 60Ω . The lamp is marked 6.0V, 0.90W.

(a) Calculate

(i) the current in the resistor,

current =[2]

(ii) the current in the power supply.

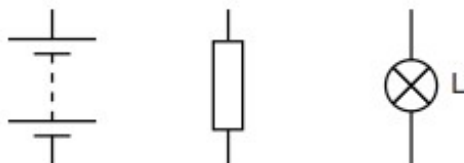
current =[2]

- (b) A second lamp is added to the circuit shown in Fig. 6.1.

The second lamp is in series with the 60Ω resistor, but is **not** in series with lamp L.

- (i) In the space below draw a circuit diagram of this new circuit.

The power supply, 60Ω resistor and lamp L have been drawn for you.



[1]

- (ii) The two lamps are identical.

Explain why the second lamp is dimmer than lamp L.

A calculation is not required.

.....
.....
.....[1]

36. O/N/14/22/Q7

The circuit of Fig. 7.1 includes an ammeter, a 6.0Ω resistor, a length of metal resistance wire and three 1.5V cells connected in parallel.

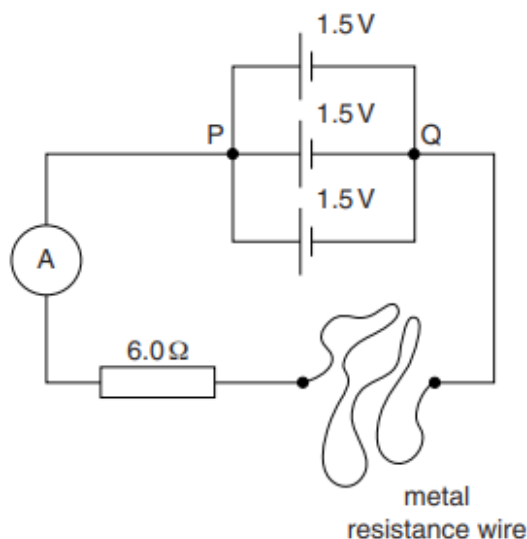


Fig. 7.1

(a) State

(i) one advantage of using cells in parallel rather than using a single cell,

.....
[1]

(ii) the potential difference (p.d.) between points P and Q in the circuit of Fig. 7.1.

p.d. =[1]

(b) (i) The ammeter in Fig. 7.1 reads 0.075A .

Calculate the resistance of the resistance wire.

resistance =[3]

(ii) The temperature of the metal resistance wire increases.

State and explain the effect of this temperature increase on the ammeter reading.

.....

[2]

37. O/N/14/21/Q11

A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 11.1.

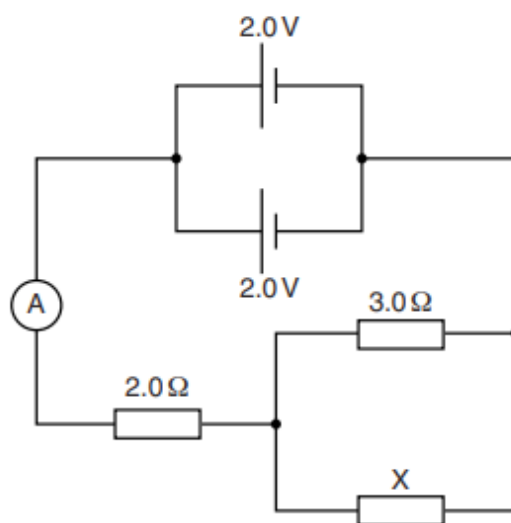


Fig. 11.1

- (a) (i) Explain what is meant by *electromotive force*.

.....

[2]

- (ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....
[1]

- (b) Resistor X and the 3.0Ω resistor have a combined resistance that is equal to 2.0Ω.

Calculate

- (i) the total resistance of the circuit,

total resistance =[1]

- (ii) the resistance of X.

resistance of X =[2]

- (c) (i) Determine the reading of the ammeter.

reading =[2]

- (ii) Suggest a suitable range for the ammeter.

.....[1]

- (d) The current in the 2.0Ω resistor is I_2 .
The current in the 3.0Ω resistor is I_3 .
The current in X is I_X .

State the equation that relates I_2 , I_3 and I_X .

.....[1]

- (e) State the potential difference (p.d.) across

- (i) the 2.0Ω resistor,

p.d. =[1]

- (ii) the 3.0Ω resistor.

p.d. =[1]

- (f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 11.2.

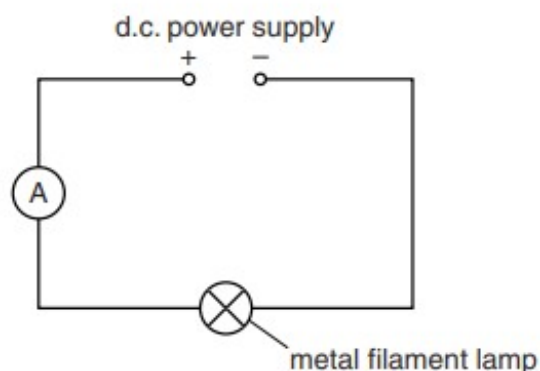


Fig. 11.2

The d.c. power supply is set to 12V and the ammeter reading is 1.5 A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

- (i) State and explain what happens to the resistance of the metal filament of the lamp.

.....
.....
.....[2]

- (ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

38. M/J/14/22/Q11

A student sets up the circuit shown in Fig. 11.1.

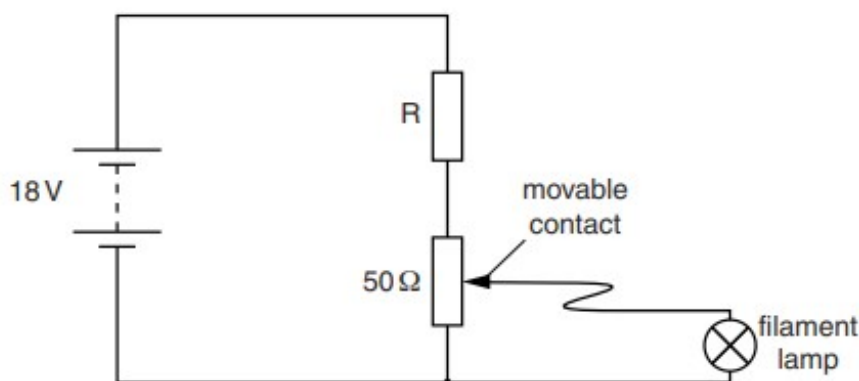


Fig. 11.1

R is a fixed resistor in the circuit. The filament lamp is marked 12V, 0.25A.

The circuit is used to produce a current/voltage graph for the filament lamp. The ammeter and voltmeter needed are not shown.

To obtain different readings, the student changes the position of the movable contact.

(a) On Fig. 11.1, draw the symbols for an ammeter and a voltmeter in the correct positions. [3]

(b) Explain why it is sensible to include the resistor R in this circuit.

.....

.....

..... [2]

(c) (i) On Fig. 11.2, sketch a current/voltage graph for the lamp.

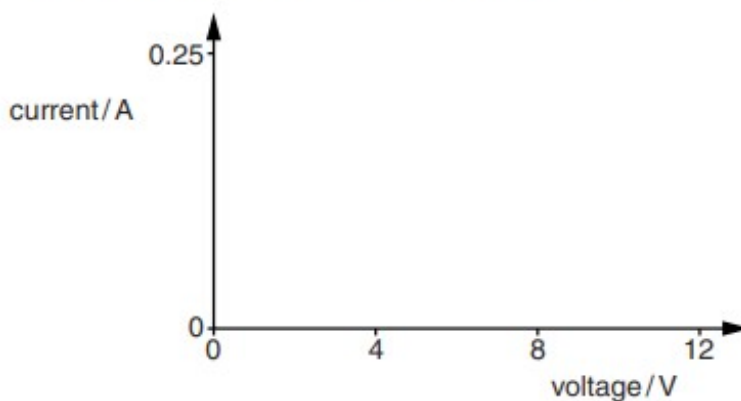


Fig. 11.2

[3]

- (ii) State and explain how a current/voltage graph for a fixed resistor is different from the graph for a filament lamp.

.....

 [2]

- (d) Fig. 11.3 shows the position of the movable contact when the voltage across the lamp is 12 V and the current in the lamp is 0.25 A.

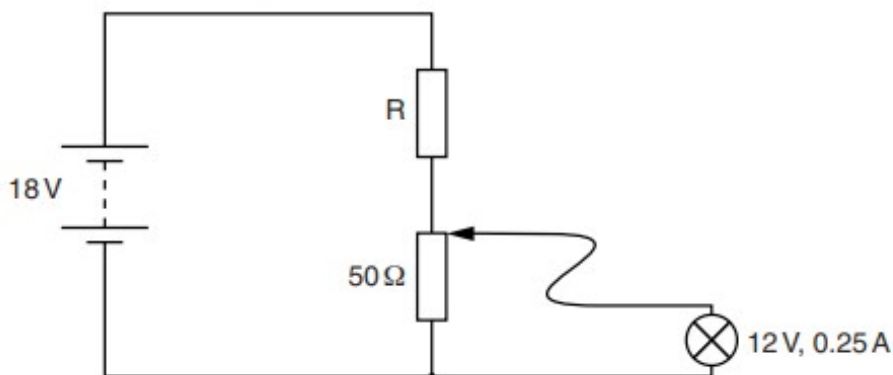


Fig. 11.3

Determine

- (i) the current in the $50\ \Omega$ resistor,

current = [2]

- (ii) the current in R,

current = [1]

- (iii) the potential difference (p.d.) across R,

p.d. = [1]

- (iv) the resistance of R.

resistance = [1]

39. M/J/14/21/Q7

(a) Fig. 7.1 shows a variable potential divider (potentiometer) connected to a 6.0 V battery.

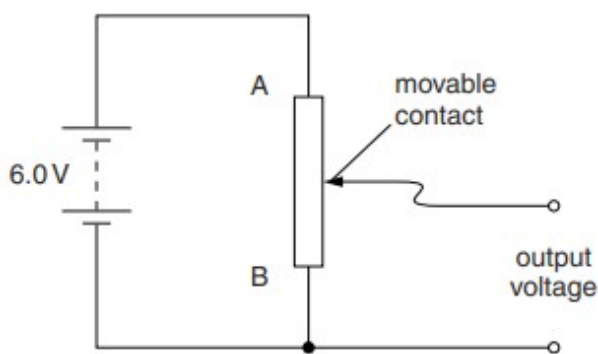


Fig. 7.1

The movable contact can slide between A and B. As it moves, the output voltage changes.

Complete the table below to show the output voltage when the contact is at A and at B.

position of movable contact	A	B
output voltage		

[2]

(b) Fig. 7.2 shows a thermistor in a potential divider circuit that is used to monitor temperature. The fixed resistor has a resistance of $2000\ \Omega$.

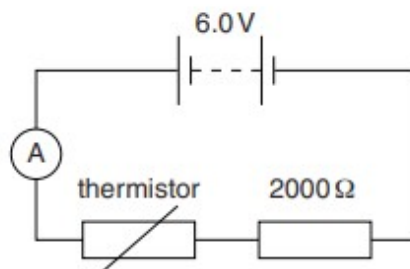


Fig. 7.2

(i) Explain why the voltage across the thermistor decreases as its temperature rises.

.....
 [2]

(ii) At one temperature, the thermistor has a resistance of $1000\ \Omega$. Calculate the current in the thermistor.

current = [2]

40. M/J/14/21/Q11

- (a) A torch contains a filament lamp connected to a battery. The battery contains four 1.5V cells connected in series.

Fig. 11.1 shows the current/potential difference (voltage) graph for the filament lamp.

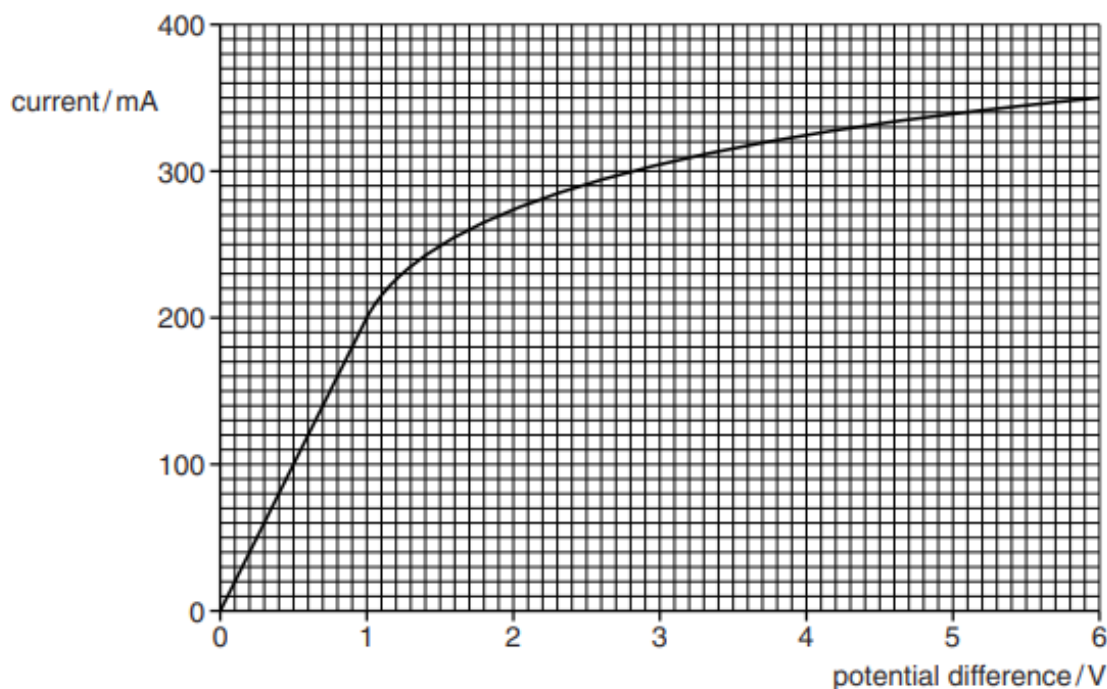


Fig. 11.1

- (i) State what is meant by the *potential difference* (p.d.) across the lamp.

.....
 [2]

- (ii) Fig. 11.1 shows that the current in the lamp is not proportional to the p.d.

1. State how Fig. 11.1 shows this.

.....
 [1]

2. Suggest why the current is not proportional to the p.d.

.....
 [1]

(iii) The lamp is at normal brightness and there is a p.d. of 6.0V across the lamp.

1. Use Fig. 11.1 to find the current in the lamp.

current = [1]

2. Calculate the charge that passes through the lamp in 2.0 hours.

charge = [2]

3. The battery in the torch is able to keep the lamp lit for 2.0 hours.
Estimate the initial energy stored in the battery.

energy = [2]

4. Explain why your value is only an estimate.

.....
..... [1]

- (b) Suggest and explain an advantage of using two cells in parallel rather than a single cell.

.....
.....
..... [2]

- (c) Another torch contains a light-emitting diode (LED) instead of a filament lamp.
The battery in the torch uses four 1.5V cells, some in series and some in parallel.
The p.d. across the LED when in operation is 3.0V.

In the space below, draw a circuit diagram showing how the four cells, a switch and the LED are connected.

[3]

41. O/N/13/22/Q11

A metal filament lamp is connected to a power supply. The electromotive force (e.m.f.) produced by the supply can be varied. Fig. 11.1 is the circuit diagram.

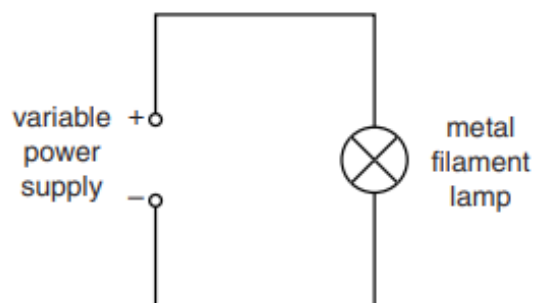


Fig. 11.1

(a) State what is meant by *electromotive force (e.m.f.)*.

.....

.....

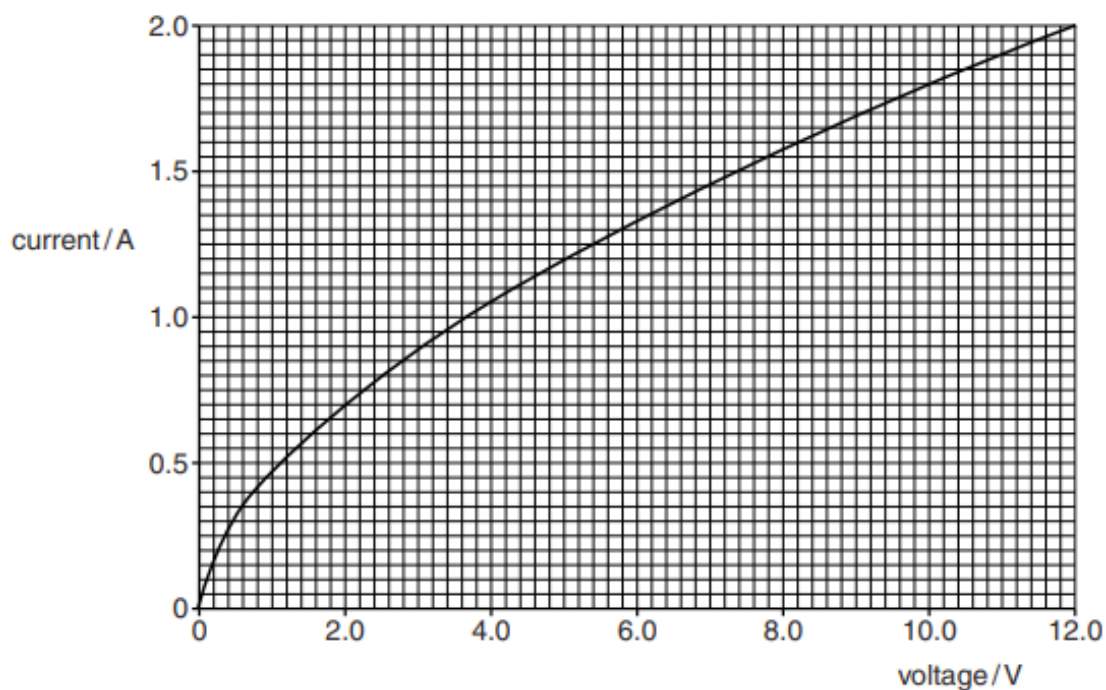
.....[2]

(b) Add appropriate circuit symbols to Fig. 11.1 to show the position of

(i) an ammeter that measures the current in the circuit, [1]

(ii) a voltmeter that measures the voltage across the lamp. [1]

(c) A student adds meters to the circuit and makes measurements of the current and voltage. He then plots the current/voltage graph shown in Fig. 11.2.



- (i) Using values from Fig. 11.2, calculate the resistance of the lamp when the current is 0.70 A. Give your answer to an appropriate number of significant figures.

resistance = [3]

- (ii) State what, if anything, happens to the resistance of the lamp as the voltage increases.

.....

..... [1]

- (d) In normal use, the lamp is connected to a 12 V supply.

Use Fig. 11.2 to determine the power of the lamp when it is used in this way.

power = [2]

42. O/N/13/21/Q11

An ammeter, a variable resistor and a 4.0Ω resistor are connected in series with a 6.0V battery. A cathode-ray oscilloscope (c.r.o.) is connected in parallel with the 4.0Ω resistor through a switch. Fig. 11.1 is the circuit diagram.

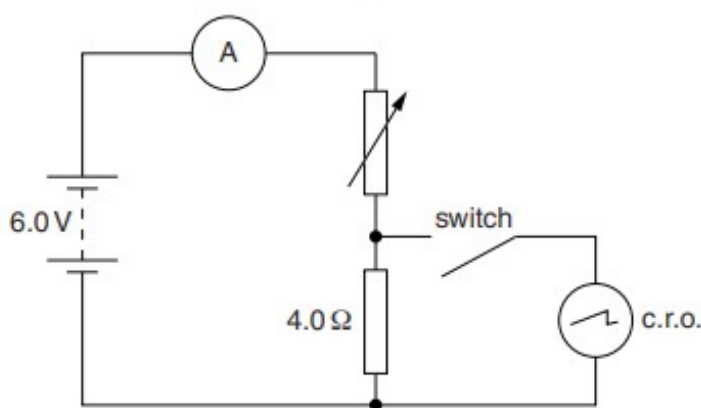


Fig. 11.1

The switch is open.

- (a)** The variable resistor is adjusted so that it has a resistance of 8.0Ω .

Calculate

- (i)** the current measured by the ammeter,

current =[2]

- (ii)** the potential difference (p.d.) across the 4.0Ω resistor.

p.d. =[1]

- (b)** The resistance of the variable resistor is adjusted until the p.d. across the 4.0Ω resistor is 5.0V .

State the effect that this adjustment has on

- (i)** the current measured by the ammeter,

.....[1]

- (ii)** the resistance of the variable resistor.

.....[1]

43. M/J/13/22/Q6

Fig. 6.1 shows the current-voltage graph for a filament lamp.

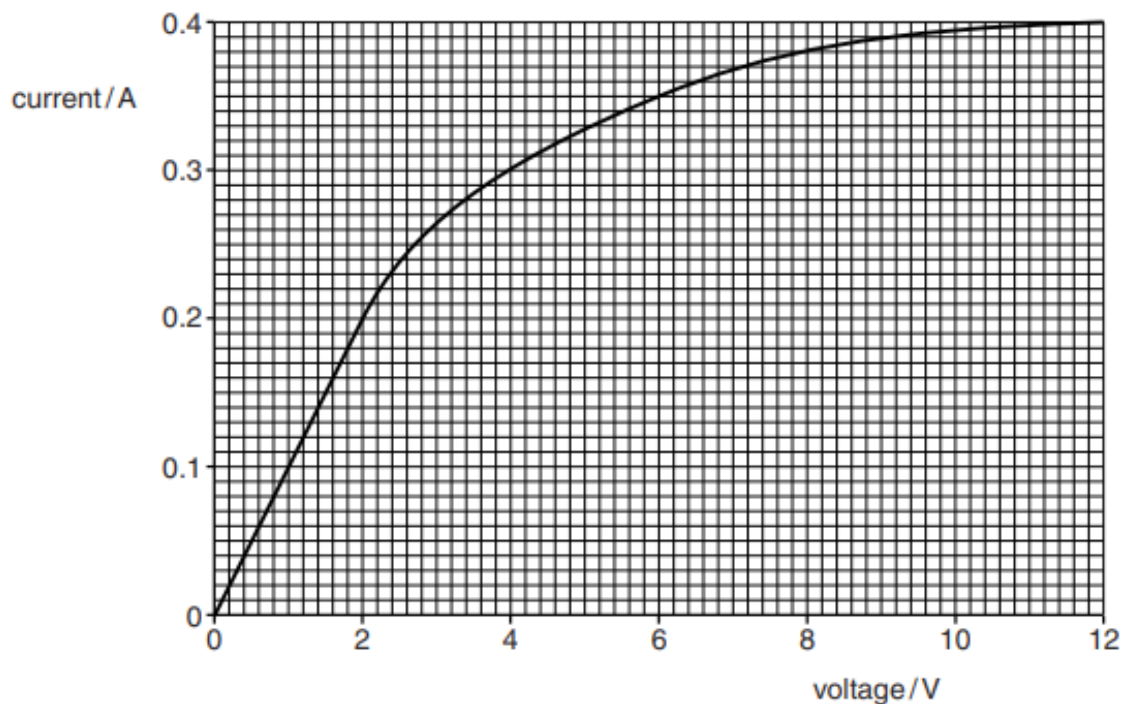


Fig. 6.1

- (a) (i) State the range of voltages where the resistance increases.

..... [1]

- (ii) State why the resistance of the lamp increases as the voltage increases.

.....
..... [1]

- (b) The filament lamp is connected in the circuit shown in Fig. 6.2.

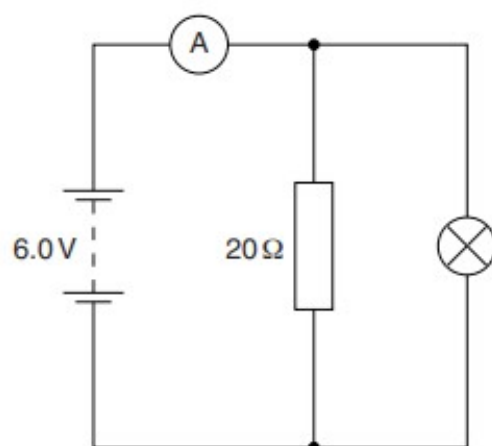


Fig. 6.2

There are currents in the lamp, the 20Ω resistor and the ammeter.

- (i) State what is meant by an *electric current*.

.....
..... [1]

- (ii) Use Fig. 6.1 to determine the current in the filament lamp.

current = [1]

- (iii) Determine the current in the ammeter.

current = [3]

44. M/J/13/21/Q6

Fig. 6.1 is a graph of current against potential difference (p.d.) for a length of metal wire.

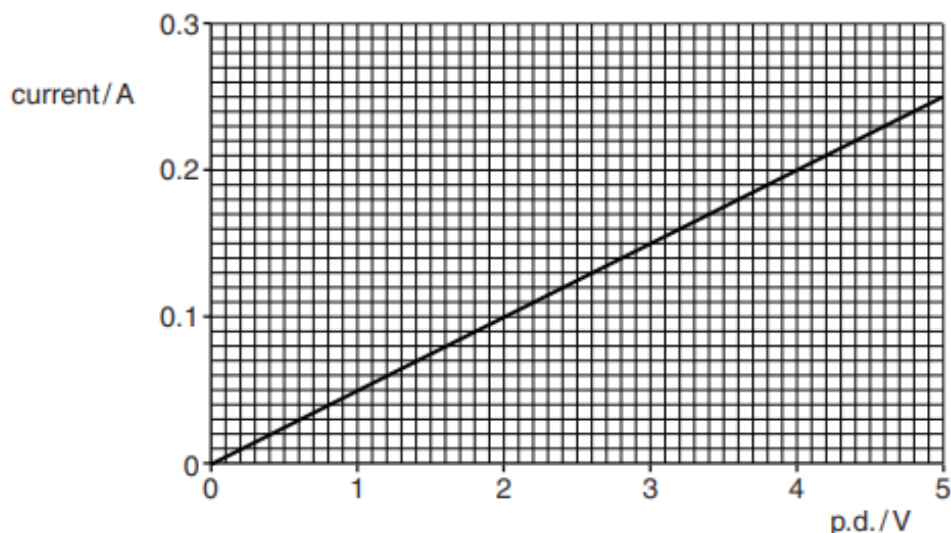


Fig. 6.1

- (a) The metal wire obeys Ohm's law. State Ohm's law in words.

.....

 [2]

- (b) Calculate the resistance of the metal wire.

resistance = [2]

- (c) A new wire is made from the same metal as the original wire. The two wires have the same length. The cross-sectional area of the new wire is half that of the original wire.

- (i) Calculate the resistance of the new wire.

resistance = [1]

- (ii) On Fig. 6.1, draw a line to show how current varies with p.d. for the new wire. [2]

45. M/J/13/21/Q7

A student connects a thermistor in series with a power supply and a fixed resistor.

(a) In the space below, draw the circuit diagram of the apparatus.

[2]

(b) Values for potential difference (p.d.) and current are given in Fig. 7.1.

p.d. across the thermistor/V	p.d. across the fixed resistor/V	e.m.f. of power supply/V	current in the thermistor/A	current in the fixed resistor/A
8.0	4.0		0.018	

Fig. 7.1

The maximum power rating of the thermistor is 150mW.

(i) In the blank spaces in Fig. 7.1, insert the values of the electromotive force (e.m.f.) of the power supply and the current in the fixed resistor. [2]

(ii) Calculate the power produced in the thermistor.

power = [2]

(iii) The e.m.f. of the power supply is doubled. Describe what happens to the thermistor.

.....
 [1]

46. O/N/12/21/Q8

A $600\ \Omega$ resistor and a thermistor are connected in series with an ammeter and a 20V d.c. (direct current) power supply. A voltmeter is in parallel with the resistor.

Fig. 8.1 is the circuit diagram.

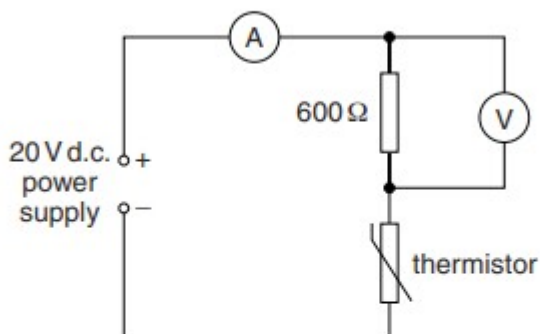


Fig. 8.1

The ammeter reads 0.025 A.

(a) Calculate

(i) the reading on the voltmeter,

reading = [2]

(ii) the resistance of the thermistor.

resistance = [2]

(b) The temperature of the thermistor increases.

State what, if anything, happens to

(i) the resistance of the thermistor,

..... [1]

(ii) the ammeter reading and to the voltmeter reading.

ammeter reading:

voltmeter reading:

[1]

47. M/J/12/22/Q7

Fig. 7.1 shows two charged metal spheres.

One sphere has a positive charge and the other sphere has a negative charge.



Fig. 7.1

- (a) On Fig. 7.1, draw the electric field between the two spheres. [2]
- (b) The negative charge on the sphere is removed and a wire is used to connect the positive sphere to earth. The charge on the positive sphere decreases from $4.8 \times 10^{-9} \text{ C}$ to zero in a time of $2.0 \times 10^{-6} \text{ s}$.

Calculate the average current in the wire.

current =[2]

48. M/J/12/22/Q8

A student investigates how the resistance of a wire depends upon its length. The student uses an ammeter, a voltmeter, a battery, a fixed resistor and the wire under test, all connected in an electrical circuit.

- (a) (i)** In the space below, draw a circuit diagram of the apparatus. Label the wire under test with the letter W.

[3]

- (ii)** Describe how the student obtains one complete set of results.

.....
.....
.....[2]

- (iii)** During the experiment, the student keeps the temperature of the wire constant.

- 1.** Suggest why it is sensible to keep the temperature of the wire constant.

.....
.....[1]

- 2.** Suggest how the student keeps the temperature of the wire constant.

.....
.....[1]

- (b) Fig. 8.1 shows part of a circuit containing three resistors X, Y and Z.

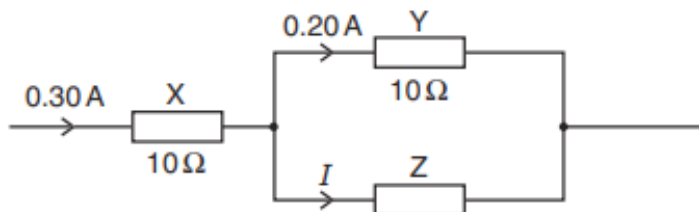


Fig. 8.1

The resistance of X and the resistance of Y are each $10\ \Omega$. The current through X is 0.30 A . The current through Y is 0.20 A .

- (i) Calculate the potential difference (p.d.) across Y.

p.d. =[2]

- (ii) Calculate the current I in resistor Z.

I =[1]

- (iii) Explain in words, rather than by calculation, how Fig. 8.1 shows that the resistance of Z is larger than the resistance of Y. Use ideas about p.d. and current in your answer.

.....
.....
.....[2]

- (iv) Calculate the total resistance of the resistors X, Y and Z in this circuit.

resistance =[3]

49. M/J/12/21/Q7

Fig. 7.1 is the circuit diagram of a circuit set up by a student.

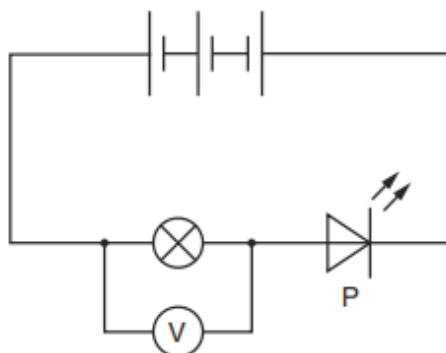


Fig. 7.1

The student uses three cells, each of electromotive force (e.m.f.) 1.5V. The cells are connected in series, as shown.

- (a) State the name of component P.

.....[1]

- (b) State what is meant by the *electromotive force* of a cell.

.....
[1]

- (c) The student expects the reading on the voltmeter to be 4.5V. The actual reading is 0.2V.

Suggest **two** reasons why the voltmeter reading is much less than 4.5V.

.....

[2]

- (d) (i) In the space below, draw three cells connected in parallel.

[1]

- (ii) State one advantage of connecting cells in parallel.

.....
[1]

50. O/N/11/22/Q6

An electrical engineer measures the potential difference across a length of metal wire and the current in the wire. He does this for different values of the current.

- (a) (i) Draw a labelled circuit diagram of a circuit that enables the engineer to do this.

[3]

- (ii) Describe how the circuit is used.

.....

.....

.....

.....[2]

- (iii) State the equation that defines resistance.

.....[1]

- (b) The engineer performs the experiment and notices that the potential difference across the wire is directly proportional to the current in it. He calculates the resistance of the wire and plots a graph of the resistance against the current.

On Fig. 6.1, sketch the shape of this graph.



Fig. 6.1

[1]

51. M/J/11/22/Q5

Fig. 5.1 shows part of a low-voltage lighting circuit.

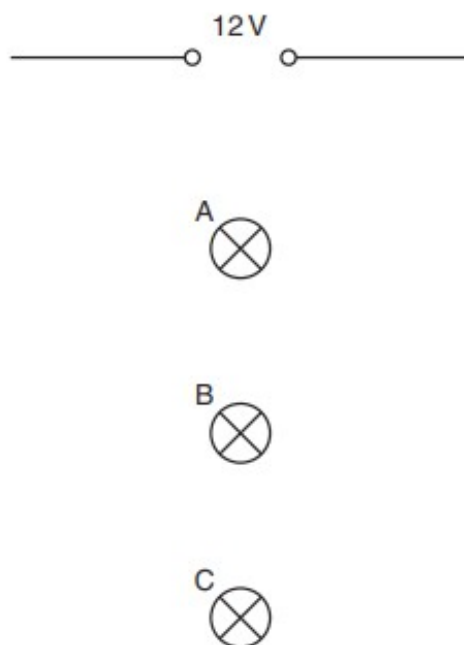


Fig. 5.1

The power supply voltage is 12V.

(a) On Fig. 5.1, complete the circuit, adding components as necessary, so that:

- the total current in the circuit can be measured,
- lamp A is on all the time,
- lamps B and C are in series with each other and are switched on or off together.

[3]

(b) All the lamps are on. The potential difference (p.d.) across lamp B is 8.0V and the current in lamp B is 50 mA.

(i) Calculate the resistance of lamp B.

resistance = [2]

(ii) State the current in lamp C.

current = [1]

52. M/J/10/22/Q11

- (a)** Describe an experiment, including an ammeter and a voltmeter, to show how the resistance of a wire varies with temperature. Draw a circuit diagram.

.....

.....

.....

.....

.....

.....[4]

- (b)** The filament in a lamp is made from metal wire.

- (i)** Describe the variation with temperature of the resistance of the filament lamp.

.....

.....

.....[2]

- (ii)** On Fig. 11.1, sketch a graph of the variation of the current in the filament against the potential difference (p.d.) across it.



Fig. 11.1

[2]

(c) Fig. 11.2 is a circuit used to monitor changes in room temperature.

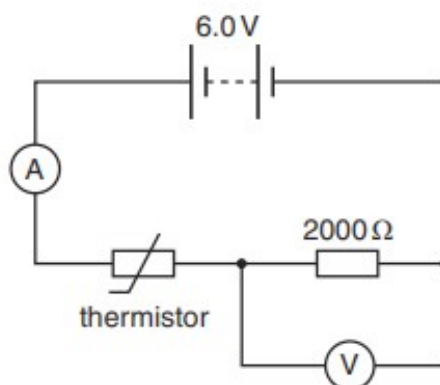


Fig. 11.2

A thermistor is connected in series with a 6.0V battery and a 2000Ω resistor.

(i) The temperature of the room increases. State and explain what happens to

1. the reading on the ammeter,

.....
.....[2]

2. the reading on the voltmeter.

.....
.....[2]

(ii) At a certain temperature, the reading on the voltmeter is 3.8V. Calculate the resistance of the thermistor at this temperature.

resistance =[3]

53. M/J/10/21/Q6

Fig. 6.1 shows a circuit used in an experiment to measure the current in a lamp for different values of the potential difference (p.d.) across the lamp.

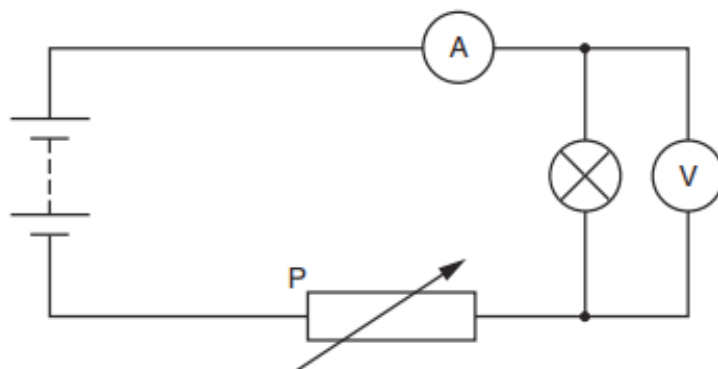


Fig. 6.1

(a) State the name of component P.

.....[1]

(b) On the axes below, sketch a graph of current against p.d. for a filament lamp.



[2]

(c) (i) State how the resistance of the lamp changes as the p.d. increases.

.....[1]

(ii) Explain how the graph in (b) shows the change you have described in (c)(i).

.....

[1]

54. O/N/08/Q7

A motorcycle battery consists of six 2.0V cells in series. The battery supplies energy to the headlight.

- (a) State the total electromotive force (e.m.f.) of the battery.

..... [1]

- (b) The motorcycle headlight contains two identical filament lamps F and G. Filament lamp F is always lit but filament lamp G is turned on and off by switch D. Fig. 7.1 is the circuit diagram.

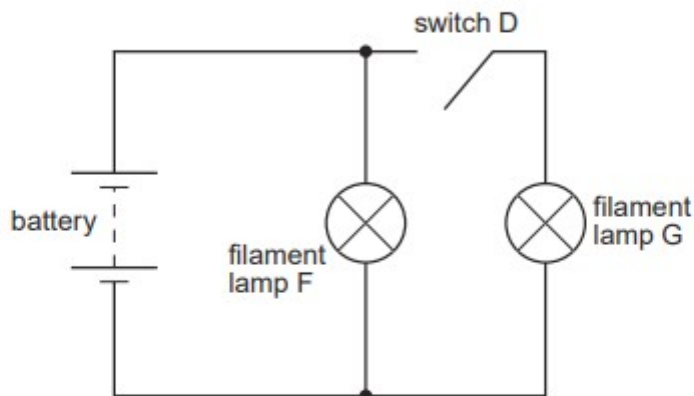


Fig. 7.1

When switch D is open, the battery supplies a current of 4.6 A.

Complete the table of Fig. 7.2.

	current supplied by battery / A	current in filament lamp F / A	current in filament lamp G / A
switch D open	4.6		
switch D closed			

Fig. 7.2

[3]

- (c) Calculate the energy supplied by the battery as an electric charge of 200C moves through the circuit.

energy supplied = [2]

55. M/J/08/Q6

A filament lamp is labelled 240V, 0.20 A.

- (a) Calculate the resistance of the lamp at normal brightness.

resistance = [2]

- (b) The lamp is connected to a 240V supply and switched on at time $t = 0$.

Fig. 6.1 shows the variation with t of the resistance of the lamp.

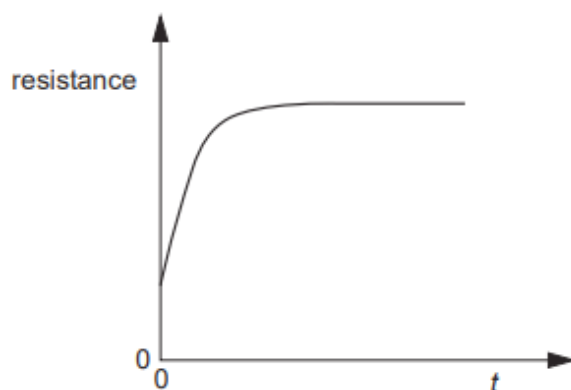


Fig. 6.1

Describe how the current in the lamp varies with t .

.....

 [2]

- (c) A second filament lamp has higher resistance.

State **two** differences between the two filaments that may cause the change in resistance.

- 1** The filament in the second lamp is

2 The filament in the second lamp is
 [2]

56. M/J/07/Q7

Fig. 7.1 shows an electrical circuit containing a 12V power supply and a number of resistors.

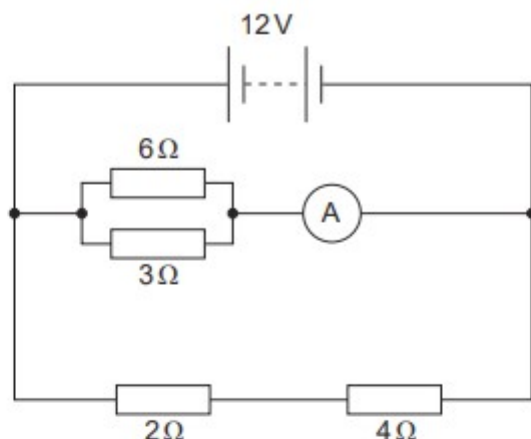


Fig. 7.1

(a) Calculate the combined resistance of

(i) the 2Ω and 4Ω resistors in series,

resistance = [1]

(ii) the 3Ω and 6Ω resistors in parallel.

resistance = [2]

(b) Calculate the reading of the ammeter in Fig. 7.1.

ammeter reading = [2]

(c) Determine the potential difference across the 4Ω resistor.

p.d. = [2]

57. O/N/06/Q8

Fig. 8.1 shows an electrical circuit using two resistors.

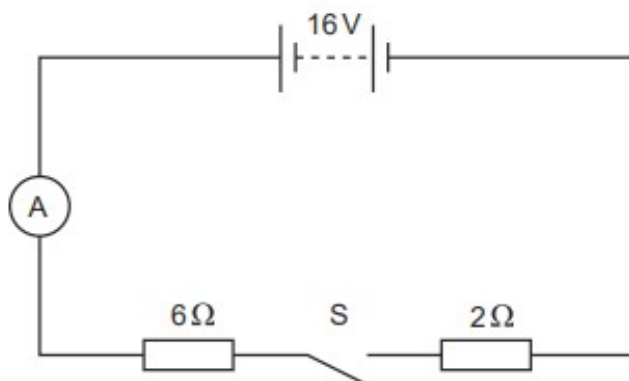


Fig. 8.1

- (a)** The switch S is open and the ammeter reading is zero.

State the value of the potential difference across the 6Ω resistor.

potential difference = [1]

- (b)** Switch S is now closed.

- (i)** State the value of the total resistance of the circuit.

resistance =[1]

- (ii)** Calculate the current in the ammeter. State clearly the formula that you use.

current =[2]

- (iii)** Calculate the potential difference across the 6Ω resistor.

p.d. =[2]

58. M/J/05/Q7

Fig. 7.1 shows an electrical circuit.

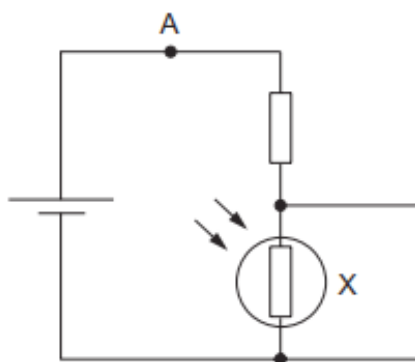


Fig. 7.1

- (a) On Fig. 7.1, draw an arrow at A to show the direction of flow of the electrons in the wire. [1]

- (b) What is the name of component X?

..... [1]

- (c) State and explain how the potential difference across X varies as the light shining on it becomes brighter.

.....

.....

.....

.....

..... [2]

59. M/J/05/Q8

Three cells are connected in series making a battery, as shown in Fig. 8.1. The e.m.f. of each cell is 1.5 V. A resistance of $15\ \Omega$ is connected to the battery.

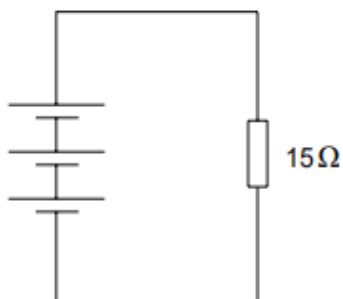


Fig. 8.1

- (a) What is the total e.m.f. of the battery?

..... [1]

- (b) Calculate the current in the circuit.

State the equation that you use.

current = [3]

- (c) A battery can be made from cells connected in parallel, as shown in Fig. 8.2.

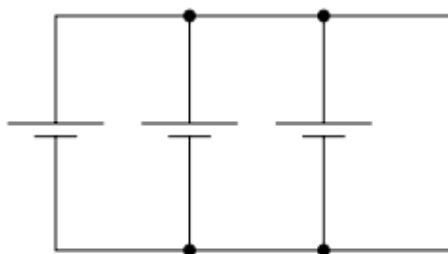


Fig. 8.2

State one advantage of connecting the cells in parallel.

.....

 [1]

60. M/J/04/Q6

Fig. 6.1 shows a circuit in which a voltmeter is placed across a resistor.

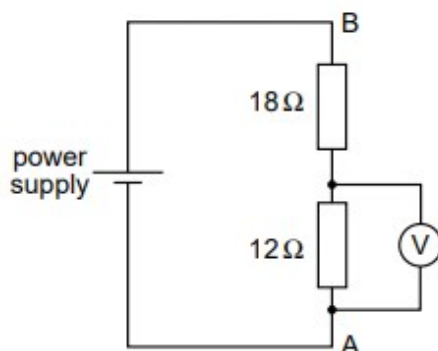


Fig. 6.1

The potential difference across the $12\ \Omega$ resistor is $4.0\ \text{V}$.

The voltmeter has three different ranges: 0 to $3.0\ \text{V}$, 0 to $6.0\ \text{V}$ and 0 to $30\ \text{V}$. The best range for use in this circuit is 0 to $6.0\ \text{V}$.

(a) Explain why

(i) using the voltmeter on the range 0 to $3.0\ \text{V}$ is unsuitable,

.....

(ii) using the voltmeter on the range 0 to $30\ \text{V}$ is unsuitable.

.....

[2]

(b) **(i)** Calculate the current in the $12\ \Omega$ resistor. State the formula that you use.

current =

(ii) Calculate the p.d. between A and B in Fig. 6.1.

p.d. =

[5]

ANSWERS

1.

Question	Answer	Marks
7(a)	 and drawn between T_1 and T_2	B1
7(b)(i)	$(I =) V / R$ or 0.40 / 2.5 0.16 (A)	C1 A1
7(b)(ii)	$(P =) VI$ or 0.40×0.16 or $(P =) I^2 R$ or $0.16^2 \times 2.5$ 0.064 (W)	C1 A1
7(c)	resistance (of the thermistor / circuit) decreases increase in current in circuit / R takes a larger proportion of the total resistance / voltage and voltmeter reading / it increases.	B1 B1

2.

Question	Answer	Marks
6(a)	(because) the current at every point in a series circuit is the same	B1
6(b)(i)	$(R =) V / I$ or 1.5 / 0.20 or 7.5 (Ω) $(R_T =) 7.5 - 1.5 - 3.5$ 2.5 (Ω)	C1 C1 A1
6(b)(ii)	resistance (of T / thermistor) decreases (as the temperature increases)	B1
	current in circuit increases or smaller fraction of total resistance	B1
	p.d. across R and S increases or (p.d. is) smaller fraction of e.m.f.	B1

3.

Question	Answer	Marks
6(a)(i)	ammeter A_2 and A_3 both 0.25 (A) voltmeter V_2 9(.0 V)	B1 B1
6(a)(ii)	$(R =) V / I$ in any form 36 (Ω)	C1 A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference provided the temperature is constant	B1 B1
6(b)	0.35 (A) seen or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$ voltage (across R) 0.35×36 or 12.6 (V) seen 19 V	B1 B1 B1

4.

Question	Answer	Marks
3(a)	($I = $) V/R in any form or (total resistance =) $700\ (\Omega)$	B1
	$0.017\ (A)$	B1
3(b)	($P = $) $I^2 R$ or VI and either V^2/R or $V=IR$ numerical or algebraic in any form	C1
	or voltage across $400\ \Omega$ resistor = (a) $\times 400$ ($6.857\ V$)	A1
	$0.12\ (W)$	
3(c)	circuit drawn with all 3 resistors in parallel with cell / power supply	B1
	ammeter in series with $100\ \Omega$ resistor or in series with resistor with largest current in arrangement drawn	B1

5.

Question	Answer	Marks
5(a)	energy / work done per unit charge or $\frac{\text{energy}}{\text{charge}}$ or $\frac{\text{work done}}{\text{charge}}$	B1
	property of a source / cell / battery / power supply or energy used driving charge around a (complete) circuit or energy transferred to electrical energy	B1
5(b)	it lasts for a longer time or more (electrical) energy available or power supply still functions if one cell fails / runs flat or can replace one cell without stopping circuit / stopping current	B1
5(c)(i)	resistance (of thermistor / circuit) decreases c.a.o.	B1
	current change or voltage change across thermistor matching the change in the resistance of the resistor	B1
	reading of voltmeter increases c.a.o.	B1
5(c)(ii)	($V_1 = $) $V_0 \times R_1 / R_0$ or $1.5 \times 4000 / (4000 + 8000)$ or $1.5 \times 4000 / 12\ 000$ or $I = V / R$ or $1.5 \times 8000 / (4000 + 8000)$ or $I = 1.5 / 12\ 000$ or $0.125\ \text{mA}$	C1
	$0.50\ V$	A1

6.

Question	Answer	Marks
8(a)	<u>electrons</u> move from the cloth / to the rod	B1
	cloth becomes positively charged	B1
8(b)	mention of <u>free</u> / <u>mobile</u> / <u>delocalised</u> / <u>sea</u> of electrons	B1
	(free) electrons present in metal and not present in plastic	B1
8(d)(i)	(region of space) where an (electrically) charged particle / charge experiences an (extra) force	B1
8(d)(ii)	at least five straight lines outside Y and none curved and perpendicular to the surface	B1
	at least one arrowhead pointing inwards and none wrong	B1
	field lines uniformly spaced at surface of sphere and no crossing / touching anywhere	B1

7.

Question	Answer	Marks
5(a)	middle box of 1st three boxes <u>and</u> top box of 2nd three boxes	B1
5(b)(i)	$(P =) VI$ or I^2R algebraic or numerical seen in any form	C1
	0.0024 W	A1
5(b)(ii)	$V = IR$ algebraic or numerical in any form or potential divider formula seen	C1
	4.8 (V) seen or $6.0 - 1.2 / 0.002$ or $3000 - 600$ or $6 = 0.002 \times (600 + R_V)$ or $1.2 = 6 \times 600 / (R_V + 600)$	C1
	2400 Ω	A1
5(b)(iii)	resistance of thermistor / X increases (on cooling)	B1
	current (in thermistor) decreases or potential divider argument used to explain decrease in pd across Y or pd across X <u>increases</u>	B1
6(a)(i)	top magnet N pole at bottom and S pole at top	B1

8.

Question	Answer	Marks
6(a)(i)	correct symbol 2 or 3 arrows onto resistor symbol	B1
6(a)(ii)	resistance (of LDR) decreases	B1
	(voltmeter reading) increases and more current in circuit or potential divider argument comparing voltage across resistor / LDR / voltage of cell	B1
6(b)(i)	$V = IR$ seen in any form or (voltage across 4 or 5 Ω) 2.4×4.0 seen	B1
	$(I =) 9.6 / 5 = 1.9(2 \text{ A})$	B1
6(b)(ii)	(current in 6.0 $\Omega =$) 4.3(2) (A) or (voltage across 6 $\Omega =$) 25.8 or 25.9(2) (V) seen or resistance of parallel resistors = $20 / 9$ (Ω) or total resistance = $8 \frac{2}{9}$ (Ω)	C1
	35 V	A1

9.

Question	Answer	Marks
5E(a)	$(t =) Q / I$ or $7.2 / 0.0016$ or $7.2 / 1.6$ or 4.5×10^N	C1
	4500 s	A1
5E(b)(i)	does not change and p.d. / voltage across resistor N does not change / ammeter 2 is not in the branch that contains the LDR	B1
5E(b)(ii)	resistance of LDR increases or current (in M / in ammeter 1) decreases	B1
	time taken increases	B1

10.

Question	Answer	Marks
6(a)	1.5 V	B1
6(b)(i)	work done / energy transferred when charge passes through a component	C1
	$\frac{\text{work done / energy transferred}}{\text{charge passed}}$	A1
6(b)(ii)	4.5 – 2.5 or 2.0 (V) seen	C1
	$(R =) V / I$ or $2.5 / (2.0 / 28)$ or $R / 28 = 2.5 / 2.0$	C1
	35 Ω	A1

11.

Question	Answer	Marks
6(a)	work done / energy (transfer) per unit time	B1
6(b)(i)	$(I =) P / V$ in any form numerical or algebraic	C1
	1.7 A	A1
6(b)(ii)	6.9–7.2 Ω	B1
6(c)	resistance (of wire) is larger	B1
	power is less and either current is smaller or $P = V^2 / R$ or $P = VI$ and $V = IR$	B1

12.

Question	Answer	Marks
8(a)	top row 3(.0) V	B1
	potential divider formula / working shown e.g. $1200 / 2000 (\times 6)$ or $800 / 2000 (\times 6)$ or split 6 (V) in ratio 1200:800 or $I = 6 / 2000$	C1
	second row 3.6 (V)	A1
8(b)(i)	correct symbol with arrows <u>towards</u> resistor 	B1
8(b)(ii)	resistance decreases and output voltage increases	B1
8(b)(iii)	thermistor	B1

13.

Question	Answer	Marks
7(a)	lower temperature / cooling of thermistor	B1
	less light on LDR	B1
7(b)(i)	energy / work per unit charge	B1
7(b)(ii)	Any 2 from: - voltage across diode - voltage across ammeter - voltage across internal resistance (of battery) (e.m.f. / voltage of battery =) $5.4 + 0.6$ (or larger)	B2
7(b)(iii)	$(E =) VQ$ or VIt in any form algebraic or numerical	C1
	3200 J	A1
7(b)(iv)	no current / no ammeter reading and diode has been reversed / diode does not conduct / has (very) high resistance / blocks current / reference to current / voltage graph	B1

14.

Question	Answer	Marks
9(a)	correct symbol for lamp	B1
	correct symbol for voltmeter placed across lamp	B1
	correct symbol for variable resistor in series with lamp and battery or used as a potential divider	B1
9(b)	start on 10 A scale (and lower if reading small) or uses 10 A scale for points from 1 to 1.8 A or uses 10 A scale for voltages above 2 V (and for 2 V)	B1
	for lower readings switch to (0-) 1 A scale	B1
9(c)	graph curved or not a straight line (through origin) or current not (directly) proportional to voltage	B1
9(d)(i)	$(R =) V / I$ in any form algebraic or numerical	C1
	at 0.40 V, resistance = $0.80 (\pm 0.03) \Omega$	A1
9(d)(ii)	at 6.0 V, resistance = $3.3\text{--}3.4 \Omega$	B1
9(e)(i)	temperature not constant / increases	C1
	resistance increases as temperature increases	A1
9(e)(ii)	(resistance is directly) proportional to length	B1
	(resistance is) inversely proportional to area or proportional to $1 / \text{area}$	B1
9(e)(iii)	any clear use of proportionality using 0.80Ω	B1
	0.050 m	B1

15.

Question	Answer	Marks
10(a)(i)	(resistance) is <u>directly</u> proportional to length	B1
10(a)(ii)	(resistance) is <u>inversely</u> proportional to (cross-sectional) area	B1
10(b)	$(R =) 6.4 \times 7.5 \times 10^{-4} / 100$ or 4.8×10^{-n} (n is an integer)	C1
	$4.8 \times 10^{-5} \Omega$	A1
10(c)(i)	energy / work done in driving a charge round a circuit or energy transferred to electrical energy	B1
	energy / work done per unit charge or energy / charge	B1
10(c)(ii)	$(R =) 6.4 + 9.6$ or 16	C1
	$(V =) 1.2 \times 6.4 / 16$ or current = $1.2 / 16$ or current = 0.075	C1
	0.48 V	A1
10(c)(iii)	trace moves vertically away from centre line	B1
	moves distance of 2.4 cm	B1
	remains horizontal or distance moved by trace directly proportional to distance moved by jockey or comment about tape	B1
10(d)(i)	the battery lasts twice as long as the single cell	B1
10(d)(ii)	no effect or trace moves in an identical manner	B1
	e.m.f. of identical cells in parallel equals the e.m.f. of a single cell	B1

16.

Question	Answer	Marks
8(a)(i)	resistance decreases as temperature increases	B1
	current increases (and voltage constant) as temperature increases	B1
8(a)(ii)	volume (of liquid/gas), expansion (of solid/liquid or gas), length (of solid), pressure (of gas) length (of bimetallic strip), e.m.f. (of thermocouple), colour (of liquid crystals or very hot objects)	B1
8(b)(i)	<u>temperature</u> of melting ice or freezing water	B1
8(b)(ii)	<u>temperature</u> of boiling water (at 1 atmosphere)	B1
8(c)(i)	work done / energy change $\div$ charge (passed through a component)	B1
8(c)(ii)	$(R =) V / I$ in any form	C1
	$(I =) 0.014(0)$ (A)	C1
	260 Ω	A1
8(c)(iii)	(V across R =) 25-3.6 or 21(.4 V) OR (total resistance =) $25 / 0.014$	C1
	1500 Ω	A1
8(d)(i)	(a thermometer where the quantity) changes a lot per unit temperature rise / $^{\circ}\text{C}$ / same temperature rise	B1
8(d)(ii)	at high temperatures	B1
	large(r) change in current / meter deflection for unit temperature rise / $^{\circ}\text{C}$ / same temperature rise or large(r) gradient	B1
8(e)	heat (thermal energy) produced in the thermistor / resistor	B1

17.

Question	Answer	Marks
8(a)(i)	electrical in first box	B1
	heat	B1
	light	B1
8(a)(ii)	energy output / energy input	C1
	<u>useful</u> output energy / <u>total</u> energy input	A1
8(b)	two decrease	C1
	all three decrease	A1
8(c)(i)	$(R =) V / I$ in any form numerical or algebraic	C1
	total current = 2.5 A or both resistances 24 (Ω) and 6(.0 Ω)	C1
	4.8 Ω	A1
8(c)(ii)	smaller area / diameter / radius	B1

18.

Question	Answer	Marks
6(a)(i)	4.0 V c.a.o.	B1
6(a)(ii)	$(V =) 12 - 4.0$ or 8.0 or $4.0 / 700$ or 5.7×10^{-3}	C1
	$(R =) V / I$ or $8.0 \times 1 / (4.0 / 700)$ or $700 \times (8.0 / 4.0)$ or $12 / (4.0 / 700)$ or 2100 (Ω)	C1
	1400 Ω	A1
6(b)	resistance of LDR decreases	B1
	current increases or p.d. across 700 Ω resistor / oscilloscope increases or p.d. across LDR decreases	M1
	trace moves up the screen	A1

19.

Question	Answer	Marks
6(a)(i)	$(I =) P / V$ or 60 / 240	C1
	0.25 A	A1
6(a)(ii)	$(R =) V / I$ or 240 / 0.25	C1
	960 Ω	A1
6(b)(i)	3.0 V	B1
6(b)(ii)	resistance is smaller	B1
	(filament at a) lower temperature	B1
6(b)(iii)	current surge (due to lower resistance when cold)	B1

20.

Question	Answer	Marks
6(a)	$(I =)V / R$ algebraic or numerical	C1
	0.4(0) A	A1
6(b)	current in Q is 0.25 (A) or (total resistance) $6 / 0.25$ or $6 / 0.65$ or $9.2(3 \Omega)$ seen	C1
	24Ω	A1
6(c)(i)	correct circuit with two lamps ammeter and battery in series	B1
6(c)(ii)	(total) resistance increases or (each) lamp has lower p.d. across it	B1

21.

Question	Answer	Marks
5(a)	adjusts the variable resistor / rheostat or change the number of cells	B1
5(b)(i)	current proportional to voltage or $V = kI$ where k is a constant	M1
	provided temperature / physical conditions unchanged	A1
5(b)(ii)	at least two values of V/I or I/V shown to be the same;	B1
5(b)(iii)	use 0–10 A range for 220 and 320 mA readings and 0–200 mA range for 100 mA reading or start with 10 A / highest range and either ∞ reduce to 200 mA for last reading ∞ reduce range and keeping high value on scale / not over scale	B1

22.

Question	Answer	Marks
10(a)	electrons c.a.o.	M1
	towards the ammeter or away from the negative terminal or towards the positive terminal	A1
10(b)(i)	thermistor c.a.o.	B1
10(b)(ii)	$1 / R_T = 1 / R_1 + 1 / R_2$ or $1 / R_T = 1 / 1.5 + 1 / 6.0$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.5 \times 6.0 / (1.5 + 6.0)$	C1
	$1.2 (\Omega)$	C1
	2.5Ω	A1
10(b)(iii)	$(I =) V / R$ or $12 / 2.5$	C1
	4.8 (A)	A1
10(b)(iv)	$I_A = I_R + I_Z$	B1
10(c)	resistance of Z / thermistor decreases	B1
	resistance of parallel combination decreases or total resistance (of circuit) decreases or current increases	B1
	voltage (across 1.3Ω) increases	B1
	trace moves towards top of screen / upwards	B1
10(d)(i)	8	B1
10(d)(ii)	1.5 V	B1

23.

Question	Answer	Marks
7(a)(i)	$1/R_T = 1/R_1 + 1/R_2$ or $1/R_T = 1/3.6 + 1/1.8$ or $(R_T =) R_1 R_2 / (R_1 + R_2)$ or $1.8 \times 3.6 / (1.8 + 3.6)$	C1
	1.2 (Ω)	C1
	4.0 Ω	A1
7(a)(ii)	$(I =) V/R$ or 6.0 / 4.0	C1
	1.5 (A)	A1
7(b)(i)	work done / energy (released) per <u>unit</u> charge (passed through component)	B1
7(b)(ii)	voltmeter symbol and across 2.8 Ω resistor	B1
7(b)(iii)	<u>total</u> resistance increases or resistance of <u>circuit</u> / <u>parallel combination</u> increases	B1
	current (in 2.8 Ω resistor) decreases	B1

24.

Question	Answer	Marks
10(a)	work done / energy transferred	B1
	work done / energy transferred in taking unit charge (through resistor)	B1
10(b)(i)	$(I =) V/R$ in any form algebraic or numerical	C1
	0.24 A	A1
10(b)(ii)	(total resistance of) 50 (Ω) seen or 2.4 (V) p.d. across 10 Ω resistor	C1
	12 V	A1
10(c)	voltmeter chosen is 0–20 V	B1
	correct explanation of why 0–200 V meter is unsuitable OR correct explanation of why 0–2 V meter is unsuitable	B1
10(d)(i)	$(P) = VI$ or $I^2 R$ or V^2 / R in any form	C1
	0.58 W	A1
10(d)(ii)	$\frac{1}{2}P$ rated resistor blows / too hot / melts	B1
10(e)	current in 10 Ω increases <u>resistance</u> decreases (of parallel combination or total resistance)	B1
	p.d. across 10 Ω increases p.d. is proportional to current or (p.d.) increases as current increases or potential divider argument	M1+A1
	p.d. across 40 Ω decreases sum of p.d.s constant / 12 V / value in (b)(ii) or p.d. across 10 Ω increases or resistance (40 Ω and R) a smaller fraction of total	B1

25.

Question	Answer	Marks
5(a)	correct circuit symbols for battery, resistor, <u>variable resistor</u> , ammeter and voltmeter	B1
	ammeter in series with resistor and voltmeter in parallel in correct series circuit	B1
5(b)(i)	$(R =) V / I$ in any form numerical or algebraic	C1
	3.5 Ω	A1
5(b)(ii)	voltage or current or power too large	B1
	calculation of power as 5(.04) W or 5.6 W or max voltage as 3.4(2) V or 3.2(4) V or max current as 0.88 A or 0.93 A	B1
5(b)(iii)	no and half voltage (2.1 V across each) or half current (0.54 A through each) or quarter power (1.13 W in each)	B1

26.

Question	Answer	Marks
7(a)(i)	$1/R = 1/R_1 + 1/R_2$ or $R_1 R_2 / (R_1 + R_2)$ or $1/R = 1/1800 + 1/9000$ or $9000 \times 1800 / 10\ 800$ or 0.00066667	C1
	1500 Ω	A1
7(a)(ii)	$(I =) V / R$ or 4.5 / 1500	C1
	0.0030 A or 3.0×10^{-3} A or 3.0 mA	A1
7(b)(i)	increases and resistance of LDR decreases	B1
7(b)(ii)	does not change and resistance / e.m.f. does not change / not affected by LDR	B1

27.

11(b)(i)	$1/R = 1/R_1 + 1/R_2$ or $R_1 R_2 / (R_1 + R_2)$ or $1/R = 1/15 + 1/60$ or $15 \times 60 / 75$ or $15 \times 60 / (15 + 60)$	C1
	12 (Ω) or 0.083 (Ω)	C1
11(b)(ii)	30 Ω	A1
	$(I =) V / R$ or 7.5 / 30	C1
11(b)(iii)	0.25 A	A1
	18 Ω resistor underlined 2 and 60 Ω resistor underlined	B1
11(b)(iv)	five cells in series and all in same direction	B1
11(b)(v)	resistance increases and current decreases	B1

28.

Question	Answer	Marks
8(a)	equal (numbers of) positive and negative charges	B1
8(b)(i)	negative charge moves from cloth to rod	C1
	electrons move from cloth to rod	A1
8(b)(ii)	apparatus needed, e.g. (small) pieces of paper / water stream / (gold leaf) electroscope / suspended or pivoted other <u>charged</u> rod / charged object / conducting object	B1
	correct statement of what is seen / felt with apparatus	B1

29.

Question	Answer	Marks
10(a)(i)	directly proportional	B1
10(a)(ii)	straight line or does not curve or constant gradient	B1
10(a)(iii)1	greater or twice as large	B1
10(a)(iii)2	straight line with half the gradient	B1
10(b)(i)1	$1/R_t = 1/R_1 + 1/R_2$ in any form e.g. $1/R = 1/20 + 1/80$ or 16 (Ω) seen	C1
	40 Ω	A1
10(b)(i)2	$(I=)V/R$ in any form e.g. 6 / 40	C1
	0.15 A	A1
10(b)(i)3	0.15 $\times$ 16 or 0.15 $\times$ 24 or 3.6 (V) seen or current split in ratio 1:4, e.g. 0.03 A and 0.12 A seen or clear attempt at potential divider formula	C1
	2.4 V	A1
10(b)(ii)1	work done $\div$ charge	B1
10(b)(ii)2	correct circuit symbol for a cell and positive correct	B1
	four cells, correct symbol, correctly in series	B1
10(b)(ii)3	four cells in series and another four in parallel or any other series and parallel arrangement of 8 cells with connections to and from battery	B1
10(c)	lasts longer or if one cell fails it still works or contains more energy	B1

30.

Question	Answer	Marks
5(a)	$1/R_t = 1/R_1 + 1/R_2$ or $R_t = R_1 R_2 / (R_1 + R_2)$ formulae or numbers using 20 and 40 Ω or 13(.3 Ω)	C1
	$R_t = R_1 + R_2$ or 20 + any attempt at parallel calculation	C1
	33(.3) Ω	A1
5(b)	$V_2 = V_3$	B1
	V_1 largest or larger than either V_2 or V_3	B1

31.

Question	Answer	Marks
11(a)	battery and ammeter symbols correct	B1
	thermistor symbol correct	B1
	all connected in series	B1
11(b)(i)1	0.05 (A) seen	C1
	$(V =) IR$ or 0.05×240	C1
	12 V	A1
11(b)(i)2	$(R =) V / I$ or 0.12 (A) seen	C1
	100 Ω	A1
11(b)(ii)	description of change in resistance in two equal changes in temperature, e.g. 240–100 and 100–48	C1
	difference values obtained are not the same, with one value shown e.g. 140 and 52 (Ω)	A1
11(c)(i)	two different metals connected together	B1
	voltmeter, ammeter or galvanometer in series	B1
	clear junction of two different metals used as the place to measure temperature	B1
11(c)(ii)	ANY 2 lines from measures high(er) temperatures more sensitive to changes in temperature measures fast(er) changing temperatures / reading can be taken more rapidly produce electronic output / output may be stored or processed / may be read remotely	B2

32.

- (a) (amount of) energy/work (dissipated by source)
by unit charge (around a circuit) M1
A1
- (b) (i) 1 they are the same B1
or $I_B = I_1 = I_2$
- 2 $E = V_1 + V_2$ B1
- (ii) $(I =) V / R$ in any form algebraic or numerical C1
0.25 A A1
- (iii) 4.5 V B1
- (iv) $(P =) VI$ or $(P =) I^2 R$ or $(P =) V^2 / R$ C1
in any form algebraic or numerical
1.1(25) W A1

(c)	current is (directly) proportional to voltage or voltage / current is a constant law holds for constant physical conditions/ constant temperature / constant pressure / for metals	B1 B1
(d) (i)	(directly) proportional or $(R) \propto 1$	B1
(ii)	inversely proportional or $(R) \propto 1/A$	B1
33.		
(a) (i)	same/equal or $I_B = I_1 = I_2$	B1
(ii)	(p.d. of) battery is sum of (p.d. across) fixed resistor and (p.d. across) the variable resistor or $V_B = V_1 + V_2$	B1
(b)	(I=) V/R numerical or algebraic in any form 0.006(0)A	C1 A1
34.		
(a) (i)	$(I =)V/R$ or $12/(6000 + 2000)$ or $12/8000$ or $12/2000$ or $12/6000$ or in (ii) $(V =)IR$ or 0.0015×6000 1.5mA	C1 A1
(ii)	9.0V	B1
(b)	(reading / it) increases resistance of LDR falls	B1 B1
(c)	light meter / sensor or automatic light switch or something sensible	B1
35.		
(a) (i)	$(I =)V/R$ or $6/60$ 0.1(0) A	C1 A1
(ii)	$(I =)P/V$ or $0.9/6$ or 0.15 (A) seen 0.25A	C1 A1
(b) (i)	lamp correctly drawn in series with resistor but not the lamp	B1
(ii)	less voltage (across lamp) because some voltage across resistor / shares voltage with resistor or less current because of effect of resistor	B1

- 36.
- (a) (i) lasts longer **or** one cell can be replaced without switching off the circuit **or** less (internal) resistance **or** if one fails the others still work B1
- (ii) 1.5 V B1
- (b) (i) $(R =)V/I$ **or** $1.5/0.075$ C1
 $20\ (\Omega)$ **or** $1.5/(0.075 - 6.0)$ C1
 $14\ \Omega$ A1
- (ii) decreases B1
 resistance of wire increases B1
- 37.
- (a) (i) energy to drive charge around a circuit **or** terminal p.d. on open circuit B1
 energy to drive **unit** charge around a circuit **or** energy/charge B1
- (ii) lasts longer **or** lower internal resistance **or** can replace a cell without switching off **or** continues to work if one cell is flat **ignore** more current (not greater e.m.f./voltage) B1
- (b) (i) $4.0\ \Omega$ B1
- (ii) $(1/R_{\text{tot}} =)1/R_1 + 1/R_2$ **or** $1/3 + 1/X$ **or** product/sum **or** $(3 \times X)/(3 + X)$ **or**
 $\frac{1}{X} = \frac{1}{2} - \frac{1}{3}$ C1
 $6.0\ \Omega$ A1
- (c) (i) $(I =)V/R$ **or** $2.0/4.0$ C1
 $0.50\ \text{A}$ A1
- (ii) (from) 0 **and** (to) 0.50 to 5.0 A B1
- (d) $I_2 = I_3 + I_X$ B1
- (e) (i) 1.0 V B1
- (ii) 1.0 V B1
- (f) (i) temperature decreases B1
 resistance decreases B1
- (ii) greater than 0.75 A (e.c.f. resistance increases in (f)(i)) B1

38.

- (a) ammeter and voltmeter correct symbols [B1]
 ammeter in series with lamp [B1]
 voltmeter in parallel with lamp [B1]
- (b) R limits or reduces the current/voltage [B1]
 otherwise lamp blows
 or more of the $50\ \Omega$ can be used to adjust voltage / current [B1]
- (c) (i) 12 V, 0.25 A correctly plotted (by eye) [B1]
 curved line from origin [B1]
 correct curvature – decreasing slope [B1]
- (ii) straight line (for fixed resistor) [B1]
lamp has changing temperature or changing resistance
 or fixed resistor has constant temperature or constant **resistance** [B1]
- (d) (i) ($I =$) V/R in any algebraic or numerical form, e.g. 12/50 [C1]
 0.24 A [A1]
- (ii) 0.49 A [B1]
- (iii) 6(.0) V [B1]
- (iv) 12(.24) Ω [B1]

39.

- (a)

A	B
6(.0) V	0 (V)

 [B1, B1]
- (b) (i) resistance (of thermistor) decreases [B1]
 current increases **and** larger voltage across $2000\ \Omega$ /fixed resistor
 or smaller **fraction** of voltage across thermistor **or** potential divider explained [B1]
- (ii) ($I =$) V/R in symbols or numbers [A1]
 0.002(0) A; 2 mA [C1]

40.

- (a) (i) work done **or** energy produced/needed [M1]
per unit charge **or** per coulomb (passing through lamp) [A1]
- (ii) 1. not straight **or** curves **or** gradient changes **or** data used correctly [B1]
2. gets hotter **or** temperature changes **or** resistance increases [B1]
- (iii) 1. 350 mA **or** 0.35 A [B1]
2. $Q = It$ in any form algebraic or numerical **or** $2 \times 60 \times 60$ **or** 7200 (s) seen [C1]
2520 C **or** 2500 C [A1]
3. ($E =$) QV or VIt in any algebraic or numerical form, e.g. $0.35 \times 6 \times 2$ [C1]
15100 J **or** 15120 J **or** 15000 J **or** 4.2 kWh [A1]
4. current and/or voltage falls/varies
or some energy remains (in cell)
or some energy/heat produced **in cell**
or correct argument involving internal resistance of cell [B1]
- (b) lasts longer [M1]
(because) larger energy (initially) **or** smaller current (in each cell) [A1]
or
avoids failure if one cell fails (M1)
(because) other (parallel) cell takes over (A1)
or
larger current/power (in external resistor/lamp) (M1)
(because) smaller (internal) resistance of combined cells (A1)
- (c) symbol for LED clear and in correct direction for cells shown [B1]
4 cells (not all in parallel or in series), switch and LED (labelled or shown) that
switches on and off properly [B1]
cells connected correctly to give 3.0 V total e.m.f., e.g. two sets of parallel cells
connected in series or three in parallel and one in series [B1]

41.

- (a) work done per (unit) charge/coulomb/C **or** energy transformed per (unit) charge/coulomb/C B1
property of a source (of electricity) **or** energy transformed to electrical energy per (unit) charge/coulomb/C B1
- (b) (i) ammeter in series B1
(ii) voltmeter in parallel with lamp **or** lamp and ammeter B1
- (c) (i) ($V =$) 2.0 (V) C1
($R =$) V/I **or** $2.0/0.70$ C1
 $2.9/2.86 \Omega$ (i.e. 2 **or** 3 s.f. only) A1
(ii) (resistance) increases B1
- (d) ($P =$) VI **or** ($P =$) V^2/R **or** $I^2 R$ **or** 12×2.0 **or** 12×0.70 C1
24 W A1
- (e) (i) emission of electrons M1
from heated metal/named metal/filament/wire A1
(ii) 1. prevents collision with air (molecules) **or** prevents deflection **or** lets electrons/particles reach screen/travel unimpeded B1
2. moves vertically (e.g. up/down/above/below **or** vertical line) **not** with horizontal movement due to this voltage B1
attracted by positive **or** repelled by negative **or** attracted by one plate and repelled by the other **or** electric field (acts on charge) B1

42.

- (a) (i) ($I =$) V/R **or** $6.0/12.0$ **or** $6.0/(4.0+8.0)$ **or** (in (ii)) ($V =$) IR **or** 0.50×4.0 C1
0.50 A A1
(ii) 2.0 V (scores C1 in (a)(i) if not already scored) A1
- (b) (i) increased **or** becomes 1.25 A B1
(ii) decreases **or** becomes 0.8Ω B1
- (c) moves up or down **or** 5.0/2.0 C1
moves up or down by 2.5 cm A1

- 43.
- (a) (i) 2 to 2.1 (V) to any value between 11 and 12 (V)
or above 2/2.1(V) B1
- (ii) temperature increases / gets hotter B1
- (b) (i) (rate of) flow of charge/electrons B1
- (ii) 0.35 A cao B1
- (iii) $(I =) V/R$ algebraic C1
or $6/20$
or $6/0.35$
- 0.3(0) (A) C1
or $1/R_T = 1/20 + 1/17.1$
or $(R_T =) 9.2 (\Omega)$ seen
- 0.65 A A1
- 44.
- (a) current is directly proportional to voltage (**accept** voltage/current = constant, but
not just = R) B1
if temperature/physical conditions constant B1
- (b) $(R =) V/I$ in any form algebraic or using any value of V and I from graph C1
 20Ω A1
- (c) (i) 40Ω or $2 \times (b)$ B1
- (ii) straight line graph through origin below given line ecf (b) (e.g. **accept** above
line if $R < 20$) M1
goes through 0.1 A at 4 V ecf (b) (e.g. allow through 0.2 A at 2 V if $R = 10 \Omega$) A1
- 45.
- (a) correct symbol for thermistor B1
in series circuit with any power supply (e.g. cell or two circles) and a fixed resistor B1
- (b) (i) 12 (V) B1
0.018 (A) B1
- (ii) $(P =) VI$ in any form algebraic or numerical with any voltage (4, 8 or 12) C1
0.14(4) W A1
- (iii) above maximum power or gets too hot or blows up or fails B1

- 46.
- (a) (i) $(V =)IR$ or 0.025×600 C1
15 V A1
- (ii) 5(V) or 5/0.025 or 800 or 800–600 C1
200 Ω A1
- (b) (i) decreases B1
- (ii) ammeter: opposite to (i) B1
voltmeter: same as ammeter (both changes correct)
- 47.
- (a) three lines from one sphere to the other **and** some lines should spread out as they leave one sphere **and** come together nearing the other B1
- correct direction on at least one line and none wrong B1
- (b) $(I =) Q/t$ in any form numerical or algebraic C1
 $2.4 \times 10^{-3} \text{ A}$ A1
- 48.
- (a) (i) **correct circuit symbols** containing, in any circuit, a
- battery/cell/d.c. power supply
 - ammeter
 - voltmeter
 - fixed resistor
- B1
- ammeter clearly measures current through W B1
- voltmeter clearly across W if W shown or a resistor if not B1
- (ii) Any 2 from
- resistance (calculated from) V/I or $V = IR$ seen
 - length (of wire), V and I **all three** measured
 - change length **and** V and I measured
- B2
- (iii) 1. resistance/resistivity changes (with temperature)
OR wire gets hot **and** melts/burns/catches fire/dangerous
OR V only proportional to I at constant temperature B1
2. use of a water bath/heat sink
OR use small currents
OR take reading (quickly) and switch off B1

(b) (i) $(V =) IR$ in any form numerical or algebraic 2(.0)V	C1 A1
(ii) 0.1(0)A	B1
(iii) (Z) has the same potential difference/voltage	B1
(Z) has less/small(er) current (thus larger resistance)	B1
(iv) (p.d. across X =) 0.3×10 (V) OR $(R_Z =) 2/0.1$ OR 20 (Ω) seen	C1
(total p.d.) 5 (V) OR 6.7 (Ω) seen OR $1/R_T = 1/R_1 + 1/R_2$ in any form numerical or algebraic OR 20/3 seen	C1
16.7 Ω ; 17 Ω ; 16.67 Ω ; 16.66 Ω	A1
49.	
(a) LED; light-emitting diode	B1
(b) energy/work done per unit charge/coulomb	B1
(c) voltage/p.d. across P one cell connected the wrong way/acting against the others	B1 B1
(d) (i) correct arrangement	B1
(ii) cells last longer; cells run down slower; one cells fails the others still work; reduces (internal) resistance; if cell removed circuit not broken	B1
50.	
(a) (i) power supply, (wire/resistor/bulb) and ammeter in series voltmeter across wire/resistor/bulb labelled/clear variable power supply or rheostat in series or potentiometer correct symbols or labelled throughout	B1 B1 B1
(ii) read ammeter and voltmeter / measure voltage and current vary power supply/rheostat/current	B1 B1
(iii) $(R =) V/I$ (ign. V/A)	B1
(b) horizontal line and above axis	B1

- 51.
- (a) ammeter in series with supply // ammeters in series with A **and** in series with B & C B1
 A across cell with no switch (condone closed switch) **not**  B1
 B and C in series with switch (closed or open) and cell B1
- (b) (i) ($R =$) V/I in any form numerical or algebraic, e.g. $8/50$, $8/0.05$ C1
 $160\ \Omega$ A1
- (ii) $50\ \text{mA}$ // $0.05(0)\ \text{A}$ B1
- 52.
- (a) circuit diagram with cell and ammeter in series with resistor/wire/lamp and B1
 voltmeter across resistor/wire/lamp B1
 $R = V/I$ in any form or gradient of V, I graph B1
 thermometer/thermocouple used or shown B1
 statement of how different temperatures obtained,
 e.g. water bath/oven/heat room/change supply voltage or current or series resistor
 to change temperature B1
- (b) (i) resistance increases with temperature M1
 (approximately) linear, proportional, straight line increase A1
- (ii) curved line starting at origin C1
 correct curvature from origin with decreasing gradient
 allow zero gradient not negative gradient A1
- (c) (i) 1 (current) increases B1
 thermistor resistance decreases B1
- 2 (voltmeter reading) increases B1
 greater fraction of voltage across resistor or potential divider equation
 explained
 or greater current through fixed/constant/ $2000\ \Omega$ resistor B1
- (ii) (voltage across thermistor) $2.2\ (\text{V})$ or attempt to use potential divider formula C1
 (current) $3.8 / 2000$ or $1.9 \times 10^{-3}\ (\text{A})$
 or $3.8 = 6 \times 2000 / (R + 2000)$ or other correct potential divider equation C1
 $1200\ \Omega$ allow $1157 - 1160$ A1

- 53.**
- (a) variable resistor **or** rheostat B1
- (b) curved line starting at origin **allow** straight at first **not** two straight lines C1
correct curvature **from origin** with decreasing gradient A1
- (c) (i) (resistance) increases (as p.d. increases) B1
- (ii) explanation, e.g. lower current than expected for given p.d. **or** given current needs larger p.d. **or** correct explanation involving $R = V/I$ **accept** gradient decreases B1
- 54.**
- (a) 12(.0) V B1
- (b) top row: 4.6 **and** 0 B1
bottom row: square 1 = square 2 + square 3 **or** 9.2 B1
bottom row: 4.6 in squares 2 **and** 3 **cao** B1
- (c) ($E=$) QV **or** VIt **or** 200×12 C1
2400 J **accept** 2370–2410 J e.c.f. A1
- 55.**
- (a) $R = V/I$ in any algebraic (e.g. $V = IR$) or numerical form C1
1200 Ω A1
- (b) decreases M1
to constant value/to 0.2 A A1
- (c) longer **or** thinner **or** hotter **or** material/made of poorer conductor (higher resistivity) B2
- 56.**
- (a) (i) 6 Ω B1
- (ii) $1/R = 1/R_1 + 1/R_2$ algebraic or numerical C1
2 Ω A1
- (b) $I = V/R$ algebraic or numerical C1
6 A ecf (ii) A1
- (c) ($I =$) 2 (A) **or** proportionality idea/potential divider idea seen C1
8 V A1

57.		
(a)	0 (V)	B1
(b)	(i) 8 Ω (i.e. accept 1 sig.fig.)	B1
	(ii) $R = V/I$ any algebraic form in (ii) or (iii)	B1
	2 A (i.e. accept 1 sig.fig.) ecf (i)	B1
	(iii) 16/8 in (ii) or (ii) $\times$ 6	C1
	12 V ecf (ii)	A1
58.		
(a)	arrow anticlockwise anywhere near top line of circuit	B1
(b)	LDR or light dependent resistor	B1
(c)	less resistance of X	B1
	same change in voltage as resistance	
	(voltage decreases alone B1)	B1
59.		
(a)	4.5 V	B1
(b)	$I = V/R$ in any form using symbols or words	B1
	4.5/15	C1
	0.3 A	A1
(c)	provides smaller (internal) resistance or lasts longer or less lost voltage or one (cell) fails others work or less heat/energy lost	B1
60.		
(a)	(i) voltage past maximum or 3V / off scale / outside range	B1
	(ii) reading less accurate or sensitive / not far up scale or smaller deflection	
(b)	(i) $V = I R$ in any algebraic format	B1
	4/12	C1
	0.33 A (accept 1/3 A)	A1
	(ii) (i) $\times$ 30 or (i) $\times$ 18 + 4 or $30 \times 4/12$	C1
	9.9 – 10 V (e.c.f (i), e.g. if (i) = 0.3, $0.3 \times 30 = 9V$ or $0.3 \times 18 + 4 = 9.4 V$)	A1
	only 1 unit error in this question	